

Program overview

14-Dec-2022 13:06

Year 2022/2023
Organization Civil Engineering and Geosciences
Education Master Civil Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
CIE Master 2022 (New Master Programme)							
CIE Year 1							
CIE Faculty Core Module (MUDE)							
CEGM1000	Modelling, Uncertainty and Data for Engineers	12					
CIE Programme Base: Mechanics and Interdisciplinary Perspectives							
CIEM0000	Interdisciplinary Mechanics and Design for CE	9					
CIE Track Construction Materials (CM)							
CIEM1000	Base: Fundamentals of Construction Materials	15					
CIEM1110	Measuring and Modelling Construction Material Behaviour	9					
CIEM1210	Construction Materials Research	15					
CIEM1220	Design and Engineering of Construction Materials	15					
CIE Track Geotechnical Engineering (GE)							
CIEM2000	Base: Geotechnical Engineering	15					
CIEM2110	Geotechnical Modelling	9					
CIEM2210	Geotechnical Structures	15					
CIEM2220	Advanced Soil Mechanics	15					
CIEM2230	Delta Geotechnics	15					
CIE Track Hydraulic Engineering (HE)							
CIEM3000	Base: Hydraulic Engineering Fundamentals and Environments Part I	15					
CIEM3110	Hydraulic Engineering Fundamentals and Environments Part II	9					
CIEM3210	Coastal Engineering	15					
CIEM3220	River Engineering	15					
CIEM3230	Advanced Design of Ports and Waterways Systems and Interventions	15					
CIE Track Hydraulic & Offshore Structures (HOS)							
CIEM4000	Base: Hydraulic and Offshore Structures	15					
CIEM4110	Hydraulic Structures	9					
CIEM4120	Offshore Structures	9					
CIEM4210	Marine Renewables	15					
CIEM4220	Dams, Dikes and Breakwaters	15					
CIEM4230	Floating and Submerged Structures	15					
CIE Track Structural Engineering (SE)							
CIEM5000	Base: Structural Engineering	15					
CIEM5110	Structural Mechanics and Dynamics	9					
CIEM5120	Design of Structural Components	9					
CIEM5130	Design of Civil Structures and Infrastructures	9					
CIEM5210	Applied Mechanics of Structures	15					
CIEM5220	Applied Dynamics of Structures	15					
CIEM5230	Concrete Structures	15					
CIEM5240	Steel and Composite Structures	15					
CIEM5250	Building Engineering	15					
CIEM5260	Transportation Infrastructures	15					
CIE Track Traffic & Transport Engineering (TTE)							
CIEM6000	Base: Traffic and Transport Engineering	15					
CIEM6110	Methods in Traffic and Transport Engineering	9					
CIEM6210	Transport Networks and Systems	15					
CIEM6220	Road Traffic Systems	15					
CIEM6230	Public Transport and Railway Systems	15					
CIEM6240	Road and Railway Engineering	15					

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Master 2022 (New Master Programme)

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Year 1

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Faculty Core Module (MUDE)

CEGM1000	Modelling, Uncertainty and Data for Engineers	12
Responsible Instructor	Dr.ir. A.A. Verhagen	
Module Manager	Dr.ir. A.A. Verhagen	
Instructor	Dr.ir. W. Daamen	
Instructor	Dr.ir. E.C. Slob	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	Dr. A. Amiri Simkoeei	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. H.R. Schipper	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Tsouvalas	
Instructor	Dr.ir. O. Morales Napoles	
Instructor	Dr.ir. R.W. Hut	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. G. Homem de Almeida Correia	
Instructor	Dr. M.F.S. Tissier	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	P.K. Krishnakumari	
Instructor	Dr. A. Cabboi	
Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Dr. D.V. Voskov	
Instructor	Dr. H. Wang	
Instructor	Dr.ir. M. Nogal Macho	
Instructor	Dr.ir. R.E.F. Lindeboom	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	E. Ragno	
Instructor	Dr. R. Taormina	
Instructor	Dr. A. Cicirello	
Instructor	J.O. Colomes Gene	
Instructor	M. Rinaldi	
Instructor	A. Jagadeesh	
Instructor	Dr. P.M.N. Mares Nasarre	
Contact Hours / Week x/x/x/x	4HC, 4W, 4PJ	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	All students have common needs at the start of their programme to further develop knowledge and skills to meet the standards of the TU Delft and its research- and design-oriented MSc programmes. The MUDE is one integrated module in which topics related to data, modelling and uncertainty quantification are applied to real engineering challenges. This module comprises two interlinked parts. The Theory, Application and Coding (TAC) part focuses on teaching and applying the fundamental concepts on modelling, uncertainty and data (MUD), as well as coding skills. In the Q1 Project part students work in multidisciplinary teams on cases in the context of a smart society, where they will apply the knowledge and skills from the TAC. In the Q2 project, the students work at the interface areas where the three topics overlap, creating opportunities for more integrated applications and the ability to focus on fields of interest per programme (when needed) while satisfying the same set of learning objectives. A gradually increasing complexity and openness of inquiry will be applied.	
Study Goals	After successfully completing the MUDE a student General - is able to describe and formulate a research question (or alternatively, design requirements) given a set problem and select the appropriate methodology and tools - is able to present a fitting work plan to investigate a set of research questions or design requirements - is able to compose a technical document using appropriate academic language and citation with references - is able to work in a collaborative group environment effectively - is able to code according to basic coding standards (e.g., consistency, readability, conciseness, structure, etc.) and collaborate with their peers via distributed control software (e.g., git) - can perform spatial / temporal / multivariate analysis of data to extract knowledge via physics-based modelling tools, data-driven approaches, and uncertainty quantification methods - can present and communicate with peers in Civil Engineering and Geosciences, results of analyses using specific modelling, uncertainty and data approaches with appropriate metrics and visualization techniques Modeling - can design a modelling framework (from problem conceptualization to governing equation setup) for a physical/engineering process - can translate the modelling framework into discretized equations and computer code - can mathematically formulate and solve an optimization problem and discuss its properties - can assess optimization and simulation models performance using a set of indicators Uncertainty - can derive relevant models and complete probability and statistic calculations for faculty-wide applications - can construct relevant models for probabilistic dependence, for example using bivariate distributions or discrete Bayesian	

	Networks - use deterministic models with probabilistic inputs to evaluate engineering questions to quantify model output uncertainty (in time and/or space) - use risk and reliability analyses concepts to describe systems, their characteristics and behaviour in time and space to support decision making under uncertainty Data - can identify parameters of interest, describe the functioning of common field/remote sensors used for measuring them, and explain resolution implications in space and time - is able to apply parameter estimation using observations and perform a quality assessment - can illustrate the procedure of gathering data from different sources, pre-processing it for analysis purposes, and sharing it according to FAIR principles
Education Method	Each week starts with a plenary session to engage students by introducing the topic and showing the relevance. There will be three blocks per week to work on Theory, Application and Coding. In each block, students will have to prepare themselves by watching videos / reading some documents / doing exercises. Support will be available. Next follows an interactive class session, where teachers and teaching assistants are available to discuss and review the materials, and to work on assignments. In addition, students spend one block on the project assignment.
Assessment	Formative assessment based on Jupyter Notebooks, in-class exercises and project deliverables. Summative assessment: - Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis, Python programming
Academic Skills	Analysis, coding, communication, visualization, project management, team skills
Literature & Study Materials	See Brightspace
Judgement	Formative assessment based on Jupyter Notebooks, in-class exercises and project deliverables. Summative assessment: - Exam at end of Q1 (resit in Q2) (25%) - Exam at end of Q2 (resit in Q3) (25%) - Project portfolio (50%) You will pass the module if you obtained at least a 5.8 for the three individual components. There will be a resit opportunity for the exams. If the grade for your project portfolio is below 5.8, you will get the opportunity to submit a revised version for which you will get a 6.0 if judged sufficient.
Permitted Materials during Exam	Pocket calculator, formula sheet
Collegerama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Programme Base: Mechanics and Interdisciplinary Perspectives

CIEM0000	Interdisciplinary Mechanics and Design for CE	9
Responsible Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Ir. R. Binnekamp	
Instructor	Dr.ir. K.N. van Dalen	
Instructor	Dr.ir. G.A. van Nederveen	
Instructor	Prof.dr.ir. A.R.M. Wolfert	
Instructor	Dr. P.J. Vardon	
Instructor	Prof.dr.ir. L.A. Tavasszy	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. R.J. Labeur	
Instructor	L.M. Leclercq	
Instructor	J.A. Arriaga Garcia	
Contact Hours / Week x/x/x/x	6 (lecture room)/ 4 (instruction room)/ 2 (assignment support)/x	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Unit 1 (4EC): Provides an overview of how the surrounding social-technical-natural-economic environment challenges CE designs and teaches students how to take this into account in design by applying principles of Systems Engineering in an Open Design Learning approach. Unit 2 (5EC): Provides students with basic concepts, equations and mathematical tools of continuum mechanics relevant to all disciplines of civil engineering, teaches them to reduce the basic problems to simpler engineering forms (i.e., Laplace-type, wave-type, diffusion-type and 29 transport-type equations), equips them with a unified and coherent framework for analysis of the obtained PDEs to analyse the system behaviour, and enables them to recognize the differences and similarities between the various mechanical problems across civil engineering.	
Study Goals	On completion of this module, the student will be able to [LO1] Know how to develop Civil Engineering Systems (CES) within the broader social, economic, natural and built environment [LO2] use the basic concepts of continuum mechanics to define a mathematical model for a CE problem. [LO3] reduce the model to a simpler (engineering) one, explain the corresponding assumptions and classify the obtained result. [LO4] solve the reduced model/problem using the appropriate technique, interpret the behaviour and its relevance to CE practice.	
Education Method	Lectures and assignments	
Assessment	- Group assignment (44%) - Written exam (56%)	
Expected prior Knowledge	Basic knowledge of mechanics and mathematics. Prior experience with programming (preferably using Python)	
Academic Skills	Unit 1: Critical and interdisciplinary design thinking, problem-based learning and solving, continuous investigation, information collection and synthesis, communication and responsiveness, time management and anticipation. Unit 2: application of theory to new situations using the relevant equations, explaining differences and similarities between problems and their physical behaviours encountered in the various domains of CE.	
Literature & Study Materials	Unit 1: Reader. Unit 2: lecture pots	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Unit 2: Closed book exam. Formula sheet will be provided	
Collegerama	Yes	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Construction Materials (CM)
--

CIEM1000	Base: Fundamentals of Construction Materials	15
Responsible Instructor	O. Copuroglu	
Instructor	Dr. F. Di Maio	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Dr. X. Liu	
Instructor	Dr.ir. F.A. Veer	
Instructor	G. Ye	
Instructor	Dr.ir. D.A. Koleva	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	Prof.dr.ir. P.C. Louter	
Instructor	O. Copuroglu	
Instructor	F. Messali	
Instructor	A.T. Gebremariam	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. M. Pavlovic	
Instructor	Dr. Y. Mosleh	
Instructor	Dr. M. Fotouhi	
Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	x/10/8/x	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The field of construction materials explores the chemical roots of mechanical behaviour. Therefore, the fundamentals of material chemistry and mechanics form the base of the CM-track. In the first part of the track base (during Q2), two units will be offered; Science of Construction Materials (5EC) and Fracture Mechanics(4EC), which provide the students with composition, manufacturing, processing, and mechanical behaviour aspects of construction materials. These fundamental subjects will form a sound basis for the construction materials selection (2.5EC) and technology (3.5EC) in the second part of the track base (during Q3). The details of the module content are as follows: MODULE: FUNDAMENTALS OF CONSTRUCTION MATERIALS Module coordinator: Dr. O. Copuroglu Unit 1: Science of Construction Materials (Q2, 5EC) Unit coordinator: Dr. O. Copuroglu U1: Introduction; Chemistry of Construction Materials; Phases and Microstructures U1: Thermodynamic concepts; First and Second Law of Thermodynamics U1: Earth Materials; Aggregates U1: Portland Cement U1: Electrochemistry U1: Glass; Metals; FRC U1: Wood and Timber; Bituminous Materials U1: Polymers and Organic Additives; Recycled Construction Materials; Sustainability Aspects Unit 2: Fracture Mechanics (Q2, 4EC) Unit coordinator: Dr. F. Messali U2: Introduction: Theory of Elasticity U2: Theory of Griffith; Linear Elastic Fracture Mechanics U2: Linear Elastic Fracture Mechanics U2: Mixed Mode Fracture U2: Elastic/Plastic Fracture Mechanics U2: Quasi-Brittle Failure; The Cohesive Zone Model U2: Fracture in Timber; Environmental Influences on Fracture U2: Fracture Mechanics in Steel; Implementation of the Prediction Methods in Real Case Situations Unit 3: Construction Materials Selection (Q3, 2.5EC) Unit Coordinator: Dr. ir. F. Veer U3: Engineering Materials in the Build Environment 1800-2050; Properties of Engineering Materials, the Logic of the Ashby Diagram U3: Criteria for Materials Selection U3: Materials Selection based on Optimized Performance U3: The Shape Factor as a parameter and Integrated Materials U3: Optimizing using Multiple Criteria, Combined structural and Eco-Impact Optimization U3: Finding if a design concept works U3: Finding if a concept material is useful. Designing concept materials using the synthesizer U3: Ethics of materials selection.	

	Unit 4: Construction Materials Technology (Q3, 3.5EC) Unit Coordinator: Dr. G. Ye U4: Introduction to Concrete Technology; Microstructural of Concrete; Introduction to Asphalt Technology U4: Pore Structure in Concrete.; Fresh concrete properties; Aggregates properties U4: Early Age Concrete Properties; Basic Properties of Asphalt cement U4: Volume Stability: Shrinkages; Creep and Relaxation; Admixtures for (asphalt) concrete U4: Transport and Concrete Durability. Concrete Mix Design. Asphalt concrete mix design U4: Concrete Construction Practices; Curing and Quality Control of Concrete U4: Concrete Recycling. Concrete Sustainability. Reuse and Recycling in maintenance of Asphalt pavements U4: Special Concrete Types and Applications; Special asphalt concrete types and applications.
Study Goals	On completion of this module, the student will be able to 1. Explain the chemical roots of mechanical behaviour of construction materials (1, 4-7) 2. Categorize construction materials based on their composition, engineering properties and function (1, 8) 3. Design and predict engineering properties of construction materials based on composition and production methods (1-7) 4. Quantify sustainability of construction materials. (1, 5) 5. Analyse, model and predict failure of construction materials (1) 6. Apply fracture mechanics principles to develop materials more resistant to cracking (1,7) 7. Select construction materials based on functional requirements, economic, environmental and ethical considerations (3, 4) 8. Design mixture of concrete, asphalt and other composite materials for application-related requirements (1, 2, 3, 6) (with link to final attainments of CM track in parentheses)
Education Method	Lectures, in class interactive experiment/demo, lecture with in-class exercises, lab practical, lectures on theory and interactive lectures on case studies
Assessment	You complete the module successfully when passing the following summative assessments: Unit 1: Science of construction materials (BASE-1) written exam (33%) Unit 2: Fracture mechanics (BASE-2) written exam (27%) Unit 3: Materials selection in civil engineering (BASE-3) written exam (17%) Unit 4: Construction materials technology (BASE-4) written exam (23%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Expected prior Knowledge	Prerequisites to be able to start the full module BSc Civil Engineering or Applied Science (or equivalent). Faculty base and CE programme base.
Academic Skills	Fundamental Civil Engineering skills required to enter the CE MSc. program
Literature & Study Materials	Lecture notes of the units will be provided via Brightspace. Unit 2: Textbook (CT Sun and Z-H Lin, Fracture Mechanics, 2012, or alternative) complemented with handouts on selected topics Unit 3: Materials selection in mechanical design, M. F. Ashby, 2016
Judgement	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Unit 1: Formula sheet and calculator. Unit 2: Formula sheet and calculator. Unit 3: Laptop and the course book (by Ashby) Unit 4: Formula sheet and calculator
Collegerama	The unit coordinators will inform the students in case they may use additional materials during the exam. No

CIEM1110	Measuring and Modelling Construction Material Behaviour	9
Responsible Instructor	F.P. van der Meer	
Instructor	Ir. C. Kasbergen	
Instructor	Ir. P.A. de Vries	
Instructor	G. Ye	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	Dr.ir. D.A. Koleva	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	R. Esposito	
Instructor	O. Copuroglu	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. Y. Mosleh	
Instructor	Dr. M. Fotouhi	
Contact Hours / Week x/x/x/x	12/x/x/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Use of materials for structural applications requires quantification of the material performance. Such quantification is related to two scientific disciplines: experimental characterization and numerical modelling. Safe engineering with construction materials requires that material behaviour has been experimentally tested under controlled conditions and that the performance of the material under expected conditions can be predicted with simulations. In materials science, the two disciplines meet when numerical simulations are used to interpret laboratory experiments and when experiments are used to calibrate and validate models. This module provides students in the construction materials track with a solid basis in both disciplines of measuring and modelling. Students will learn about using advanced experimental techniques that can be used in the laboratory and on-site to monitor and characterise construction material performance. Furthermore, they will be trained to interpret measured data and determine consequences for their short and long-term performance in the specific application. This module will also prepare students for the development and application of multi- scale and multi-physics modelling techniques to study the behaviour of construction materials and to explain deterioration and damage accumulation.	
Study Goals	On completion of this module, the student will be able to 1. Set up a computational model for simulation of the behaviour of materials 2. Perform linear and nonlinear finite element analysis for mechanical and transport problems 3. Calibrate and validate computational models with experimental observations 4. Set up a testing plan for characterisation of materials to define input parameters for structural analysis 5. Analyse experimental measurements to characterise the performance of construction materials 6. Advise structural engineers on characterisation and modelling strategies for describing material behaviour 7. Quantify uncertainty in models and measurements	
Education Method	Videos, interactive lectures, computer lab sessions, lectures, lab experiments.	
Assessment	Assessment of the modules is done with three examinations: - Written exam 1: Numerical modelling (31%) - Written exam 2: Experimental characterization (39%) - Presentation based on combined assignments (30%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Academic Skills	BSc Civil Engineering or Applied Science (or equivalent). Faculty base and CE programme base.	
Literature & Study Materials	Analytical thinking, coding, data collection and processing, reporting.	
Judgement	Lecture notes will be provided via Brightspace.	
Permitted Materials during Exam	See Assessment	
Collegerama	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
	None	
	No	

CIEM1210	Construction Materials Research	15
Responsible Instructor	Dr.ir. B. Šavija	
Instructor	Ir. C. Kasbergen	
Instructor	Prof.dr.ir. L.J. Sluijs	
Instructor	G. Ye	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	Dr.ir. D.A. Koleva	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	M. Lukovi	
Instructor	O. Copuroglu	
Instructor	Dr. K. Anupam	
Instructor	F. Messali	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. M. Fotouhi	
Instructor	N.G. Ozerkan	
Contact Hours / Week x/x/x/x	18/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Module B1 is designed to offer students the opportunity to develop indepth scientific insight into the behaviour of construction materials under various environmental and loading conditions. Specifically, students will learn about degradation mechanisms and durability (5EC), imposed deformations (5EC), and constitutive modelling (5EC).	
Study Goals	On completion of this module, the student will be able to 1. Define durability concepts of construction materials, describe material properties and their effect on durability. (1, 2, 7) 2. List and explain physical, chemical and biological degradation mechanisms of construction materials. (2) 3. Perform service life design of structures based on material properties. (2) 4. Set-up repair and maintenance strategies for degraded structures (7) 5. Analyse and predict the influence of eigenstresses in materials on the behaviour of structural elements and components (1, 2) 6. Explain how environmental effects and different gradients on material level and structural level impose deformations and stresses (1, 2) 7. Formulate constitutive models to describe the linear/nonlinear response of materials and implement them in FEM codes (1, 7) 8. Select and apply constitutive models for various physicochemical material behaviours. (1) (with link to final attainments of CM track in parentheses)	
Education Method	Lectures, practicals with hands on exercises, workshop, case studies, computer lab sessions.	
Assessment	You complete the module successfully when passing the following 4 summative assessments: B1-1 Durability: Written exam (24%) B1-1 Durability: Report (10%) B1-2 Imposed: Oral exam (33%) B1-3 Constitutive: Oral exam (33%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Prerequisites to be able to start the full module BSc Civil Engineering or Applied Science (or equivalent). Faculty base and CE programme base.	
Literature & Study Materials	Lecture notes will be provided via Brightspace.	
Judgement	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

CIEM1220	Design and Engineering of Construction Materials	15
Responsible Instructor	A. Varveri	
Instructor	Dr. F. Di Maio	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Dr. X. Liu	
Instructor	G. Ye	
Instructor	Prof.dr.ir. J.W.G. van de Kuilen	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Dr. H.M. Jonkers	
Instructor	F.P. van der Meer	
Instructor	Dr.ir. M. Ottele	
Instructor	Dr.ir. B. Šavija	
Instructor	A. Varveri	
Instructor	O. Copuroglu	
Instructor	Dr.ir. J.A. Pascoe	
Instructor	A.T. Gebremariam	
Instructor	Dr. Y. Mosleh	
Instructor	Dr. M. Fotouhi	
Contact Hours / Week x/x/x/x	14/x/x/x	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Module B2 focuses on the design, performance and engineering of smart materials, self-healing materials, composite materials, bio-based materials, and emerging new materials. Innovative technologies such as 3D printing will receive special attention as well as recycling and renewable construction materials. The module is organized in three units: (1) The first module unit B2-1 on Recycling and renewable construction materials considers the design and engineering of construction materials from the perspective of recycling and sustainability. (2) The second module unit B2-2 on Advanced construction materials and smart technologies focuses on design and performance of new high-performance or multifunctional materials including new manufacturing technologies. (3) The third module unit B2-3 on Bio-based/hybrid/composite materials in construction technology addresses the design and engineering performance of bio-based and polymer-based composite and hybrid materials.	
Study Goals	On completion of this module, the student will be able to 1. Advise on the use of construction materials and production processes for reliable and sustainable structural design (1, 3, 8) 2. Design potential routes for recycling and reusing of EoL (End-of-Life) construction materials 3. Analyse the environmental impact of construction materials in their complete life cycle (4) 4. Develop strategies for sustainable use of materials in construction (4,7) 5. Facilitate the use of new, high performance and sustainable materials (e.g. UHPFRC, SHCC, geopolymers, bio-based materials, epoxy-modified bitumen) in structural engineering (5) 6. Evaluate current trends in the development of smart and sustainable construction materials (5) 7. Explain the opportunities and challenges of additive manufacturing technologies (such as 3D printing) in construction (3) 8. Analyse the short- and long-term behaviour of polymer and bio-based composite materials (1,5,7)	
Education Method	Interactive lectures, lab hands-on practicals, lab demonstrations, industrial visit, assignments	
Assessment	Students complete the module successfully when passing the following four (4) summative assessments: B2-1 Recycling: Lab report (17%) and written exam (17%) B2-2 Smart materials: Report portfolio (33%) B2-3 Bio-base & Composite: Written exam (33%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Construction Materials track base module.	
Academic Skills	Collection and synthesis of information, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate with their colleagues, produce reports and present their work, so organizational, interpersonal, writing and presentation skills will be also developed.	
Literature & Study Materials	Lecture notes, slides and handouts will be provided in Brightspace.	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Calculator	
Colleggerama	No	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Geotechnical Engineering (GE)
--

CIEM2000	Base: Geotechnical Engineering	15
Responsible Instructor	Dr. K.G. Gavin	
Instructor	Dr.ir. R.B.J. Brinkgreve	
Instructor	Prof.dr.ir. C. Jommi	
Instructor	S. Muraro	
Instructor	M. Korf	
Responsible for assignments	M.A. Cabrera	
Contact Hours / Week x/x/x/x	Contact hours: Q2: 4+6 hr per week Q3: 6 hr per week	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The Geotechnical Engineering Track Base Module includes three units, Theory and modelling of geo-processes (3 ECTS), Testing and modelling of soil behavior (6 ECTS), and Foundations and Excavations (6 ECTS). Students will learn how to describe mechanical behavior (stress-strain response) of soils and rocks based on laboratory testing and data analysis. The concepts of constitutive modelling will be introduced, and the applications are discussed. The observed soil behavior from standard geotechnical laboratory tests will be presented and students will learn to analyze and interpret the test data to translate them into mathematical formulations and model parameters needed for engineering design. Formulation, assessment, and evaluation of the soil coupled hydro-mechanical response in engineering applications will be covered. The students will use the derived soil parameters in common design approaches to predict the behaviour of foundations and excavations. Case studies where the actual response of structure was measured will be used to promote reflective learning. Moreover, the interaction between the subsurface and geotechnical structures as well as the impact of the environment on the design, construction and performance of geotechnical structures, and vice versa will be presented. Topics such as communicating with stakeholders from all relevant backgrounds about geotechnical problems and solutions will be presented and discussed. In Track Base, the learning line of Monitoring and Data Science is covered within all three units. The basics of topics related to the monitoring of geotechnical structures will be introduced in Foundations and excavations. Additionally, all three units provide opportunities for elaborating the data, and development of models for predicting soil and geotechnical structures behavior. Such elaboration is guided throughout the module for students to learn how to process/visualize/present information for clients and stakeholders. This module will add to the learning line on Ethics related to design of geotechnical structures considering issues such as usability/safety of geotechnical systems, evaluation, and assessment of the interaction between the subsurface and geotechnical structures as well as effective Communication with stakeholders.	
Study Goals	On completion of this module, the student will be able to: C-LO1: Describe the mechanical behavior (stress-strain response) of soils and rocks based on laboratory testing and data analysis C-LO2: Recognize and apply the concept of constitutive modelling C-LO3: Analysis and interpret laboratory tests and derive and apply mathematical formulations for engineering design C-LO4: Evaluate and assess the interaction between the subsurface and geotechnical structures C-LO5: Recognize, and determine the impact of the environment on the design, construction, and performance of geotechnical structures, and vice versa C-LO6: Communicate the results with experts from other disciplines and other stakeholders C-LO7: Assess the impact of hydro-mechanical coupling on the soil response	
Education Method	Lectures, individual and group assignments, laboratory practicals and site visits. See unit descriptions on Brightspace for details.	
Assessment	The module includes in total 4 summative assessments. Three of them are unit-based. The 4th summative assessment is an overarching group assignment, which covers LOs and aspects of all three units. The student groups need to elaborate on the disciplinary connections among the different units in the module. Formative assessment moments are organized within each unit of the module. The distribution of marks is: Theory and modelling of geo-processes - Written Exam (15%) Testing and modelling of soil behaviour - Written Exam (30%) Foundations and excavations - Oral Exam (30%) GE Track Base - Group Assignment (25%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Academic Skills	Attendance of Faculty and Program Base Modules	
Literature & Study Materials	Critical thinking; application of theory to new situations using the relevant methods; scientific writing	
Judgement	Lecture Notes, Readers, Scientific papers, Lecture slides	
Permitted Materials during Exam	See Assessment	
Collegerama	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Communicated in advance by responsible teacher.	
	No	

CIEM2110	Geotechnical Modelling	9
Responsible Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Dr. P.J. Vardon	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Dr.ir. A.P. van den Eijnden	
Contact Hours / Week x/x/x/x	0/0/14/0 4 (Unit 1) + 10 (Unit 2) = 14 contact hours (variable per week) in Q3	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The module covers two complementary processes in geotechnical engineering: Engineering Geology and Numerical Modelling. A geological perspective is a prerequisite of a complete geotechnical analysis. Once this interpretation of the geology in an engineering context is established in the Engineering Geology unit, the resulting ground model provides the starting point for geotechnical analyses and predictions by numerical models. The Numerical Modelling unit covers the theory and application of these models to come to quantitative predictions, as well as the interpretation of the results in the geotechnical context, because after all a number is not an answer. Parts of this module follow the learning line on Data and Monitoring by covering 1) the manipulation of both sparse and dense data, and both soft and hard data to highlight ground related risks and opportunities, 2) the interpretation of numerical analysis results in the context of data- and model uncertainty, 3) uncertainty propagation in (numerical) models. Parts of this module follow the learning line on Ethics by considering ethical aspects in feasibility studies conducted in Engineering Geology. This allows students to apply moral frameworks to the allowable actions as a CE Engineer. The Numerical modelling unit covers modelling techniques which are at the heart of various value sensitive methods for CE designs to be understood in the learning line on Ethics. Keywords: Engineering geology; desk study; soil and rock properties; conceptual ground and ground water flow modelling; anticipated ground conditions; geo-hazards and geotechnical issues; feasibility study; finite element method; numerical models; engineering solutions; sensitivity analysis;	
Study Goals	On completion of this module, you should be able to: A-LO1: Acquire conceptual (hydro)geological knowledge on project site; A-LO2: Integrate (hydro-)geological information and knowledge into ground models and estimate resulting uncertainty for a variety of geotechnical applications A-LO3: Formulate, explain and solve the (equations of the) finite element method in the context of numerical modelling of geotechnical applications; A-LO4: Design and execute numerical simulations for geotechnical applications; A-LO5: Critically evaluate the results and the limitations of model solutions for geological and geotechnical problems, considering data and model uncertainty; A-LO6: Communicate results of geo-engineering investigation and numerical simulations to stakeholders in terms of geo-risks and opportunities, in the context of a responsible use of the subsurface for multi-purposes.	
Education Method	Lectures, tutorial sessions, question hours, exercises, assignments, oral presentation, peer and self-evaluation, coursework, field excursions	
Assessment	Numerical modelling in Geotechnical Engineering - Written Exam (4 ECTS) Engineering Geology - Written Exam (5 ECTS) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Attendance of MUDE + Program core + track base units Q2	
Academic Skills	Critical and interdisciplinary thinking, problem-based learning and solving, continuous investigation, information collection and synthesis, communication, technical writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations. Making inferences of physical situations by analysing observations/data.	
Literature & Study Materials	Lecture slides, scientific papers, software packages, accompanying lecture notes Selected chapters out of textbooks: 1) Programming the Finite Element Method, 5th edition - Smith, Griffiths & Margets - 2014 John Wiley & Sons, Ltd 2) Waltham: Foundation of Engineering Geology, 3) Fetter: Applied hydrogeology, and 4) ICE manual of geotechnical engineering, Volume I, Geotechnical Engineering Principles, Problematic Soils and Site Investigation	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Unit 1: handwritten notes and completed assignments Unit 2: all permitted, including laptop with internet connection disabled.	
Collegerama	No	

CIEM2210	Geotechnical Structures	15
Responsible Instructor	Dr. G. Giardina	
Instructor	Dr.ir. W. Broere	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Dr. G. Giardina	
Instructor	Prof.dr. M.A. Hicks	
Instructor	Dr.ir. C. Zwanenburg	
Instructor	Prof.dr.ir. C. Jommi	
Instructor	S. Muraro	
Instructor	Drs.ir. R.E.P. de Nijs	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Dr.ir. A.P. van den Eijnden	
Contact Hours / Week x/x/x/x	6,6,6+4 (18 hours of lectures, 4 hours practicals)	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module covers the design, monitoring and assessment of geotechnical structures. It addresses the measurement, characterization and numerical modelling of spatial variability, the evaluation of risk and uncertainty in geotechnical engineering, the interaction between soil layers and the geotechnical structure, and the impact of geotechnical structures on other structures in an urban environment. Applications include the design of geotechnical structural elements (e.g., tunnel lining, soil reinforcement). The learning line on Monitoring and Data Science is included through the analysis of conventional and innovative geotechnical monitoring, and the evaluation of data from site investigation and physical modelling (Site Investigation modelling, Risk and variability in geotechnical engineering). Parts of this module follow the learning line on Ethics by considering ethical aspects in geo-risk management. In the Site Investigation and monitoring unit, students will gain understanding into approaches used to reduce risk in geotechnical engineering. Ethical aspects associated with human factors and risk communication in geo-engineering will be also covered.	
Study Goals	On completion of this module, the student will be able to: B1-LO1: describe the nature, impact and characterisation of soil spatial variability. B1-LO2: explain the relationship between soil spatial variability and geotechnical design and risk assessment. B1-LO3: evaluate advantages and shortcomings of different alternative choices for the design of infrastructure. B1-LO4: evaluate the design and construction methods for different types of infrastructure. B1-LO5: identify suitable site investigation techniques and critically analyse the results of a site investigation survey. B1-LO6: analyse geotechnical problems based on physical and numerical modelling techniques.	
Education Method	Lectures, individual and group assignments, laboratory practicals and site visits. See unit descriptions on Brightspace for details.	
Assessment	The module includes in total 4 summative assessments, transversal to the LOs of the module. 3 of them are unit-based, with an attention for connecting the assessment of LOs that are common to more than one unit. Formative assessment moments are organised within each unit of the module. They include an individual assignment (30%), a report (30%) and an oral exam (30%). The 4th summative assessment, included in the unit Geotechnical Design, is a group design assessment which addresses disciplinary connections among the different units in the module. (10%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Attendance of Track Base and Module A of GE track	
Academic Skills	Critical thinking; interdisciplinary thinking; problem-based learning and solving; application of theory to new situations using the relevant equations; time management; making inferences of physical situations by analysing observations/data.	
Literature & Study Materials	See Brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Assignments may be carried out by using all the teaching materials provided during the course and further background material as needed. The report on the group assignment can be used as reference, no further materials are needed for the oral exam.	
Collegerama	No	

CIEM2220	Advanced Soil Mechanics	15
Responsible Instructor	M.A. Cabrera	
Instructor	Dr.ir. R.B.J. Brinkgreve	
Instructor	Prof.dr. M.A. Hicks	
Instructor	Dr.ir. C. Zwanenburg	
Instructor	S. Muraro	
Instructor	F. Pisanò	
Instructor	M.A. Cabrera	
Contact Hours / Week x/x/x/x	Contact hours: 16 hours lectures + 4 hours lab per week	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module includes three units, namely Mechanics of Deltaic Soils, Experimental Soil Mechanics, and Soil Dynamics. Students will dive into non-standard aspects of Soil Mechanics, both in terms of analysis/modelling techniques and geotechnical applications. From a methodological standpoint, the module covers advanced topics regarding the testing and constitutive modelling of soils, up to the solution of boundary value problems through both physical and numerical modelling techniques. Relevant applications include the analysis of geotechnical problems in soft deltaic soils, and/or in the presence of cyclic/dynamic loading conditions. In this module the learning line on Monitoring and Data Science is enriched with further content. In particular, the unit on Experimental Soil Mechanics addresses topics related to the monitoring of physical geo-models during small-scale centrifuge experiments, including the selection of the most suitable sensors for real-time measurements. Overall, the three units provide opportunities for elaborating the data, often of big size, returned by physical and numerical models. Such elaboration is guided throughout the module for students to learn how to process/visualise/present information for clients and stakeholders. Ethical aspects associated with Advanced Soil Mechanics subjects are discussed throughout the module. Given the emphasis on experimental and numerical analyses as essential input to geotechnical design, important implications are pointed out regarding the impact of (i) selection of testing methods, (ii) identification of soil model parameters, (iii) validation of physical/numerical models on the usability/safety of geotechnical systems. Particularly, conflicts of interest, and their potentially catastrophic consequences, between (financial benefit of) service providers and (satisfaction/safety of) clients is discussed throughout the three units from a mechanics-oriented perspective.	
Study Goals	On completion of this module, the student will be able to: B2-LO1: analyse the role of advanced, time-dependent features of soils' hydro-mechanical behaviour B2-LO2: identify and integrate testing and constitutive modelling strategies for soils subjected to environmental and man-induced perturbations B2-LO3: select and perform advanced analyses of geotechnical problems based on most appropriate physical/numerical modelling techniques B2-LO4: critically reflect on benefits and limitations of advanced physical and numerical modelling strategies in geotechnical engineering B2-LO5: effectively report in writing the outcome of advanced experimental and computational analyses in relation to geotechnical engineering applications	
Education Method	MECHANICS OF DELTAIC SOILS Introduction, overview and issues with deltaic soils Week 1 (6 hrs): Lectures with active teaching including practical examples, quizzes and discussions. Instr.: M. Hicks, C. Zwanenburg Time-dependent behaviour, creep; Assignment 1a Week 2 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: R. Brinkgreve, C. Zwanenburg Stress path-dependent shear strength, anisotropy, fabric, bonding; Assignment 1b Week 3 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: R. Brinkgreve, C. Zwanenburg Issues related to loose soils, static liquefaction; Assignment 1c Week 4 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: M. Hicks, R. Brinkgreve Advanced numerical modelling of large deformations; engineering and design aspects of geotechnical projects; introduction of case study Week 5 (6 hrs): Lectures with active teaching including practical examples, problem-based learning through assignments, quizzes and discussions Instr.: M. Hicks, C. Zwanenburg, R. Brinkgreve Start numerical modelling of case study Week 6 (6 hrs): Practical example, problem-based learning through numerical simulation Instr.: C. Zwanenburg, R. Brinkgreve Elaboration of case study and results; Assignment 1d Week 7 (6 hrs): Practical example, problem-based learning through numerical simulation Instr.: C. Zwanenburg, R. Brinkgreve EXPERIMENTAL SOIL MECHANICS Introduction & soil classification week 1 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, laboratory activities Instr.: M. Cabrera	

Triaxial testing
 Week 2 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies,
 Instr.: M. Cabrera, S. Muraro

Direct simple shear and ring shear test
 Week 3 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies
 Instr.: M. Cabrera, S. Muraro

Small strain stiffness
 Week 4 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises
 Instr.: M. Cabrera, S. Muraro

In situ testing
 Week 5 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, visit to geotechnical companies
 Instr.: M. Cabrera

Physical modelling
 Week 6 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies
 Instructor(s): M. Cabrera

Physical modelling
 Week 7 (8 hrs): Lectures with active teaching including demonstration videos, practical examples, problem-based learning through class exercises; inquiry based learning, laboratory activities; visit to geotechnical companies
 Instr.: M. Cabrera

SOIL DYNAMICS

Waves in Soils
 Weeks 1-2 (12 hrs): lectures with active teaching,
 Instr.: F. Pisano

Man-induced and environmental soil vibrations
 Week 3 (6 hrs): lectures with active teaching
 Instr.: F. Pisano

From linear to non-linear cyclic/dynamic soil behaviour
 Week 4 (5 hrs): lectures with active teaching
 Instr.: F. Pisano

Seismic site response
 Week 5 (6 hrs): lectures with active teaching
 Instr.: R. Brinkgreve, F. Pisano

Computational soil dynamics
 Weeks 6-7 (12 hrs): lectures with active teaching
 Instr.: R. Brinkgreve, F. Pisano

Assessment

The module includes in total 4 summative assessments, which are all transversal to the LOs of the module. 3 of them are individual and more pronouncedly unit-based, with an attention for connecting the assessment of those LOs that are common to more than one unit. The 4th summative assessment is an overarching oral exam, during which students will be challenged to elaborate on the disciplinary connections among the different units in the module. The oral exam will be taken in couples and will build around advanced geotechnical applications preliminarily suggested by the teachers. Formative assessment moments will be organized within each unit of the module.

Module assessment will be overall distributed as follows:

3 assignments - 75% of the total (3x25%)
 1 oral exam - 25% of the total

Expected prior Knowledge Academic Skills

A successful participation in the module, requires a minimum evaluation of 5.8 in all units.

Attendance of Track Base and Module A in the Geotechnical Engineering Track

Critical thinking; problem-based learning and solving; application of theory to new situations using the relevant equations; making inferences of physical situations by analyzing observations/data; scientific writing

Literature & Study Materials

MECHANICS OF DELTAIC SOILS

tbd

EXPERIMENTAL SOIL MECHANICS

- Head, K.H. (1980). Manual of soil laboratory testing. London: Pentech press. (selected chapters)
- Das, B.M. (2010). Geotechnical engineering handbook. J. Ross publishing. (selected chapters)
- Taylor, R.N. (2018). Geotechnical centrifuge technology. CRC Press.

Further suggested readings (research papers) will be provided during the course.

SOIL DYNAMICS

- Lecture slides
- Kramer, S.L. (1996). Geotechnical Earthquake Engineering. Prentice-Hall.

	- Towhata, I. (2008). Geotechnical Earthquake Engineering. Springer Science & Business Media. - van Dalen, K.N. (2015). Body Waves in an Elastic Continuum & Rayleigh Waves at the Free Surface. Lecture notes - Zienkiewicz, O.C., Chan, A.H.C., Pastor, M., Schrefler, B.A., & Shiomi, T. (1999). Computational Geomechanics -- with special reference to Earthquake Engineering. Wiley - Holscher, P. (2016). Soil Dynamics in Urban Areas. Lecture notes (Chapters 9-10) (provided through Brightspace) Further suggested readings (research papers) will be provided during the course.
Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Take-home assignments may be carried out by using all the teaching materials provided during the course. No further materials needed for the final oral exam.
Collegerama	No

CIEM2230	Delta Geotechnics	15
Responsible Instructor	Dr.ir. W. Broere	
Instructor	Dr.ir. R.B.J. Brinkgreve	
Instructor	Dr.ir. W. Broere	
Instructor	Dr.ir. D.J.M. Ngan-Tillard	
Instructor	Dr. G. Giardina	
Instructor	Prof.dr. M.A. Hicks	
Instructor	Dr.ir. C. Zwanenburg	
Instructor	Prof.dr.ir. C. Jommi	
Instructor	S. Muraro	
Instructor	Drs.ir. R.E.P. de Nijs	
Instructor	Dr.ir. A.A.M. Dieudonné	
Instructor	Dr.ir. A.P. van den Eijnden	
Instructor	M.A. Cabrera	
Contact Hours / Week x/x/x/x	x/x/Q4/x 6,6,6+4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Geotechnical Engineering in delta areas, in the Netherlands and worldwide, must face challenges associated to soft, and possibly unstable, soils. Such challenges must inevitably be tackled at different levels, i.e., from fundamental understanding and simulation of deltaic soil behaviour, to their in-situ characterisation for geotechnical design. More specifically, the module covers the design, monitoring and assessment of geotechnical structures. It addresses the mechanics of deltaic soils, the interaction between soil layers and the geotechnical structure, and the impact of geotechnical structures on other structures in an urban environment. Applications include the design of geotechnical structural elements (e.g., tunnel lining, soil reinforcement). The learning line on Monitoring and Data Science is included through the analysis of conventional and innovative geotechnical monitoring, and the evaluation of data from site investigation and physical modelling (Site Investigation & Monitoring). Parts of this module follow the learning line on Ethics by considering ethical aspects in geo-risk management. In the Site Investigation & Monitoring unit, students will gain understanding into approaches used to reduce risk in geotechnical engineering. Ethical aspects associated with human factors and risk communication in geo-engineering will be also covered.	
Study Goals	On completion of this module, the student will be able to: B3-LO1: describe testing and constitutive modelling strategies for soils subjected to environmental and man-induced perturbations. B3- LO2: explain the role of advanced, time-related features of soils' hydro-mechanical behaviour. B3- LO3: evaluate advantages and shortcomings of different alternative choices for the design of infrastructure. B3- LO4: evaluate the design and construction methods for different types of infrastructure. B3- LO5: compare the various methods for site investigation. B3- LO6: analyse geotechnical problems based on physical and numerical modelling techniques	
Education Method	Lectures, individual and group assignments, laboratory practicals and site visits. See unit descriptions on Brightspace for details.	
Assessment	The module includes in total 4 summative assessments, transversal to the LOs of the module. 3 of them are unit-based, with an attention for connecting the assessment of LOs that are common to more than one unit. Formative assessment moments are organised within each unit of the module. They include an individual assignment (30%), a report (30%) and an oral exam (30%) The 4th summative assessment, included in the unit Geotechnical Design, is a group design assessment which addresses disciplinary connections among the different units in the module. (10%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Attendance of Track Base and Module A of GE track	
Academic Skills	Critical thinking; interdisciplinary thinking; problem-based learning and solving; application of theory to new situations using the relevant equations; time management; making inferences of physical situations by analysing observations/data.	
Literature & Study Materials	See Brightspace	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Assignments may be carried out by using all the teaching materials provided during the course and further background material as needed. The report on the group assignment can be used as reference, no further materials are needed for the oral exam.	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Hydraulic Engineering (HE)

CIEM3000	Base: Hydraulic Engineering Fundamentals and Environments Part I	15
Responsible Instructor	Dr.ir. B.C. van Prooijen	
Instructor	Dr.ir. B.C. van Prooijen	
Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Prof.dr.ir. W.S.J. Uijtewaald	
Instructor	Dr.ir. B. Hofland	
Instructor	Dr.ir. W.M. Kranenburg	
Instructor	Dr.ir. S. de Vries	
Instructor	Dr. M. Hrachowitz	
Instructor	Dr.ir. A. Blom	
Instructor	Dr.ir. E. Mosselman	
Instructor	Dr.ir. M. Zijlema	
Instructor	Dr. M.F.S. Tissier	
Instructor	Dr. R.M.J. Schielen	
Instructor	C. Piccoli	
Instructor	Dr. L.M. Stancanelli	
Instructor	J.A. Arriaga Garcia	
Contact Hours / Week x/x/x/x	x/16/6-8/x	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The Hydraulic Engineering Extended Track Core of 24 EC consists of two parts. Part I (15 EC) contains three units: Modelling Flow, Transport and Bed Protection, MFTBP (6EC); Ocean Waves (5EC); and River Systems and Interventions (4EC). The units MFTBP and Ocean Waves provide the fundamental knowledge and skills. The unit River Systems and Interventions covers together Part II the Hydraulic Engineering Environments or systems. The unit MFTBP covers: (1) an in-depth analysis of the hydrodynamic and transport processes; (2) the theory and techniques to numerically solve the hydrodynamic and transport equations; (3) the theory and tools to describe and calculate the hydraulic loads and bed protection. The unit Ocean Waves covers the dynamics of wind-generated waves in the open ocean and in coastal waters. This unit River Systems and Interventions covers: (1) the characteristics of a river system, (2) the morphodynamic behaviour of these systems (including the response to human interventions) based on the underlying hydrodynamic and sediment transport processes, and (3) typical river engineering problems and methods design interventions.	
Course Contents Continuation	MFTBP: The unit is organised around three main topics: Fundamentals of free surface flows and transport processes: The first block treats the mathematical description of unsteady flow with free surface in open channels based on the principles of conservation of mass and momentum, including long waves, tidal basins and flood waves in rivers. Further to this, transport of dissolved matter due to advection and diffusion is covered in depth. Specifically, the underlying transport processes on the basis of which shear stress, bed roughness, logarithmic velocity profile, dispersion, turbulent diffusivity and mixing are described, but also the associated two-dimensional (horizontal) and three-dimensional (vertical) flow structures, are treated. Computational modelling of flow and transport In the second block elementary notions and techniques from numerical analysis are addressed for the approximation of solutions to partial differential equations (PDEs), with particular attention to the numerical accuracy and stability. This block is also an introduction to finite difference methods for constructing numerical solution of parabolic and hyperbolic PDEs. These equations describe a large variety of transport processes and wave propagation phenomena arising in hydraulic engineering. Emphasis is laid upon the assessment of the introduced uncertainty while using different numerical schemes in terms of the numerical diffusion and dispersion, along with measures to prevent wiggles. Lastly, a class of numerical methods is introduced for solving one-dimensional shallow water equations, principally aiming at the simulation of unsteady free surface flows in seas, estuaries, rivers, channels and lakes. Of crucial importance is the knowledge concerning the abilities and limitations of the commonly used practices related to approximating the equations of long waves. Loads and protection In this third block hydraulic loads and protections are introduced. Bed protections, filter layers, stability, porous flows are introduced. Also a 5-step design principle is introduced. Fundamental principles of physics are combined with data based (hydraulic) engineering principles and rules of thumb to facilitate engineering decisions on loads and protection. The Ocean Waves Unit first focusses on the description of regular (monochromatic) waves before extending the results to realistic wind-generated wave fields that consist of wave components of varying frequencies and directions. It starts with linear wave theory which gives the interrelation between physical characteristics as the surface motion, the wave-induced pressure in the water and the motion of water particles. Commonly used nonlinear wave theories are then briefly introduced. We then discuss how the main properties of regular waves change when they propagate over varying depths and how they are affected by currents. In the second part of the Unit, we transition to irregular and directionally spread wave fields. We discuss how such wave fields can be measured and subsequently characterized, both in the time- and frequency-domain. We finally discuss how the wave field develops in the open ocean and coastal waters, and how an energy balance can be used to predict spectral wave evolution. The unit River Systems and Interventions covers three main topics: (1) The characteristics of a river system. Here we address river systems, river response to change and associated time scales, catchment hydrology, rainfall-runoff, and climate aspects. (2) The hydraulic and morphodynamic behavior of these systems. Here we address flow properties (normal flow, backwater types, flood waves, flow structures), sediment transport, and initial, transient and equilibrium morphodynamic response. (3) River interventions. Here we address river-related problems, (inter)national river management, flood safety, navigation and	

	other river functions, as well as river interventions.
Study Goals	On completion of the Hydraulic Engineering Extended Track Core (Part I and Part II combined), the student will be able to: 1. Explain the physical principles behind flow, transport, turbulence, mixing, sediment transport, surface waves and hydraulic loads. 2. Derive analytical/numerical models to assess typical hydraulic engineering problems based on physical principles combined with data. 3. Design solutions for typical hydraulic engineering problems regarding scour, siltation, contamination, sediment transport, bed stability. 4. Explain the characteristics of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems. 5. Predict and assess the behaviour of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems in response to natural and anthropogenic change taking uncertainties into account. 6. Identify the range of functions of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems.
Study Goals continuation	The specific Learning Objectives for the unit MFTBP are: LO 01: Explain the physical principles of free surface flow, turbulence, mixing and transport in open channels and the associated hydraulic loads; LO 02: Assess and select appropriate numerical techniques for solving the equations for transport and free surface flows. LO 03: Derive analytical/numerical models to assess typical hydraulic engineering problems regarding based on physical principles. LO 04: Devise suitable computing methods for the physically consistent solution to flow and transport equations; LO 05: Apply mathematical and numerical methods in combination with data to predict and assess flow and transport phenomena in hydraulic engineering; LO 06: Design solutions for typical hydraulic engineering problems regarding scour, river contamination, and bed stability The specific Learning Objectives for the unit Waves are: LO01: Describe the different observation techniques of wind-generated waves and their limitations LO 02: Explain the rationale behind the definition and computation of the wave spectrum and calculate spectral wave characteristics LO 03: Characterize wind-generated waves in a statistical framework LO 04: Explain the physical processes driving wave transformation in oceanic and coastal waters and evaluate which processes dominate in a given situation LO 05: Calculate the evolution of wind-generated waves in oceanic and coastal waters . The specific Learning Objectives for the unit River Systems and Interventions are: LO 01: Explain the role of the river system in the hydrological cycle, the controls of a river system, time scales of change of the controls versus time scale of system response (catchment to reach scale), and phases (initial, transient, quasi-equilibrium, equilibrium) of river response LO 02: derive and apply simplified forms of the one-dimensional shallow-water equations (i.e., normal flow, flood waves, and backwater curves); LO 03: predict and assess (initial, equilibrium, and transient) river response to natural and anthropogenic change of controls LO 04: identify and assess the range of river functions (e.g. flood safety, navigation, nature restoration, freshwater supply, ecology, port operability, water quality, and construction in water), system-related problems (from catchment to reach scale) in national and international river management, and possible solutions or interventions LO 05: apply (inter)national safety standards in flood risk management;
Education Method	The education will be a mix of: - Plenary Lectures - Online videos - Formative assignments and exercises - group discussion sessions.
Computer Use	Numerical exercises via python Notebooks will be a substantial part of the course.
Assessment	Modelling flow, Transport and Bed Protection: written exam (40%) Numerical assignments must be completed with a pass to be allowed to participate in the exam. Ocean waves: written exam (20%), project (13.3%) River Systems and Interventions: written exam (26.7%)
Contact	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. B.C. (Bram) van Prooijen B.C.vanProoijen@tudelft.nl
Expected prior Knowledge	The module expects prior knowledge of fluid mechanics, including Open Channel Flow. It furthermore builds on the Faculty Core and Program Core. We expect the students to work with Python.
Academic Skills	Critical thinking, (mono-disciplinary and partly multi-disciplinary) problem solving and the application of theory in new engineering situations are the main academic skills that will be trained.
Literature & Study Materials	Books: Unsteady flow in open channels by Jurjen Battjes and Robert Jan Labeur (Cambridge University Press), also as free ebook available via library.tudelft.nl Waves in Oceanic and Coastal Waters by Leo H. Holthuijsen (Cambridge University Press), also as free ebook available via library.tudelft.nl Introduction to bed, bank and shore protection by Schiereck. Additional material: Slides

Judgement	Syllabus
	Reading material
	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Materials that are allowed during the exam will be provided via Brightspace.
Collegerama	Yes

CIEM3110	Hydraulic Engineering Fundamentals and Environments Part II	9
Responsible Instructor	Dr.ir. B.C. van Prooijen	
Instructor	A.J. van der Hout	
Instructor	Dr.ir. M. van Koningsveld	
Instructor	Dr.ir. J. Bosboom	
Instructor	P. Taneja	
Instructor	T.M. de Boer	
Instructor	Drs. O.C. Koedijk	
Instructor	S.G. Pearson	
Instructor	Ir. F.R.S. Vinke	
Instructor	Drs. R. ter Hofstede	
Instructor	J.A. Álvarez Antolínez	
Instructor	M.C. Onderwater	
Instructor	Ir. L. de Boom	
Contact Hours / Week x/x/x/x	x/x/12-13/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The Hydraulic Engineering Extended Track Core of 24 EC consists of two parts. Part II (9 EC) contains two units: Coastal and Estuarine Systems and Introduction to Port and Waterway Systems. Together with the unit River Systems and Interventions of Part I, they cover the Hydraulic Engineering Environments or systems. 1. Coastal and Estuarine Systems (6EC). This unit covers: (1) the characteristics of a range of coastal systems (seas, open coasts, estuaries and tidal basins), (2) the morphodynamic behaviour of these systems (including the response to human interventions) based on the underlying hydrodynamic and sediment transport processes, and (3) typical coastal engineering problems and methods for coastal protection. 2. Introduction to Port and Waterway Systems (3EC) This unit explains the importance of port and waterway systems in supporting waterborne supply chains at different scales, and discusses performance optimization from an infrastructure perspective. The unit is organized around three themes: (1) Ports and terminals: challenges for port development, port planning, port layout, terminal development, (2) waterways: challenges for waterway development, waterway dimensions, waterway elements, ship and ship-waterway interactions, traffic management, (3) system performance: analyzing the performance of systems of ports/terminals and waterways (queueing theory).	
Course Contents Continuation	Various topics will be considered in this module. An overview per unit is provided below. Topics per week for Coastal and Estuarine Systems (and corresponding chapter in textbook): Week 1: Introduction (Ch1) and large-scale geographical variation of coastal systems (Ch2) Week 2: Wind waves and tides (Ch3), Week 3: Global wave and tidal environments (Ch4), Coastal hydrodynamics wind waves (Ch5) Week 4: Coastal hydrodynamics wind waves (Ch5), Coastal hydrodynamics tides (Ch5), Week 5: Sediment transport (Ch6), Cross-shore transport and profile development (Ch7) Week 6: Cross-shore transport and profile development (Ch7), Longshore transport and coastline changes (Ch8) Week 7: Exam questions Cross-shore and Longshore, practical Ch8, Coastal inlets and tidal basins (Ch9) Week 8: Coastal inlets and tidal basins (Ch9), exam questions Tidal, Coastal protection (Ch10) Topics per week for Introduction to Ports and Waterways: Week 1: General introduction ports and waterways; integral port planning Week 2: Planning & design of port water areas Week 3: Terminal planning & design + Container terminals (example) Week 4: Queueing theory part 1 & 2 + method Kooman Week 5: Port economics, port-city interaction, port environment Week 6: Inland navigation: history + corridors + cargo + fleet & Performance of inland ships (ship-ship interaction) Week 7: Dimensioning of waterways and waterway elements; dimensions of channels Week 8: Guest lecture; Q&A session	
Study Goals	On completion of the Hydraulic Engineering Extended Track Core (Part I and Part II combined), the student will be able to: 1. Explain the physical principles behind flow, transport, turbulence, mixing, sediment transport, surface waves and hydraulic loads. 2. Derive analytical/numerical models to assess typical hydraulic engineering problems based on physical principles combined with data. 3. Design solutions for typical hydraulic engineering problems regarding scour, siltation, contamination, sediment transport, bed stability. 4. Explain the characteristics of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems. 5. Predict the behaviour of rivers, coasts, estuaries, ports and waterways in hydraulic engineering systems in response to natural and anthropogenic change taking uncertainties into account.	
Study Goals continuation	The specific Learning Objectives for the unit Coastal and Estuarine Systems are: LO 01 - Discuss the nature and complexity of typical coastal engineering problems (Ch1, but also Ch7-10). LO 02 - Explain the imprint on present-day coastal system (uninterrupted coastlines as well as tidal inlets and basins) of: - Geological processes, sea level and climate (Ch2). - Fluvial, wave and tidal processes and their relative influence (Ch2). LO 03 - Understand the global variation of: - Coastal systems (Ch2, Ch4) - Wave and tidal climate (Ch4) LO 04 - Analyse the hydrodynamics of waves and tides: - in the oceans (Ch3); and - along open coasts and in tidal basins (Ch5).	

	LO 05 - Evaluate sediment transport processes under waves, tides and currents (Ch6), distinguishing between: - cross-shore transport (Ch7) - longshore transport (Ch8) - transport in tidal basins and inlets (Ch9). LO 06 - Apply system knowledge to determine the morphological response to environmental conditions as well as interventions (groins, breakwaters, nourishments, land reclamations, sand mining) of: - open coasts (Ch7-8); and - tidal inlets and basins (Ch9). LO 07 - Assess the merits and disadvantages of various coastal interventions for protection against flooding and erosion (Ch10) The specific Learning Objectives for the unit Introduction to Ports and Waterways Systems LO 01 - general: - Explain the functions of ports and waterways in the total transport chain to a level that is necessary for functional design and capacity analysis; - Discuss the importance of performance indicators such as capacity, efficiency, safety and sustainability, and - Describe the environmental issues relevant to ports and waterways and explain the importance of integrating environmental considerations in (stakeholder inclusive) port planning and design. LO 02 - ports/terminals: - Explain the principles of port master planning and develop basic port layouts based on principles of robustness and adaptability in an uncertain future; - Develop functional designs of port terminals (number of elements and their order-of-magnitude dimensions) using state-of-the-art design methods and rules (Module A focuses on container terminals, Module B04-01 will address other terminal types); - Explain the basic elements of investment analysis and can perform such an analysis for a simple case (analysing trade-offs). LO 03 - waterways and port water areas: - Discuss the importance of inland waterway transport corridor policies/plans and perform a basic capacity analysis taking into account forecasted traffic volumes and waterway infrastructure in place or planned. - Explain the basics of ship motions, ship-ship interaction and ship-fairway interaction - Develop functional designs of inland waterways and port waterway areas using state-of-the-art design methods and rules (canals, locks, bridges, approach channels, turning basins, berths). LO 04 - system behaviour: - Analyse relevant processes influencing capacity - Compare the performance of port and waterway systems using rules of thumb and queueing theory (waiting time as a fraction of service time) and the Kooman method for lock capacity assessment.
Education Method	Coastal and Estuarine Systems: 1. Mix between video lectures, lectures, practicals with exam questions, question hours. 2. Weekly formative digital assignments. Active participation and demonstration of a learning curve in the formative assignments is required to access the exam.
Assessment	Introduction to Port and Waterway Systems: Mix between lectures and flipped class room; work sessions Exam 1: Coastal and Estuarine Systems - Written Exam (66,7%) Exam 2: Introduction to Ports and Waterways Systems (33,3%): Written Exam and Design Exercise (the written exam counts for 70% of the grade and the design exercise counts for 30%).
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Academic Skills	We build on the Faculty Core, Program Core and Part 1 of the Hydraulic Engineering Extended Track Core. This module will strengthen various academic skills, ranging from critical thinking, conceptualization, dealing with data, to problem solving.
Literature & Study Materials	The Hydraulic Engineering Track Core will use open-source material. This includes the following books: Bosboom, J. and Stive, M. J. F. (2022 or 2023 - latest version). Coastal Dynamics. TU Delft Open. ISBN (softback/paperback) 978-94-6366-370-0, ISBN (e-book) 978-94-6366-371, DOI 10.5074/T.2021.001 Lecture Notes, Van Koningsveld, M. (Ed.), Verheij, H.J., Taneja, P. and De Vriend, H.J. (2021). Ports and Waterways - Navigating the changing world TU Delft, Hydraulic engineering, Ports and Waterways, Delft, The Netherlands. Ligteringen H. (2022), Ports and Terminals TU Delft, Hydraulic Engineering, Ports and Waterways, Delft, The Netherlands.
Judgement	In addition to these books, we will provide Lecture Slides and Open Source software via Brightspace. See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Permitted Material during the exam will be announced via Brightspace.
Collegerama	Yes

CIEM3210	Coastal Engineering	15
Responsible Instructor	Dr.ir. J. Bosboom	
Instructor	Dr.ir. B.C. van Prooijen	
Instructor	Dr.ir. B. Hofland	
Instructor	Dr.ir. S. de Vries	
Instructor	M.A. de Schipper	
Instructor	A.P. Luijendijk	
Instructor	Dr. A. Antonini	
Contact Hours / Week x/x/x/x	x/x/x/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Central to this B-module are coastal engineering problems and interventions and specifically the assessment of the impact of these interventions and their technical design. Impact assessment and design require advanced system knowledge and proper application of tools. These three elements in connection (interventions, knowledge and tools) are the basis of this B-module. Ethical issues and the role of the engineer will be addressed in the model assignments part of the second two units described below by addressing conflicting perspectives of various vocal and non-vocal stakeholders. The module consists of three units of equal weights, which run in parallel (with especially strong connections between the first and second units): 1. Advanced Coastal and Estuarine Systems 2. Coastal and Estuarine Modelling 3. Shore protection & Breakwaters Advanced Coastal and Estuarine Systems treats, in-depth, coastal and estuarine processes and system behaviour (a.o. waves, tides, currents, coarse and fine sediment transport, morphodynamic change) and response to interventions (a.o. storm surge barriers, groynes, dredging and nourishments). A wide range of coastal systems is considered: deltas, estuaries, beaches and dunes, tidal plains, barrier and lagoon coasts. Not only the physical principles but also their complex interplay in the built and natural environment is discussed. This process knowledge is linked to coastal engineering problems and interventions. Coastal and Estuarine Modelling combines knowledge on prediction, modelling and monitoring concepts with a hands-on approach. Students apply various state-of-the-art models in order to solve complex coastal engineering problems taking into account its uncertainties. Students (functionally) design and evaluate coastal engineering interventions related to amongst others coastal safety or sediment management. Students collaborate in the group assignment. Shore protection & Breakwaters treats the mechanisms of (in)stability of protection layers under wave loads. Aspects needed for practical implementation of these protection layers are introduced, like construction, environment, uncertainty, maintenance, boundary conditions, and modelling techniques. The technical design of all parts of a breakwater cross-section is treated in-depth.	
Study Goals	(B1-1) Analyse and assess in depth the complex behaviour of coastal and estuarine systems (deltas, estuaries, beaches and dunes, tidal plains, barrier and lagoon coasts) including the response of these systems to human interventions (B1-2) Apply models to predict the behaviour of coastal and estuarine systems and assess the impact of coastal engineering interventions (B1-3) Apply the theories of wave loads on different types of slope protection (B1-4) In collaboration propose and design coastal engineering interventions for purposes such as protection against flooding and erosion and construction in water (B1-5) Account for uncertainties in relation to monitoring, modelling and data analysis for research and design (B1-6) Clearly present and report design and modelling results	
Education Method	Advanced Coastal and Estuarine Systems: Lectures and/or knowledge clips, Practicals (inquiry-based learning), Coastal and/or estuarine field trip (with assignment) Coastal and Estuarine Modelling: Lectures and/or knowledge clips, Practicals (inquiry-based learning), Master classes (Guests are invited to discuss ongoing projects), Modelling assignments Shore protection & Breakwaters: Lectures and/or knowledge clips, Master classes by guests, Possibly participation in field trip of Unit 1	
Assessment	Exam 1: Advanced Coastal and Estuarine Systems - Oral Exam (33%) Exam 2: Coastal and Estuarine Modelling - Project (33%) Exam 3: Shore protection and Breakwaters - Project (33%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Academic Skills	Track Base (units Flow and Transport and Ocean Waves), module A (unit Coastal and Estuarine Systems)	
Literature & Study Materials	Critical thinking, problem-based learning and solving, synthesis of system knowledge and response, the application of theory in new engineering situations, making inferences of coastal systems by analysing model results and data, communicating results by oral presentation (live or videos) and to a lesser extent by writing are the main academic skills that will be trained Advanced Coastal and Estuarine Systems: Bosboom, J. and Stive, M. J. F. (2021). Coastal Dynamics. TU Delft, Open. ISBN (softback/paperback) 978-94-6366-370-0, ISBN (e-book) 978-94-6366-371, DOI 10.5074/T.2021.001, possibly revised to include and match the new courses, Annotated lecture slides (Selected sections of) Journal papers, Selected chapters from Winterwerp et al (2021), Fine Sediments in Open Water from fundamentals to modelling. Coastal and Estuarine Modelling: Annotated slides and tutorials, Selected journal papers, In-depth material for specific problems/systems, Selected chapters from: 1) Roelvink & Reniers, A guide to modelling, Coastal Morphology and 2) Winterwerp et al, Fine Sediments in Open Water from fundamentals to modelling. Shore protection and Breakwaters: Part A: Bed, Bank, and Shore Protection book by G.J Schiereck Part B: Breakwater design lecture notes Online lectures, videos and links to literature	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

CIEM3220	River Engineering	15
Responsible Instructor	C. Piccoli	
Instructor	Prof.dr.ir. W.S.J. Uijttewaai	
Instructor	Dr.ir. B. Hofland	
Instructor	Dr.ir. W.M. Kranenburg	
Instructor	C. Mai Van	
Instructor	Dr.ir. A. Blom	
Instructor	Dr. T.A. Bogaard	
Instructor	Dr.ir. E. Mosselman	
Instructor	Dr.ir. C.J. Sloff	
Instructor	Dr.ir. M. Zijlema	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	E. Ragno	
Instructor	Dr. R.M.J. Schielen	
Instructor	C. Piccoli	
Instructor	Dr. D. Wüthrich	
Instructor	Dr. L.M. Stancanelli	
Instructor	J.A. Arriaga Garcia	
Contact Hours / Week x/x/x/x	x/x/x/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	HE Module B2 consists of three main topics: B2-1. Physics of River Systems (4EC), which focusses on the physics of river systems and their response to anthropogenic or natural change; B2-2. River Functions and Management (6EC), which provides students with a broad knowledge on the wide range of purposes or functions of a river system (flood safety, navigation, freshwater availability, nature, etc) and stakeholders, as well as river interventions; B2-3. Schematization and Modelling of River Problems (5EC), which teaches students how to schematize and simplify a river-related problem, formulate research questions and alternative solutions, assess the problem using appropriate tools (such as modelling and monitoring), propose a solution and mitigate unwanted impact. Ethics related to river engineering is addressed in the unit River functions and Management. The topic is considered in frontal lessons as well as other learning activities. During the latter, students work on a river-related problem in which the requirements or constraints regarding one (or more) of the river functions are not met. This allows students to work on learning activities regarding river engineering that relate to ethical issues. Students formulate alternative solutions based on the information available using core ethical values as a guide (see for example ASCE Code of Ethics). They assess the alternative solutions regarding, amongst other criteria, ethical impact, and select the course of action. All learning activities are carried out looking for a balance between economy, safety, and sustainability.	
Study Goals	On completion of this module, the student will be able to 1. understand and apply scientific knowledge that contributes to solving river system-related problems in an ethically sound manner; 2. analyse and assess the behaviour of river systems and catchment, and assess and predict the system response to interventions, while taking into account neighbouring/adjacent systems; 3. (a) analyze the problem, (b) formulate functional requirements and constraints, (c) formulate alternative solutions (including autonomous development), (d) design human interventions in the river or catchment environment for purposes such as flood protection, nature development or restoration, navigation, freshwater supply, ecology, port operability, water quality, and construction in water; 4. account for societal, environmental, and economic aspects, and uncertainties in research and design, at various scales; 5. account for the complexities of multidisciplinary environments and the range of stakeholders: (inter)national river management authorities, nature organisations, NGOs, dredging companies, navigation stakeholders, other governments.	
Education Method	Lectures, discussion sessions, exercises, presentations, field trip, project groups, group sessions, project group presentations	
Assessment	Exam B2-1: Physics of River Systems - Oral Exam (26,7%) Exam B2-2: River Functions & Management - Portfolio (40%) Exam B2-3: Schematization and Modelling of River Problems - Portfolio (33,3%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	MUDE, and Track Base plus Module A of CE track Hydraulic Engineering	
Academic Skills	Critical thinking, problem solving, and application of theory in new engineering situations are the main academic skills that will be trained.	
Literature & Study Materials	B2-1. Physics of River Systems: Powerpoint slides, selected journal papers and other reading material (available through links on Brightspace), home exercises B2-2. River Functions & Management: Powerpoint slides, collected reading material and journal papers B2-3. Schematization and Modelling of River Problems: Collected reading material	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during	Materials that are allowed during the exam will be provided via Brightspace.	

Exam	
Collegerama	Yes

CIEM3230	Advanced Design of Ports and Waterways Systems and Interventions	15
Responsible Instructor	O. Kirichek	
Instructor	Dr.ir. M. van Koningsveld	
Instructor	F. Baart	
Contact Hours / Week x/x/x/x	x/x/x/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The world's economy relies heavily on waterborne supply chains. Approximately 80% of all global trade is at some point transported by shipping. Overall efficiency of global supply chains is to a great extent determined by the in-port and hinterland transport networks they are connected with. A major challenge for the field of ports and waterways engineering is timely adaptation of transport networks, and their associated infrastructure, to ever changing external circumstances (changing vessel sizes, shifting trade patterns, increased attention for sustainability, digitalization etc.). In this module, we teach students advanced design methods for ports, waterways, and systems of ports and waterways in a societal, environmental, and economic context. Students will learn to analyze performance of these systems and design interventions for improvement (e.g. port accessibility, locks, waterway dimensions, breakwaters, quays), accounting for the logistical, nautical and natural processes (e.g. waves, tides, siltation) at play. Data science and advanced simulation techniques are essential for the quantification of trade-offs and therefore an integral part of this B Module. As such students that follow this B-line take a first step towards becoming research software and data engineers. Students will also become familiar with ethical issues arising around large scale engineering projects through real life case studies that involve multiple stakeholders. Unit 1 will cover the full extent of Q4 and Unit 2 will start in week 5. The students are expected to spend about 20% of their time on lectures, 30% on self-study, 20% on formative exercises and 30% on data-driven simulations. The guest lectures will be given by the key industry experts, who will share the most recent insights on innovations and developments in the ports and waterways. The guest lectures will be combined with the field visits, where the students will explore monumental engineering facilities (e.g. container terminals, smart quay walls, locks, breakwaters) in ports. Topics to be addressed in Unit 1 are: 1) Waterways Advanced dimensioning of waterways and waterway elements (e.g. locks, crossings, bends) Sediment management (e.g. accessibility design, dredging and maintenance strategies, maneuverability) Design of hard infrastructure (e.g. locks, slope protections) the latter are part of HOS B1 New developments (e.g. climate change, nature-inclusive solutions) 2) Ports and Terminals Functional design of various terminal types: liquid bulk, dry bulk (including mooring arrangements), general cargo, multipurpose terminals, Ro-Ro, ferry terminals, fishery ports, inland ports, marinas and cruise terminals Port infrastructure and environmental conditions (quay walls, breakwaters, port operability, safe navigation) Trade-offs and financial feasibility New developments (e.g. energy transition, digitalization) 3) Ports and Waterways systems Simulation: Introduction into simulation for P&W (planning and trade-offs) Simulation: Introduction into simulation models (deterministic, stochastic, discrete time/event) Data science for P&W: key concepts Topics to be addressed in Unit 2 are: 1) Advanced data-driven simulations 2) Simulation for ports and waterways (planning and trade-offs) 3) Simulation models (e.g. for analyzing the effect of climate change, sea level rise, emission mitigation measures, etc.) 4) Data science for ports and waterways: key concepts	
Study Goals	On completion of the module B3 (B03-01 and B03-02), the student will be able to: LO1. Analyse logistic and operational processes in ports and waterways systems and discuss the capacity controlling parameters; LO2. Design port/terminal and waterway infrastructure to pre-defined objectives covering economic, societal and environmental aspects; LO3. Design interventions in ports and waterways systems, model interactions ranging from simple to complex, between physical processes and port and waterways systems, and apply simulations for the evaluation of performance; LO4. Design hard and soft infrastructure interventions to improve port, terminal, and waterway performance including quantifying trade-offs and discussing these trade-offs with peers; LO5. Demonstrate the learning over intervention design and performance evaluation in real-life case studies, and LO6. Apply data science techniques and concepts for a quantitative analysis of processes in port and waterway systems in various design exercises.	
Education Method	Interactive lectures, design exercises, Q&A sessions	
Assessment	Unit 1: Ports and Waterways - Intervention, Design and Performance evaluation - Written Exam (60%), formative exercise (in groups) Unit 2: Advanced simulation and quantification methods - Individual Assignment (40%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	Module A, old curriculum: CIE4330, CIE4301, CIE5308	
Academic Skills	The student will acquire essential skills in simulation and data science and apply these to understand and quantify the interactions between the state of the water infrastructure and the (economic, environmental and societal) performance of port and waterway systems. The student will integrate multidisciplinary knowledge and apply it in design exercises addressing real life situations. The exercises will be designed to stimulate critical thinking and develop conceptual, analytical, scientific reporting and communication skills.	
Literature & Study Materials	Van Koningsveld, M. (Ed.), Verheij, H.J., Taneja, P. and De Vriend, H.J. (2021). Ports and Waterways - Navigating the changing world. Lecture notes. TU Delft, Hydraulic engineering, Ports and Waterways, Delft, The Netherlands. Ligteringen, H. (2022) Ports and Terminals. TU Delft, Hydraulic engineering, Ports and Waterways, Delft, The Netherlands Lecture Slides	

	Open Source software Handouts
Judgement	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	none
Collegerama	Yes

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Hydraulic & Offshore Structures (HOS)
--

CIEM4000	Base: Hydraulic and Offshore Structures	15
Responsible Instructor	Ir. E. Kementzetzidis	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr.ir. A.J.H.M. Reniers	
Instructor	Dr.ir. B. Hofland	
Instructor	Ir. J.W. Welleman	
Instructor	C. Mai Van	
Instructor	A. Tsouvalas	
Instructor	Dr.ir. H. Hendrikse	
Instructor	Dr. M.F.S. Tissier	
Instructor	Ir. E. Kementzetzidis	
Instructor	Dr. A. Antonini	
Contact Hours / Week x/x/x/x	14 hrs. Q2 4 hrs. Q3	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The CE HOS track addresses the challenges associated with the design, construction and operation of hydraulic and offshore structures by unifying the knowledge from structural, hydraulic, and geotechnical engineering disciplines. The role of the Track Base Module is to equip students with the fundamental knowledge necessary to tackle the engineering challenges associated with the structural design, construction and operation of structures in river, coastal and marine environments.	
Study Goals	Upon completion of this module, the student will be able to: MLO-1. Analyse fluid- and soil-structure interaction processes, as well as structural mechanics principles, Relevant for the engineering of hydraulic and offshore structures. MLO-2. Analyse scientific knowledge on fluid- and soil-structure interaction, as well as structural mechanics, that contributes to solving complex hydraulic and offshore structures. MLO-3. Perform, statistically model selection and identification for fluid- and soil-structure interaction analyses. MLO-4. Quantify uncertainty in models and measurements for fluid and soil-structure interaction analyses. MLO-5. Evaluate uncertainty propagation for fluid-and soil-structure interaction analyses.	
Education Method	Class lectures and recordings, group assignments	
Assessment	The Track Base Module is assessed both in Q2 and Q3. The gravity associated with each unit module is as follows, Waves (HOS-TC-1), Q2 : Written exam [19%] Fluid-Structure Interaction (HOS TC-2), Q2-Q3 : Group assignment [27%] Structural Mechanics (HOS-TC-3), Q2 : Written exam [27%] Soil-Structure interaction (HOS-TC-4), Q3 : Written exam [27%] A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	BSc Civil Engineering or equivalent, CE Programme Base Module.	
Academic Skills	Critical and interdisciplinary thinking, identification of problem-oriented solution procedures, information collection and synthesis, application of theory to new situations using the relevant equations and correlating mathematical solution with physical insights and design procedures, use of data for critical assessment of modelling assumptions and design considerations.	
Literature & Study Materials	Information available in each module unit.	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Information available in each module unit.	

CIEM4110	Hydraulic Structures	9
Responsible Instructor	Dr. D. Wüthrich	
Instructor	Dr. Y. Yang	
Instructor	S.C. Barbosa Nunes	
Instructor	D.J. Peters	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. D. Wüthrich	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	HOS-A1 Hydraulic Structures: 9ECs - HOS-A-1 Ultimate Strength & Fatigue: 3ECs - HOS-A-2 Design of Hydraulic Structures: 6ECs The module enables students to produce a conceptual engineering design of common hydraulic structures, building on already acquired skills. Students learn to analyse a problem to formulate a design objective, which leads to a functional-spatial design. This is followed by a structural design where next to stability and strength checks, constructability plays a major role. The student further develops skills to motivate assumptions, methods and values of design parameters, using basic sketches. Hydraulic, geotechnical and structural aspects are addressed in coherence to understand the behaviour of these structures. The content of the module contributes to the MUDE and Ethics learning lines.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Generate sound preliminary designs for hydraulic structures based on current natural and societal needs, accounting for its interaction with the environment its life-cycle aspects, using available scientific knowledge. (HOS-1,2,3) MLO-2. Analyse principles of ultimate resistance and fatigue relevant for the design of hydraulic structures, using available scientific knowledge. (HOS-2,3) MLO-3. Perform statistically model selection and identification for the analysis and verification of the structural health of hydraulic structures. (HOS-3,4) MLO-4. Effectively collaborate and communicate with the aim to generate and optimize the design of hydraulic structures. (HOS-4) MLO-5. Critically reflect on the design and reliability of hydraulic structures and its structural components. (HOS-2,4) MLO-6. Translate current natural and societal needs into possible technical designs of offshore structures by accounting for relevant moral values, norms and principles. (HOS-4)	
Education Method	LECTURES & TUTORIALS During lectures, the main theories related to hydraulic structures are described and its applications discussed. Topics needed in support of the design exercise are also introduced and further elaborated. Tutorials apply theories or techniques previously explained during lectures. During the tutorials students work in class, while the lecturer is available for answering questions. Students will implement the tutorial work within the design exercise. DESIGN EXERCISE To stimulate the active use of the lectured theory and tutorials, a design exercise has to be completed, usually in teams of two students. Team formation and subject selection for the exercise will start during the 1st teaching week of the course. Progress of the exercise will be as follow (see Brightspace for detailed deadlines): - End of week 2 - 1st check : workplan and basic design - End of week 4 - 2nd check : progress & results - End of week 6 - 3rd check : progress & results - End of week 8 - Final report	
Assessment	The assessment for this module consists of: - Individual Exam: 50 % - Group Design Project: 50 % Both the individual exam and the final deadline for the group design project take place at the end of the education period (Q3). The content of the individual exam is evenly divided over the involved units, so that 50% of the exam problems are related to unit HOS A1-1 (Ultimate Strength and Fatigue) and 50% of the exam problems are related to the content of unit HOS A1-2 (Design of Hydraulic Structures). To successfully pass the module, all components of the assessment must be evaluated with a pass-grade. A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	BSc Civil Engineering or equivalent, MUDE Module, CE Programme Base Module.	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well.	
Literature & Study Materials	Lecture notes and handouts will be available digitally on Brightspace.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Permitted Materials during Exam will be announced on Brightspace	
Collegerama	No	

CIEM4120	Offshore Structures	9
Responsible Instructor	Ir. J.S. Hoving	
Instructor	Ir. J.S. Hoving	
Instructor	S.C. Barbosa Nunes	
Instructor	Dr.ir. P. van der Male	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Parts	The Module Offshore Structures of the Hydraulic and Offshore Structures (HOS) track, is worth 9 ECTS and consists of the following units: - HOS-A-1 Ultimate Strength and Fatigue = 3 ECTS - HOS-A-3 Design of Offshore Structures = 6 ECTS	
Course Contents	Note here that the unit HOS-A-1 is shared with HOS Module A1: Hydraulic Structures. This module enables students to produce conceptual engineering designs for fixed offshore structures accounting for ultimate strength and fatigue criteria. Within this module, students gain a thorough theoretical understanding of the design process, its structuring, available design methodologies and the philosophy behind the design codes. Furthermore, this module addresses offshore, geotechnical, structural and reliability aspects in coherence to analyse the behaviour of offshore structures. The ultimate strength and fatigue part of the module specifically focuses on the ultimate strength and fatigue assessment of steel and concrete structures. The provided knowledge and skills are required to be able to perform the verification of offshore structures design and are therefore directly applied in the design component of this module. Throughout the module students work on a group design project, where each student team designs an offshore support structure from scratch up to a preliminary design by performing a full offshore support structure design-cycle. This design-cycle includes I) the assembly of the structural configuration, II) the determination and assessment of the permanent, variable and environmental loads on the structure, III) the design and analysis of its foundation and IV) the static and dynamic structural analysis of the structure based on ultimate strength and fatigue principles. By means of this group design project, students will learn to judge their design against ULS and FLS design criteria and be able to modify and improve their design based on static and dynamic structural analyses. Thereby, students will be able to analyse and evaluate the dimensioning and performance of structural components for strength and stability. Furthermore, students learn to predict how changing (parts of) their design will influence the corresponding loads and based on their insight in these loads, be able to defend their design choices. Moreover, students will learn what the practical implications of their design choices are for operations and other feasibility and life-cycle aspects. To reinforce the understanding of these aspects, the course program also includes a visit to an offshore structures fabrication yard. During this module, students will not only learn how to design an offshore structure, but they also learn how to collaborate effectively in a team and will further develop skills and competences to motivate their assumptions and provide argumentation of their design choices by accounting for relevant moral values, norms and principles.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Generate sound preliminary designs for offshore structures, accounting for its interaction with the environment and its life-cycle aspects, using available scientific knowledge. MLO-2. Analyse principles of ultimate resistance and fatigue relevant for the design of offshore structures, using available scientific knowledge. MLO-3. Perform, statistically sound and context-based, model selection and identification for the analysis and verification of the structural health of offshore structures. MLO-4. Effectively collaborate and communicate with the aim to generate and optimize offshore structures designs. MLO-5. Critically reflect on the design and reliability of offshore structures and its structural components. MLO-6. Translate current natural and societal needs into possible technical designs of offshore structures by accounting for relevant moral values, norms and principles.	
Education Method	The education in this module includes problem-based lectures, workshops, a group design project, home assignments and self-study. In total, this module has 12 contact hours per week, consisting of: - two 2-hour lectures (2x45 min) regarding methods to assess the ultimate strength and fatigue of steel and concrete structures (HOS-A-1). - two 2-hour lectures (2x45 min) focused on comprehension and application of theory and methodology of offshore structures design (HOS-A-3) and to provide feedback and reflection on the group design project, and - a 4-hour workshop (4x45 mins) focused on problem analysis and evaluation, treating and discussing example problems, as well as allowing for discussion and Q&A for the group design project. The group design project is performed in groups of 3 students each and consists of 3 phases. Each phase comes with an assessment & feedback session per group, followed by a final deadline at the end of the teaching period at which the assignment is graded. Additionally, under reservation of unforeseen circumstances, the module includes an excursion to a fabrication yard where offshore support structures are constructed. Here note that the number of participants to this excursion is generally limited by the capacity at the yard and the excursion is therefore not a compulsory part of the module.	
Prerequisites	BSc Civil Engineering (or equivalent), CEGM1000 Modelling, Uncertainty and Data for Engineers, CIEM000 Programme Base: Mechanics and Interdisciplinary Perspectives.	
Assessment	Participants of this module are summatively assessed through a Group Design Project, worth 4 credits, and an individual written exam, worth 4 credits. In addition, participants are formatively assessed through home assignments, worth 1 credit. This means that the home assignments are checked to provide feedback but are not graded. The final grade for the module is the weighted average of the Group Design Project (50%) and the individual written exam (50%). To pass the course, participants must score a pass grade, i.e. a 5.8 or higher, on both the project and the exam. Group Design Project: The final deadline for the Group Design Project is set at the end of the Q3 education period. If you fail to get a pass grade for the assignment, but the assignment grade is equal to or higher than a 5.0, you are allowed to improve your grade by submitting an addendum to your assignment report in the following education period, i.e. during Q4. Here, please note that the maximum grade that can be obtained after submitting an addendum to improve an insufficient grade is 6.0. Written exam:	

	The regular individual written exam for this course is organised during the exam period of Q3, and a re-exam is organised in the exam period of Q4. The written exam is an open-book exam, meaning that any written or printed material is permitted during the exam sessions. The content of the individual exam is evenly divided over the involved units, so that 50% of the exam problems are related to unit HOS A-1 Ultimate Strength and Fatigue and 50% of the exam problems are related to the content of unit HOS A-3. If due to any unforeseen circumstances the written exam for this course may not be given in a plenary on-campus session, alternatively the written exam for this course may be given as an online open-book exam through Brightspace. To participate in the online exam you must be enrolled for this course in Brightspace. For further practicalities regarding any online examinations, you will be informed through Brightspace. A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Expected prior Knowledge	BSc Civil Engineering or equivalent, MUDE Module, CE Programme Base Module.
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills addressed in this course. Furthermore, students will collaborate intensively with their colleagues in a module-wide group design project thereby developing organizational and interpersonal skills. In addition, students present and reflect on their designs in a compact design report thereby developing their reporting skills as well as learn to focus on the essentials of their design and their work.
Literature & Study Materials	The lecture material for this course consists of the lecture slides, lecture notes, papers and chapters from relevant literature, a basis for design document for the group design project and the stick-model hand-out, which will all be available through Brightspace. Additional reading material that provides useful information in support of the course are: > The books Handbook of Bottom Founded Offshore Structures, parts 1 and 2, by prof.dr.ir. Jan H. Vugts, available online through the website of the Publisher Eburon. > The book Hollow Sections in Structural Applications, by prof.dr.ir. J. Wardenier, available at VSSD.
Judgement	See Assessment. For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	The written exam is an open-book exam, meaning that any books, slide print-outs, personal notes, i.e. any written or printed material, is permitted during the exam sessions. Additionally, it is allowed and advised to bring a scientific calculator to the exam. Nevertheless, any devices that allow for communication are not allowed, i.e. no laptops, pads or phones.

CIEM4210	Marine Renewables	15
Responsible Instructor	Dr. G. Lavidas	
Instructor	Dr.ir. P. van der Male	
Instructor	Dr. E. Lourens	
Instructor	S. Basu	
Instructor	Dr. G. Lavidas	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	J.O. Colomes Gene	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module concerns the design, monitoring and assessment of offshore wind and ocean energy farms, in general, and different technologies (both bottom-fixed and floating), specifically, in an integrated manner, including control, installation, maintenance and economics. Regarding the development of offshore energy farms, this means that all relevant aspects and stakeholders are addressed and their implications for design assessed. From a structural perspective, this implies that the different technologies should be understood as a system as a whole and the varying environmental interactions aerodynamic, hydrodynamic and soil be considered in conjunction. The content of the module contributes to the MUDE and Ethics learning lines.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Create offshore renewable solutions based on future natural and societal needs, using available scientific knowledge (HOS- 1, 3). MLO-2. Evaluate offshore renewable structures in interaction with the surrounding environment, using available scientific knowledge (HOS-2, 3). MLO-3. Convey quantified uncertain outcomes corresponding the design of hydraulic and offshore structures (HOS-4). MLO-4. Propose new ways for dealing with ethical issues within the field of hydraulic and offshore structures from the perspective of different relevant moral frameworks (HOS-4). MLO-5. Conduct a solution-oriented debate with stakeholders about ethical issues within the field of hydraulic and offshore structures, while acknowledging real-world constraints in terms of technological possibilities, governance and institutional options (HOS-4).	
Assessment	The module is assessed at the end of Q4: - Group assignment: 80% - Individual exam: 20% Relation between assessment module units (weighing) - HOS-B-1 Probabilistic Design: 20%, - HOS-B-2 Numerical Modelling: 25% - HOS-B-3 Offshore renewable Structures: 55% A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	BSc Civil Engineering or Applied Science (or equivalent). MUDE Module, CE Programme Base Module, HOS Track Base Module and HOS Modules A1 or HOS Module A2	
Literature & Study Materials	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well.	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	

CIEM4220	Dams, Dikes and Breakwaters	15
Responsible Instructor	Dr. A. Antonini	
Instructor	Prof.dr.ir. S.N. Jonkman	
Instructor	Dr.ir. B. Hofland	
Instructor	C. Mai Van	
Instructor	Ir. R.E. Jorissen	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	J.O. Colomes Gene	
Contact Hours / Week x/x/x/x	0/0/0/12	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module aims at the engineering design of hydraulic structures that protect against the effects or impact of waves, currents and high-water levels, starting with understanding the underlying processes and related systems. It concerns the design of breakwaters, dikes and closure dams; flood risk reduction strategies, and structures to protect beds, banks and shorelines. The content of the module contributes to the MUDE and Ethics learning lines	
Study Goals	On completion of this module, the student will be able to: MLO-1. Create solutions for dams, dikes and breakwaters based on future natural and societal needs, using available scientific knowledge (HOS-1, 3). MLO-2. Evaluate dams, dikes and breakwaters in interaction with the surrounding environment, using available scientific knowledge (HOS-2, 3). MLO-3. Convey quantified uncertain outcomes corresponding to the design of dams, dikes and breakwaters (HOS-4). MLO-4. Consider ways of dealing with ethical issues within the field of dams, dikes and breakwaters from the perspective of different relevant moral frameworks (HOS-4). MLO-5. Conduct a solution-oriented debate with stakeholders about ethical issues within the field of dams, dikes and breakwaters, while acknowledging real-world constraints in terms of technological possibilities, governance and institutional options (HOS-4).	
Education Method	Frontal lectures	
Assessment	The module is assessed at the end of Q4: - Exercise portfolio: 60% - Individual exam: 40% Relation between assessment module units (weighing) - HOS-B1 Probabilistic Design: 20%, - HOS-B2 Numerical Modelling: 25% - HOS-B4 Dams, Dikes and Breakwaters: 55% A successful participation in the module, requires a minimum evaluation of 5.8 for all assessments	
Expected prior Knowledge	BSc Civil Engineering or Applied Science (or equivalent). MUDE Module, CE Programme Base Module, HOS Track Base Module and HOS Modules A1 or HOS Module A2	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well	
Literature & Study Materials	Lecture notes: Breakwater design, Van den Bos and Verhagen, 2018. Lecture notes: Closure works, Verhagen, 2017. Lecture notes: Introduction to Bed, bank and shore protection. 2nd edition, Schiereck and Verhagen 2019. Lecture notes: Flood defences, Jonkman, Jorissen, Schweckendiek and van den Bos, 2021 Manual: Hydraulic Structures, Voorendt and Molenaar, 2021 Lecture notes: Lanzafame (2023) Probabilistic Design Lecture Notes (online website)	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Lecture note and books	
Colleggerama	Yes	

CIEM4230	Floating and Submerged Structures	15
Responsible Instructor	J.O. Colomes Gene	
Instructor	Dr.ir. R.C. Lanzafame	
Instructor	J.O. Colomes Gene	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module addresses the design, monitoring and assessment of floating and (semi)-submerged civil engineering solutions for infrastructural and urban development applications, for instance floating bridges and moored tunnels, including control, installation, maintenance and economics. This implies that the structure should be understood as a system as a whole and the varying environmental interactions aerodynamic, hydrodynamic and soil be considered in conjunction. The content of the module contributes to the MUDE and Ethics learning lines.	
Study Goals	On completion of this module, the student will be able to: MLO-1. Create floating and submerged solutions based on future natural and societal needs, using available scientific knowledge (HOS-1, 3). MLO-2. Evaluate floating and submerged structures in interaction with the surrounding environment, using available scientific knowledge (HOS-2, 3). MLO-3. Convey quantified uncertain outcomes corresponding the design of hydraulic and offshore structures (HOS-4). MLO-4. Propose new ways for dealing with ethical issues within the field of hydraulic and offshore structures from the perspective of different relevant moral frameworks (HOS-4). MLO-5. Conduct a solution-oriented debate with stakeholders about ethical issues within the field of hydraulic and offshore structures, while acknowledging real-world constraints in terms of technological possibilities, governance and institutional options (HOS-4).	
Education Method	This module is based on project-based learning methodology. Students will mainly work on hands-on group activities, supported by online pre-recorded videos and lecture notes.	
Assessment	The module is assessed at the end of Q4: - Group assignment: 60% - Individual exam: 40% Relation between assessment module units (weighing) - HOS-B-1 Probabilistic Design: 20%, - HOS-B-2 Numerical Modelling: 25% - HOS-B-5 Floating and Submerged Structures: 55% A successful participation in the module, requires a minimum evaluation of 5.8 for all assessments.	
Expected prior Knowledge	BSc Civil Engineering or Applied Science (or equivalent). MUDE Module, CE Programme Base Module, HOS Track Base Module and HOS Modules A1 or HOS Module A2.	
Academic Skills	Critical thinking, analytical thinking and problem solving are the main academic skills which will be trained. In addition, students will need to collaborate intensively with their colleagues in a module-wide project so organizational and interpersonal skills will be developed as well	
Literature & Study Materials	Probabilistic design unit: - Lecture notes: Lanzafame (2023) Probabilistic Design Lecture Notes (online website) - Recommended textbooks: - Der Kiureghian, A. (2022). Structural and System Reliability. Cambridge University Press. - Moss, Robb Eric S. (2020) Applied civil engineering risk analysis. Springer. Computational Modeling unit: - Lecture notes and online tutorials - Recommended textbooks: - Numerical Methods for Engineers by Steven C. Chapra and Raymond P. Canale. Floating and Submerged Structures unit: - Lecture notes	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	No additional material is expected to be allowed in exams. If the instructors consider necessary the use of additional material, this will be communicated to students in sufficient time in advance.	
Collegerama	Yes	

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Structural Engineering (SE)
--

CIEM5000	Base: Structural Engineering	15
Responsible Instructor	A. Tsouvalas	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Prof.dr. A. Metrikine	
Instructor	Ir. J.W. Welleman	
Instructor	Dr. H.M. Jonkers	
Instructor	Dr.ir. M. Ottele	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr.ir. H.R. Schipper	
Instructor	A. Tsouvalas	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	S.C. Barbosa Nunes	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. F. Kavoura	
Instructor	A. Jagadeesh	
Instructor	S. Hiller Bernal	
Contact Hours / Week x/x/x/x	xx/14/8/xx Q3 week 1-4 = 8 hours per week Q3 week 5-8 = 6 hours per week	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	In this module, the learning line on Modelling, data monitoring and sensing and Uncertainty and risk analysis is continued from the Faculty base module. Modelling of structures is central in the SE Track, since both mechanics and codes are in their essence models of real structural behaviour. Data and sensing is addressed sideways when it comes to the determination of mechanical properties and behaviour of materials and structures from laboratory experiments. Experimental results are discussed to illustrate e.g. failure phenomena during lectures and to explain the basis of codes and design models. Ethics is reflected through sustainability and circularity that are the major aspects that future structural engineers need to develop in order to preserve environment. Designing skills are important for all structural engineers and in this module the goal is that students embrace principles and good feeling of structural design with traditional construction systems and materials. The module is composed of 2 units: Unit 1 - Sustainable Construction Members and Systems (8EC); Unit 2 - Mechanics of Slender Structures (7EC). Unit 1 focuses on loads and sustainability aspects of structural design and design and analysis of steel, timber and concrete members. More specifically, the following topics are covered: Identification and quantification of loads on structures. This includes short-term, long term, and moveable loads, and the combination of different types of loads. Design of steel cross sections and members. Influence of properties of base material, production routes and execution tolerances on verification of critical steel cross sections loaded in bending and shear, members loaded in compression and steel bolted and welded connections. Design of timber cross sections and members. Identification of material properties and their use in the design and verification for strength, stiffness and stability at member level. Design and verification of timber joints and connections. Analysis and verification of response of prestressed concrete members exposed to forces, i.e.: axial, bending and shear, in order to meet requirements for ultimate and serviceability limit state. Application of the principles of sustainability and circularity in civil engineering structures. The content on sustainability covers general LCA approaches and system boundaries supported by specific environmental product data, fabrication processes used in construction of structures, railways and roads. Main principles of circularity and reusability will be covered supported by examples of real structures. The content of Unit 1 is divided into 6 themes offered over 8 weeks in Q2 and 8 weeks in Q3 with a workload of 5EC and 3EC, respectively. In Q2, loads and sustainability aspects of structural design and design and analysis of steel and timber members are covered. In Q3, the design and analysis of prestressed concrete systems and connections in steel and timber structures are covered. Unit 2 treats the mechanics of slender structures. In Q2, simple cases are treated in which analytical solutions of single elements are possible. The goal is to explain physics and the basics of the methods. In Q3, the focus is shifted towards more design-type assignments in which the students apply the solution methods in more complex systems consisting of combined one-dimensional structural elements and plates together with complex boundary/interface conditions. The unit covers: Formulation of governing equations of one-dimensional and two-dimensional slender structures. Statics and dynamics of cables, rods, shear beams and beams in bending. Statics and dynamics of plates in bending. Statics and dynamics of combinations of one-dimensional slender structures. Statics and dynamics of plates undergoing in-plane deformations. Statics and dynamics of combinations of one-dimensional and two-dimensional slender structures.	
Study Goals	After completion of this module, students will be able to: 1. Identify and quantify the most important loads on structures, roads and railway and schematize and combine them as input for the determination of the force distributions and deformations. 2. Design critical steel cross sections, members and joints with optimized dimensions and reliable performance against structural and environmental requirements under the influence of single and combined bending, shear, torsion and axial loads. 3. Design and verify structural performance of timber members and connections subjected to single and combined bending, shear and axial loads through understanding of fundamental material properties and force flows in timber joints. 4. Design, analyse and optimize prestressed concrete structures, considering influence of internal stress distribution, time dependent effects, type of pre-tensioning, boundary conditions, prestress losses, cracks and reinforcement details. 5. Understand and apply the principles of sustainability to identify, analyse and evaluate negative effects of civil engineering structures at component and system level on natural environment and society and to use circularity strategies to reduce the environmental impact of structures, railways and roads. 6. Formulate the governing equations of motion of a class of one-dimensional slender structures (beams, cables, rods) and plates subjected to static and dynamic loads in- and out- of plane. 7. Analyse 1D slender structures subjected to static loads and dynamic loads and evaluate the structural response. 8. Analyse plates subjected to static and dynamic loads in- and out- of plane and evaluate the structural response. 9. Create mathematical models of complex structures consisting of one-dimensional elements and plates subjected to static and	

Education Method	dynamic loads and evaluate their structural response.
Computer Use	Lectures. Practical sessions. For more information please check the Brightspace page of the module.
Assessment	Maple, Python. Please check the Brightspace page of the module for more detailed information.
	Unit 1: There are two open-book written exams, one at the end of each quarter related to content of those quarters. Knowledge on theoretical background and problem-solving skills are tested by open-ended (non-repeating) questions. The assessment of design and analysis skills are tested through analysis and calculation exam questions that are related to home assignments, but not limited to.
	Unit 2: Material delivered in Q2 will be assessed on the basis of a written exam. In Q3 the assessment will be on the basis of final report related to the collection of (group) assignments/project in Q3.
	The weight for the assessments above are as follows: Sustainable Construction Members and Systems - Written Exam (Q2) 33.3% Sustainable Construction Members and Systems - Written Exam (Q3) 20% Mechanics of Slender Structures - Written Exam (Q2) 27.7% Mechanics of Slender Structures - Report (Q3) 20%
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. The design of Sustainable Construction Members and Systems relies on pre-knowledge of students related to: Uncertainty quantification, probabilistic and stochastic processes. Basic principles of reliability of structure and partial safety factors. Basic knowledge of building principles and construction methods. Basic knowledge of mechanics of structures to determine sectional forces and deformation of schematized structural systems (simple beam, cantilever, frame, etc.) from engineering tables. Knowledge related to problems of stability of 1D members (basic flexural buckling Euler cases). Basic knowledge of mechanics of materials: elastic, plastic behaviour, stress-strain relations, relaxation. Basic design and verification of steel-cross sections subjected to single bending and shear loads. Knowledge of basic material properties of steel, concrete and timber. Basic design of reinforced concrete members (beams and slabs) and verification of ULS and SLS. The required pre-knowledge of students to follow the unit on Mechanics of Slender Structures is based on base faculty module, program base module and existing Bachelor level courses: - Differential Equations (CTC2100) - Dynamics & Modelling (CTC1210) - Dynamics of Systems (CTC2300) - Structural Mechanics 1-3 (CTC1110, CTC1310, CTC2210). More specifically for the structural mechanics part: - Knowledge on computing statically determinate structures, including support reactions and force/moment diagrams. - Basic knowledge on computing structures using the Principle of Virtual Work. - Knowledge on kinematic, constitutive and equilibrium relations for basic deformation cases (extension, bending, torsion). - Knowledge on stress distributions on basic prismatic members and on deformations of trusses and frames. - Knowledge on computing statically indeterminate structures, including the force and displacement methods.
Academic Skills	Analytical thinking; critical appraisal
Literature & Study Materials	Several textbooks and reading notes will be distributed by the lecturers. For more information, please check the Brightspace page of the module.
Judgement	See assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Any written text brought by the students is permitted.
Collegerama	Yes

CIEM5110	Structural Mechanics and Dynamics	9
Responsible Instructor	Dr. E. Lourens	
Instructor	Prof.dr. A. Metrikine	
Instructor	Dr.ir. C.C.J.M. Tiberius	
Instructor	F.P. van der Meer	
Instructor	Dr. E. Lourens	
Instructor	C. Shen	
Instructor	Dr.ir. I. Barcelos Carneiro M Da R	
Instructor	Dr. A. Cicirello	
Contact Hours / Week x/x/x/x	x/x/9-12/x Contact hours from 9-12 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The A1 module on Mechanics and Dynamics of structures and structural components provides the students with advanced knowledge on mechanics and dynamics of structures which are typically encountered in civil engineering. The structure of the module balances a strong theoretical basis with application cases relevant to the structural engineer. By following this module, students will learn how to analyse the stability of structures to both static and dynamics loads. They will get acquainted with the various computational techniques used in the static and dynamic analysis of complex structural systems and they will learn how to deal with measurements and analysis of vibrations of engineering structures. The learning line on Modelling, Data and Uncertainty is continued from the Faculty Base and Track Base module. In the third unit of this module (SE-A1-3), the student will extract modal parameters from spectral estimates obtained using measurement data from self-designed set-ups in the laboratory, quantify and interpret the uncertainty in these estimates and modal parameters, and statistically judge consistency between mode shapes. Also comparisons will be made between the estimates and their modelled equivalents as obtained in the second unit. The learning line is addressed explicitly in the LO's 2, 3, 7 and 8 of the third unit. In the first unit, the learning line on design will be continued from the preceding modules. In this unit students will learn to explain the criteria for designing stable structures. A range of instabilities that needs to be considered in a conceptual design phase is considered, including buckling, plastic collapse of frames, galloping, flutter, and vortex-induced vibrations.	
Study Goals	On completion of this module, the student will be able to: 1. Analyse the stability of structures subjected to both static and dynamics loads. 2. Explain the theoretical background of finite element formulations for structural analysis. 3. Perform finite element analysis to analyse the force distribution, stability and dynamic response of frame structures. 4. Analyse and interpret vibration measurements to identify modal parameters, including their associated uncertainty. 5. Design a measurement set-up for vibration analysis, with the purpose of quantifying modelling and measurement uncertainties.	
Education Method	Differs per unit. Please consult the Brightspace page.	
Assessment	Stability of structures - Written Exam (33%) Finite Elements for Structural Analysis - Written Exam (33%) Measurement and Analysis of Vibrations Report (34%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. University units, or equivalent, in Calculus, Linear Algebra, Probability & Statistics, Numerical Analysis. Coding skills: Python (MUDE level) Partial Differential Equations, Statics, Beam theory (BSc level) Numerical methods, Introduction to Finite Elements (MUDE level) Signal processing (MUDE level)	
Academic Skills	Skills that students will be working on in this course are: - Analytical and critical thinking - Interpretation of results - Performing laboratory experiments - Data analysis - Writing professional reports	
Literature & Study Materials	1. V.V. Bolotin. The dynamic stability of structures. San Francisco, Holden-Day (1964). 2. M. Paidoussis. Fluid-structure interactions: Slender Structures and Axial Flow. Academic press (1998). 3. Thomsen JMT. Vibrations and Stability, Advance Theory, Analysis and Tools. 2nd Edn. Verlag: Springer (2003). 4. Dewey H. Hodges, G. Alvin Pierce: Introduction to Structural Dynamics and Aeroelasticity. Cambridge University Press (2011) Noise and Vibration Analysis, Anders Brandt; Fundamentals of Signal Processing for Sound and Vibration Engineers, Kihong Shin & Joseph Hammond; Modal testing, Ewins	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Unit-specific.	
Colleggerama	No	

CIEM5120	Design of Structural Components	9
Responsible Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	S.C. Barbosa Nunes	
Instructor	Prof.dr.ir. P.C. Louter	
Instructor	A.T. Gebremariam	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. F. Kavoura	
Instructor	Dr. M.A. Popescu	
Contact Hours / Week x/x/x/x	x/x/4/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The A2 Module focuses on the Design of Structural Components. Potential for reducing environmental impact of structures and achieving an optimum structural design lies in integrating construction materials in structures where they best fit, given their mechanical and functional properties. By successfully completing this module, you will learn to understand the possibilities and limitations of both conventional and innovative construction materials, select and or adjust their properties according the functionality of the structures. You will learn to design structural members which combine the advantages of various materials such as steel, concrete, FRP or timber. Designing future structures, using innovative construction materials, optimised for circularity (prefabricated and demountable constructions) will be emphasised. The module consists of two units: (1) Prefabricated and composite structures and (2) Design with innovative materials. You will learn to design prefabricated and composite structures made of conventional construction materials, and will also explore improvements with novel construction materials, (composite) systems and fabrication methods. A2-1: Prefabricated and composite structures: The content is divided into 4 themes offered in Q3. More specifically the envisioned content is as follows: Theme 1: STEEL Design verification of steel members (beam-columns) exposed to interaction, axial force and transversal load Hollow section steel structures; Main topology of joints Steel-concrete composite structures; Principles of design; prerequisites and consequences of full and partial composite action; Theme 2: TIMBER Engineered wood products Massive timber buildings and timber frame housing Connections in prefabricated timber structures Timber trusses and portal frames Timber concrete composite structures Theme 3: CONCRETE External prestressing Strut-and-tie models and detailing Prefabricated elements for building and infrastructure Theme 4: RAILWAY Conventional track structures and innovations, such as prefab concrete slab tracks; prestressed concrete sleepers, timber sleepers, composite sleepers; composite ballast bed Modular in railway infrastructure: current experiences and perspectives A2-2: Design with innovative materials: The content deals with a wide overview of innovative construction materials, their applications, functional, sustainability and safety benefits and basic principles of design. The unit is offered in Q3. More specifically the envisioned content is as follows: Theme 1: HIGH PERFORMANCE CONCRETE Introduce new types of cementitious materials ((SHCC), UHPFRC, Geopolymer concrete, auxetic, phase change, self-sensing, self-healing, 3d printed, recycled) Theme 2: RECYCLED MATERIALS Introduction of recycled construction materials and their circularity Adaptive design for easy dismantling and reusing structural materials Challenges and opportunities. Theme 3: HIGH PERFORMANCE STEEL AND COMPOSITE STRUCTURES Weathering and stainless steel; High strength steel Composite materials and members Theme 4: STRUCTURAL GLASS basic information on how structural glass is used and what the material properties are, basic calculations Theme 5: DIGITAL MANUFACTURING 3D printed steel (material, members, connections) 3D printed concrete; Wood -plastic composites. Theme 6: INNOVATIONS IN PAVEMENT ENGINEERING Introduction of pavement structure with emerging materials Emerging materials to enhance performance and functionality of pavement Theme 7: INNOVATIONS IN RAILWAY ENGINEERING Conventional track structures and innovations, such as embedded rail (elastomeric materials), polymer sleepers, coated ballast.	
Study Goals	On successful completion of this module, you will be able to: LO1. Explain the principles to design prefabricated and structural (composite) components in a circular manner for buildings and infrastructure by using conventional and novel civil engineering materials. (SE1, SE3, SE5, SE8) LO2. Apply the advanced design aspects in structural engineering, focusing on stability problems, understanding and designing composite members, connections, detailing, external prestressing and prefabricated members with conventional construction materials (timber, steel, concrete). (SE1, SE3, SE6) LO3. Design, analyse and optimize composite structural members, e.g. steel-concrete, timber-concrete, timber-steel beams, decks and columns, with respect to short and long term loads and environmental conditions. (SE1, SE3, SE7) LO4. illustrate the influence of connections and detailing aspects in prefabricated and composite structures and apply this in the design and verification. (SE1, SE3, SE5) LO5. Identify the innovative, high performance and sustainable civil engineering materials and simulate their behaviour. (SE4, SE5, SE6, SE8, SE9) LO6. Discuss limitations and advantages of applying novel materials, advanced manufacturing and innovative systems in design of structural members and optimize design accordingly. (SE7, SE8, SE9) LO7. Discover and justify the possibilities of applying innovative materials in real engineering problems to improve the design and behaviour of the structure or the road and railway system. (SE1, SE3, SE5, SE6, SE7, SE8, SE9)	
Education Method	-Lectures in lecture halls, based on lecture notes, parts of lecture books, readers etc. - Quizzes and discussions in lectures -Self study of lecture material -Writing group reports, cooperating in groups. - Performing literature search and interpreting this literature for group reports	
Assessment	Summative assessments are all done on unit level. First unit has two summative assessments, one assignment (which accounts for 20% of the module grade) and a written exam (which accounts for 45% of the module grade). Assessment of the second unit is done on basis of a project (which accounts for 35% of the module grade). The total grade for the module is calculated based on total contribution of these 3 summative assessments. A successful participation in the module, requires a minimum evaluation of 5.8 in all units.	
Expected prior Knowledge	The module relies on pre-knowledge of Sustainable Construction Members and Systems (from CE-SE Track Base in Q2) A2-1: Prefabricated and composite structures: Assessment is based on Witten exam and final assignment reports. In the written exam, knowledge on theoretical background and problemsolving skills are tested by open-ended (non-repeating) questions. The assessment of design and analysis skills and collaborative work are tested through the assignments that is done partially individually (calculation and analysis) and partially in the group (design aspects). The summative assessment will be on the basis of final report related to collection of individual and group weekly assignments. The written exam can contain questions related to analysis and calculation exam questions that	

are related to home assignments, but not limited to.

A2-2: A2-2: Design with innovative materials:

The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam and the quality of the final assignment reports. Individual knowledge on theoretical background and problem-solving skills are tested by open-ended (non-repeating) questions in oral exam (group presentation). The assessment of design and analysis skills and collaborative work are tested through group project. The assessment will be on the basis of final report related to collective of (group) project.

Academic Skills

- To be able to discuss content with teachers and fellow students
- To be able to search for literature on specific topics and be able to interpret these
- To be able to write clear and consistent reports
- To be able to prepare for exam questions that require insight on an academic level

Literature & Study Materials

- Lecture notes
- (parts of) text books
- papers

Judgement

See Assessment

Permitted Materials during Exam

- calculator
- equation sheet

CIEM5130	Design of Civil Structures and Infrastructures	9
Responsible Instructor	Dr. K. Anupam	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Ir. G.J.P. Ravenshorst	
Instructor	Dr. Y. Yang	
Instructor	Dr.ir. H.R. Schipper	
Instructor	Dr. K. Anupam	
Instructor	Prof. M. Veljkovic	
Instructor	C. Noteboom	
Instructor	M.P. Felicita	
Instructor	Ir. J.P.G. Ramler	
Instructor	S. Hiller Bernal	
Contact Hours / Week x/x/x/x	0/0/14/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The SE_A3 module teaches you how to design engineering (infra)structures within a realistic context: from scratch, investigating multiple variants and materials to find the most suitable design. You will make a functional and technical design, with considerable level of detail. A3 will be organised as an engaging module: during the first weeks of the module keynote lectures from industry and expert staff members will be organised; broad and advanced content on design methodology, and the design of buildings, bridges, rail track and roads/pavements. in parallel you will be trained in various (soft/hard) skill-related lectures/workshops (teamwork, parametric design, presentation, setting up a clear calculation, visual communication) the central task is a group assignment around a realistic clients brief, containing buildings, bridges, rail and/or road, and including an individual contribution of each team member in the form of a preliminary design of one structural type. for deepening your knowledge on subjects, which is necessary for your individual contribution to the team design, various self-study content will be offered from a learning pool. In A3, attention will be given to ethical aspects of design of engineering structures by means of addressing a realistic design context and working in a realistic team atmosphere. Considerations related to ethics are: Which team member will take responsibility for which part of the design? Can we guarantee that the design is safe? Is this design addressing the societal need in a correct way? To what extent is our global environment loaded by the presented design? To what extent will people be affected by the construction process of this design? Can the design be constructed safely and how? The Core faculty module (CFM) will provide the fundamental basis for ethics, on which the course will build forth. Structural Design is also about optimisation and graphical representation CFM content that prepares for this will be built upon. This module also considers topics that will continue the line of both modelling and coding from the MUDE module. MUDE: Coding skills (Python, Matlab etc.) and MUDE on modelling, data and uncertainty analysis are pre-requisites. This module will follow the lines of MUDE by analysing advanced methods with application on bridges, building, road, railway structures and circularity. The MUDE course in this aspect prepare students to understand reliable design parameters, criteria and reliable design choices out of several options.	
Study Goals	On successful completion of this module, you will be able to 1. Integrate steps of conceptual and preliminary design phases of (infra)structures to design a (infra)structure from scratch taking into account the demand of the client and the society, the interaction between the structure and its environment and the need of lifetime maintenance, including design for circularity, while quantifying uncertainties in the brief, the boundary conditions, the loads, the design, and the construction phase (SE3, SE5). 2. Select optimal structural systems (traditional or innovative) for specific design task of buildings and bridges and combination of construction materials steel, timber and concrete, to be able to realize which options can be chosen for the structural system, taken into account the uncertainty in the end-use, lifespan and degradation mechanisms.(SE1, SE4, SE7) 3. Explain how to analyse the interfaces and interaction of track infrastructure and train and their effects on the railway system and the various influencing structural pavement design parameters on different supporting structures (SE1, SE4, SE7) 4. Explain principles of parametric geometry design and the uncertainty in tolerances that have to be taken into account, and apply these principles with appropriate software tools cooperatively in a design team.(SE1, SE7, SE9). 5. Examine transfer of forces in different structural systems and with different structural materials starting from the applied load down to the foundation. 6. Design the structural systems consisting of structural members and joints with appropriate dimensions and materials in a realistic design context (SE1, SE2, SE9) 7. Learn to work in a design team and to make and visualize and present a functional design of a site including buildings, bridges, rails, and roads (SE9) 8. Develop general design of a building, bridge, railways and road structural system to a detailed design level through cooperation within a design team. (SE1, SE2, SE3, SE4, SE5, SE9)	
Education Method	A3-1: Traditional lectures, possibly with use of blended-learning materials or flipped-classroom pedagogy A3-2: Self-study during week 3-10 Availability of lecturers during Q&A-sessions or office hours A3-3: Problem-based learning, discussion in teams, tutor sessions (individual or with full group)	
Assessment	The assessment is as follows: 1. General design principles of (infra)structural systems - Written Exam (35%) 2. Specific design aspects of buildings, bridges, rail and road structural systems - Report (25%) 3. Design challenge - Project (40%) A successful participation in the module, requires a minimum evaluation of 5.8 in all units. A3-1:	

	Written exam (digital midterm exam via Möbius) A3-2: The level of the submitted work for the design assignment (SE-A3- Learning Unit 2) will be assessed against the frame of reference and learning objectives of the available self-study content. In this assessment, students will be assessed by means of some prescribed questions about their learnings of A3-2. The assessment of A3-2 will be done both in formative and in summative way. In the formative assessment, students will be asked to peer-review each others answer sheet and prepare a report on the assessment. In the second step, students will be asked to submit these questions (individual summative assessment and the peer review report). This means that the final report that the students will submit will contain two parts (a chapter containing A3-2 & peer report; and second report of A3-3). While carrying out the final assessment, attention will also be given on the improvements that the students made on the A3-2 questionnaire on the basis of peer review comments. The description of A3-3 assessment has been provided later in the document. A3-3: A3-3 module will be evaluated as a project on the basis of a final report and group presentations. The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam (presentation) and the quality of the final assignment reports. This submodule will be assessed against the frame of reference and learning 259 objectives of A3-3. Final report (individual and group part) and presentation will be rubrics in one summative assessment format. Final report (80 points): It is expected that each team will provide a report containing chapters both from their individual learning together with their teams final design. Please note that since one student per group will be a leader of a specific topic, this will provide the assessor a mean to judge the performance of the individual student. 25 points will be given on the basis of the overall report, whereas 55 points will be given on the basis of the individual chapter. Group presentation (20 points): A group presentation per team will be scheduled at the end of A3-3 sub module. This will be an overall evaluation of the teams final design choices. Students are expected to defend these choices in from of team of assessors.
Expected prior Knowledge	Basic and advanced structural mechanics (from BSc and SE Track Base Q2) - advanced knowledge on loads on structures (from BSc and SE Track Base Q2) - advanced knowledge of reinforced concrete members and basic knowledge of prestressed concrete and joints/connections (from BSc and SE Track Base Q2) - advanced knowledge of steel members and basic knowledge of joints/connections (from BSc and SE Track Base Q2) - advanced knowledge of timber members and basic knowledge of joints/connections (from BSc and SE Track Base Q2) - basic knowledge of circularity and sustainability (from BSc and SE Track Base Q2) - basic skills in CAD, teamwork, report writing, framework software, hand sketching, information literacy (from BSc ctc1000) - basic skills in structural design (BSc ctc1410 and ctc2320) - basic knowledge of road & railway design (BSc ctc3320) - knowledge of integral design and management of civil engineering projects (BSc ctc1220) - MUDE: Coding skills(Python, Matlab etc.) and MUDE on modelling, data and uncertainty analysis
Academic Skills	Team-based work, planning a design task, setting up a clear calculation, convincing visual presentation, oral presentation, report writing, sketching, self-study (problem-based-learning) that is functional to role and responsibility in team
Literature & Study Materials	Content on convincing visual presentation Content on setting up a clear calculation Content on team-based working Content on sketching Content on example details and drawings Quick Reference: Content on principles of design through phases Delft design Method"
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	Calculator and a Formula sheet.

CIEM5210	Applied Mechanics of Structures	15
Responsible Instructor	F.P. van der Meer	
Instructor	Ir. C. Kasbergen	
Instructor	Prof.dr.ir. L.J. Sluijs	
Instructor	Dr.ir. P.C.J. Hoogenboom	
Instructor	A. Varveri	
Instructor	R. Esposito	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	Dr. K. Anupam	
Instructor	Dr. A.A. Mehrotra	
Contact Hours / Week x/x/x/x	0/0/0/18	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module provides students with in-depth expertise on the analysis of the mechanical response of structures. Analytical and computational techniques are explained and applied. The module offers students the possibility to become applied mechanics specialists with the ability to use state-of-the-art computational tools and analytical tools across the structural engineering field. The module also prepares students to contribute to academic research that advances the limits of computational and analytical tools for analysis of the mechanical response of structures. Monitoring & data science learning line: This module builds on the modelling and coding line from the MUDE, with Unit 1 focused exclusively on modelling and Unit 2 addressing a specific aspect of modelling of structures including python coding workshops. The broader theme of uncertainty is addressed in discussions of the reliability of different models. Ethics learning line: Ethics is given attention in this module when the limits of different analysis approaches are discussed. From research perspective, being critical towards the accuracy and reliability of available methods and tools is a first step towards advancing their limits. In a pragmatic engineering environment, having the expertise to be critical of models may lead to moral dilemmas: What level of uncertainty about model reliability is acceptable? Having discussed the uncertainties of relatively high-fidelity models, what is a the good way to deal with even larger uncertainties of simpler modeling approaches? Moreover, ethical aspects of reporting in academia and engineering practice will be discussed in the module, such as acknowledgement of sources and fair representation of results. Design: This module equips students to analyse the behaviour of materials, structures and structural elements. Analysis is an important aspect of the design cycle for civil engineering structures. Units Unit 1 and Unit 2 focus primarily on methods and their underlying formulation. Unit 3 is more applied in the sense that it provides insight in the mechanical response of structures. It will be illustrated with practical examples how this insight can be used in the design of structures. Pitfalls and best practices in structural design will be discussed in the light of the theory provided in the unit. Unit 1 Computational modelling of structures: Introduction to the use of the finite element method; Nonlinear finite element analysis of steel structures; Nonlinear finite element analysis of structures made of a quasi-brittle material; Application of the finite element method to solve engineering problems. Unit 2 Advanced constitutive modelling: Mathematical and mechanical preliminaries; Computational plasticity; Fracture models and continuum damage mechanics; Large deformation mechanics; Computational modelling of visco-elastic materials; Large deformation elasto-plasticity; Introduction to thermodynamics. Unit 3 Advanced mechanics of structural elements: Theory of elasticity: energy principles, closed form solutions and limitations; Theory of plasticity: frame analysis and plate analysis	
Study Goals	On completion of this module, the student will be able to [1] Analyze the mechanical response of structures where that is complicated by the presence of geometric and material nonlinearity [2] Calculate the ultimate load of plates and frames accounting for plasticity [3] Explore the limits of analytical tools for evaluation of elastic stress and strain [4] Formulate, implement and test constitutive laws for detailed numerical analysis of the response of structures [5] Perform nonlinear finite element analysis for structural design verification Unit 1: 1. Select an appropriate mechanical model that idealizes the physical problem of analysis of a structure (boundary conditions, loading, dimension reduction, geometric order of approximation, material model) [1,4] 2. Build an appropriate finite element model that discretizes the selected mechanical model [1, 5] 3. Carry out nonlinear finite element analysis for a range of different applications (steel/concrete structures, seismic response of buildings) [1,5] 4. Interpret finite element solution results to solve engineering problems [1,5] 5. Create a technical report of a performed finite element analysis [5] Unit 2: 1. Select an appropriate constitutive model to account for physical phenomena of time-dependent behavior, direction dependent behavior, permanent deformations and failure in the analysis of material response [1,4] 2. Derive and apply the stiffness relations for orthotropic elasticity and classical laminate theory [1,4] 3. Understand and apply the concepts of large deformation mechanics [1] 4. Explain the concepts of thermodynamics and apply it to large deformation elasto-thermal material modelling [1,4] 5. Explain and apply concepts of elasto-plasticity, visco-elasticity, viscoplasticity in small and large deformation formulations [1,4] 6. Formulate constitutive models to describe the nonlinear response of materials 7. Implement constitutive models as user material in finite element code [1,4] 8. Verify consistency of a user material implementation with its intended behavior [1,4] Unit 3: 1. Reproduce the main steps in the derivation of analytical elasticity solutions, for example the point load on a half space, point	

	load on a half disk, stresses around a spherical cavity in a continuum, stresses around a circular opening in a plate, moments in circular plates, stresses in curved beams, the differential equations for torsion [3] 2. Demonstrate understanding of the physical meaning of these analytical solutions [3] 3. Argue about the usefulness and limitations of these solutions and of elasticity theory in general [3] 4. Explain the upperbound and lowerbound theorems for plastic collapse [1,2] 5. Calculate, by hand, the ultimate load of elastic/plastic frames with or without accounting for moment-normal force interaction [1,2] 6. Calculate, by hand, the ultimate load of elastic/plastic plates [1,2]
Education Method	Unit 1: Lectures, computer lab sessions Unit 2: Lectures, computer lab sessions Unit 3: Lectures
Assessment	Unit 1: The unit will be assessed in an oral exam covering the content of the lectures as well as the assignments (27% of the module grade). The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam and the quality of the final assignment reports. Reporting skills are also taught and assessed. Unit 2: The unit will be assessed in an oral exam covering the content of the lectures as well as the assignments (33% of the module grade). The final grade will be determined in a single rubric based on the level of understanding displayed at the oral exam and the quality of the final assignment reports. Unit 3: Students receive a grade based on peer feedback from presentation sessions (4% of the module grade). Final assessment is done with a written exam (36% of the module grade). A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Expected prior Knowledge	University units, or equivalent, in Calculus, Linear Algebra, Numerical Analysis. Coding skills: Python (MUDE level) Mechanics of slender structures (SE track base) Numerical methods, Introduction to Finite Elements (MUDE level)
Academic Skills	Analytical thinking, coding, reporting, presenting.
Literature & Study Materials	Unit 1: Lecture slides, DIANA user manual Unit 2: - R. de Borst and L.J. Sluys, (2001) Computational methods in non-linear solid mechanics - G.A. Holzapfel (2000), Nonlinear Solid Mechanics: A Continuum Approach for Engineering, John Wiley & Sons Ltd, ISBN 978-0-471- 82319-3 - Brannon, R.M. (2003), Functional and Structured Tensor Analysis for Engineering, UNM Book Draft, (https://my.mech.utah.edu/~brannon/public/Tensors.pdf) Unit 3: - Vrouwenvelder, A.C.W.M. and Witteveen, J. (2003) "Plastic Analysis of Structures, The plastic behaviour and the calculation of beams and frames subjected to bending", Lecture notes Delft University of Technology. - Vrouwenvelder, A.C.W.M. and Witteveen, J. (2003) "Plastic Analysis of Structures, The plastic behaviour and the calculation of plates subjected to bending", Lecture notes Delft University of Technology. - Blaauwendraad, J. (2001) Theory of Elasticity Direct methods. Lecture notes Delft University of Technology - Blaauwendraad, J. (2001) Theory of Elasticity Energy principles and variational methods. Lecture notes Delft University of Technology - Handouts on selected topics
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	None
Collegerama	No

CIEM5220	Applied Dynamics of Structures	15
Responsible Instructor	Prof.dr. A. Metrikine	
Instructor	Ir. C. Kasbergen	
Instructor	Dr.ir. M.J.M.M. Steenberghe	
Instructor	Dr.ir. K.N. van Dalen	
Instructor	A. Tsouvalas	
Instructor	Dr. K. Anupam	
Instructor	Dr. Z. Yang	
Instructor	Dr. G. Lavidas	
Contact Hours / Week x/x/x/x	0/0/0/18	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Study Goals	Overall Module ILOs: 1. Describe conceptually the earthquakes, wind, wave and vehicle loads on structures (SE1, SE3). 2. Formulate and interpret the physical foundations of mathematical models for the dynamics of structures under earthquakes, wind, wave and vehicle loads (SE1, SE3). 3. Describe, model and analyse the earthquakes, waves, wind, and rail and road irregularities as random signals (SE1, SE3). 4. Understand and apply analytical methods of mathematical analysis of the dynamic response of structures to earthquakes, wind, waves and moving loads (SE1, SE3). 5. Describe and explain the main mechanisms of vibration excitation of structures by moving loads (SE1, SE3). 6. Analyse the dynamic response of structures in earthquake-prone regions and use it in design considerations (SE2, SE5, SE6). 7. Design structures for renewable energy generation based on their response to wind and waves (SE2, SE5, SE6). 8. Compute and evaluate the dynamic response of bridges to moving loads and use it in design considerations (SE2, SE5, SE6). 9. Compute and evaluate the dynamic response of rigid and flexible pavements at the tire/pavement interface and use it in design considerations (SE2, SE4, SE5, SE6). 10. Compute and evaluate the dynamic behaviour of a railway track and its environment under train loading and use it in design considerations (SE2, SE4, SE5, SE6) B2-1: 1. Explain basic seismological concept (SE_B2_1,2). 2. Analyse recorded ground motions and distinguish local site effects (SE_B2_3). 3. Formulate the equations of motion of NDOF systems subjected to ground excitation and critically assess their correctness (SE_B2_2). 4. Apply different methods of seismic analysis in structures (linear and non-linear response) and analyse the results (SE_B2_4,6). 5. Consider soil-structure interaction and fluid-structure interaction in seismic analysis of structures (SE_B2_4,6). 6. Analyse and design structures in earthquake-prone regions using code provisions (SE_B2_4,6). B2-2: 1. Describe wind and waves stochastically (SE_B2_1,2,3). 2. Formulate and interpret the physical foundations of mathematical models for the structural wind and wave loads on structures (SE_B2_2). 3. Formulate and evaluate the equations of motion of NDOF systems subjected to wind and wave excitations (SE_B2_3,4). 4. Define wind and wave interactions with basic structural concepts (SE_B2_3,4). 5. Analyse the dynamic response of simple structures to wind and wave excitations (SE_B2_3,4). 6. Describe the phenomena of galloping, flutter and vortex-induced vibrations (SE_B2_1,4,7). 7. Design and analyse the dynamic response of structures for renewable energy generation from wind and waves (SE_B2_7). B2-3: 1. Describe and formulate mathematical models for the structural elements and structures in interaction with moving loads (SE_B2_1,2,3,4). 2. Describe and explain the major types of structural and environmental responses to moving loads (SE_B2_1,5). 3. Derive and analyse the response of 1D structures and 3D layered continua to moving loads (SE_B2_7). 4. Design a concept bridge model under moving loads and evaluate its dynamic response (SE_B2_8). 5. Design rigid and flexible pavements under moving loads and evaluate their dynamic responses and of their environment, taking into account the stochastic nature of the loading (SE_B2_9). 6. Compute and evaluate the dynamic behaviour of a railway track under train loading and design environmentally friendly tracks, taking into account the stochastic nature of the loading (SE_B2_10). 7. Describe the degradation mechanisms of road pavements and railway tracks (SE_B2_10) based on targeted monitoring.	
Education Method	B2-1: Lectures, discussion of assignments, lecturer-assisted computational exercises. B2-2: Lectures, discussion of assignments, lecturer-assisted computational exercises B2-3: Lectures, discussion of assignments, lecturer-assisted computational exercises.	
Assessment	One summative assessment on module level (all units) is based on an assignment that encompasses the basics of the random vibrations and numerical modelling (both subject matters are taught to all students of the module). In addition, at the end of the quarter, one written exam per unit is held. The total number of summative assessments in this module is four. The assessment of the module is as follows: 1. Structural Response to Earthquakes - Written Exam (25%) 2. Dynamics of structures subjected to wind & waves - Written Exam (25%) 3. Dynamics of structures under moving loads - Written Exam (25%) 4. Applied Dynamics of Structures - Group Assignment (25%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. - structural mechanics at the BSc level; - formulation of equations of motion for NDOF systems and onedimensional continuous structural systems (cables, beams, rods); - solution of ordinary differential equations at the BSc level; - modal and Fourier analysis of undamped and damped discrete and continuous systems.	
Academic Skills	The following skills will be further developed in this module: critical and interdisciplinary thinking, identification of problem-oriented solution procedures, information collection and synthesis, application of theory to new situations using the relevant equations and correlating mathematical solution with physical insights and design procedures, information collection and synthesis, use of data for critical assessment of modelling assumptions and design considerations.	
Literature & Study Materials	B2-1: Material distributed by the lecturer: readers of CIE5260 and CIE5340 (adjusted to the new learning unit); Dynamics of Structures: Theory and Application to Earthquake Engineering by Anil K. Chopra; Maple files and Matlab (Python) scripts for seismic analysis (prepared by the teaching team). B2-2: Lecture slides and notes, relevant book chapters B2-3: Lecture slides and notes, book chapters (Fryba, Vibration of solids and structures under moving loads; Yang, Yau and Wu, Vehicle-bridge interaction dynamics).	

Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
------------------	---

CIEM5230	Concrete Structures	15
Responsible Instructor	Dr. Y. Yang	
Instructor	G. Ye	
Instructor	Dr.ir. H.E.J.G. Schlangen	
Instructor	Dr.ir. H.W.M. van der Ham	
Instructor	Dr. Y. Yang	
Instructor	R. Esposito	
Instructor	M. Lukovi	
Instructor	Ir. E.O.L. Lantsoght	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	M. Poliotti	
Contact Hours / Week x/x/x/x	0/0/0/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Concrete is the worlds mostly used construction material. The module Concrete Structures offers a comprehensive cluster of courses that, through the application of existing knowledge and insights in new scientific developments, prepares students for the current and future challenges of society relating concrete structures. Various sectors of Civil Engineering and the entire life-span of concrete structures are covered: from the understanding and design of concrete material, the advanced modelling of concrete members taking into account the unique behaviour of concrete, to the design, maintenance of complex concrete structures, such and concrete bridges. The course will include 5 units, namely: B3-1: Advanced modelling of concrete structures. B3-2: Concrete science and technology. B3-3: Numerical modelling of concrete structures. B3-4: Concrete structures under special loadings. B3-5: Concrete bridges. On completion of this module, the student will be able to: 1. Design concrete structures and members using plasticity theory and fracture mechanics for concrete. 2. Explain and design cement-based building materials and the advanced concrete mixture design methodology, further apply to design, control, and execute durable and sustainable concrete in practice. 3. Create, carry out, interpret, and document nonlinear finite element analysis of reinforced concrete structures or elements. 4. Model the behaviour of concrete members and structures subject to various environmental and anthropogenic loads, and to create appropriate design accordingly. 5. Design and assess precast and cast-in-situ concrete bridges and bridge elements considering the construction methods and the loading conditions.	
Study Goals	A breakdown of the learning objectives in each unit given in the course is explained as following: B3-1: 1. Interpreted simplified theoretical models with limit state theorems of the theory of plasticity 2. Design and assess various structural types using limit analyses, the types of structures include beam, wall, slabs, and joints. 3. Point out the fracture process of concrete and its difference between linear elastic fracture mechanics. 4. Outline the influence of the local fracture process of concrete on the global structural behaviour in terms of fatigue behaviour, shear and punching shear behaviour, etc. B3-2: 1. Outline the fundamentals of concrete materials 2. Explain the mechanisms and the chemical process relate to the strength development and the resultant mechanical properties 3. Control volume change during the hardening process 4. Design concrete mixture and advise corresponding execution methodology. B3-3: 1. Create appropriate finite element models for reinforced concrete structures or elements 2. Interpret nonlinear finite element analysis results 3. Explain how to account for uncertainties when using finite element analysis 4. Compile adequate finite element analysis reports. B3-4: 1. Explain the behaviour of concrete structures under temperature effects in the early stage of hardening and hardened concrete, including (technological and structural) measures to influence this behaviour. 2. Discuss the difference between structural response under external loads and imposed deformations, and design concrete structures subjected to imposed deformations. 3. Design and execution of storage systems and reservoirs in reinforced and prestressed concrete, focusing on tightness criteria. 4. Design of concrete protective structures under extreme loads, e.g. impact, blast, fire, cryogenic loads. 5. Design concrete structures for cyclic loading with various amplitude and duration such as seismic loading and fatigue loading. 6. Outline the main environmental loads and possible deteriorations of concrete structures. B3-5: 1. Outline the considerations of designing a concrete bridge in the Netherlands. 2. Evaluate the advantages and disadvantages between different construction methods (cast in situ) and able to explain the relationship between the construction method and the alignment of the reinforcement and/or prestressing cables. 3. Apply the different traffic load models specified in the Eurocode. 4. Apply the methods of load distribution. 5. Evaluate the different alternatives and analyse and calculate roughly the main dimensions of the bridge. 6. Choose the different types of bearings and expansion joints properly.	
Education Method	The course will be given though various education methods, including conventionally in person lectures, self-study with prerecorded video and reading materials, and workshops in the lecture room.	
Assessment	B3-1: 100% from the combined final written exam with unit SE-B3-2 B3-2: 100% from the combined final written exam with unit SE-B3-1 B3-3: 100% from oral exam B3-4: 100% from oral exam B3-5: 100% from oral exam	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Knowledge of material properties of concrete, concrete mix design, mechanics of materials. Design principles and calculation methodes of reinforced and prestressed concrete structural members in both ULS and SLS. Structural analysis and matrix analysis of structures. Understand the safety phylosofies and theoretical background of Eurocode.	
Academic Skills	Evaluate and interpretation of results and development of judgemental skills, writing scientific reports, learn how to solve problems, work and discussing in a group, present your results orally, and simplify practical problems into physical models.	
Literature & Study Materials	B3-1: Learning materials, slides and prerecorded videos publishehd on Brightspace. B3-2: Suggested text books and Lecture notes published on Brightspace. B3-3: Lecture slides and selected papers published on Brightspace.	

Judgement	B3-4: Textbook and slides published on Brightspace.
	B3-5: Learning materials, slides and videos published on Brightspace.
	See Assessment
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Permitted Materials during Exam	None
Collegerama	Yes

CIEM5240	Steel and Composite Structures	15
Responsible Instructor	Prof. M. Veljkovic	
Instructor	Ir. P.A. de Vries	
Instructor	F.P. van der Meer	
Instructor	Prof.dr.ir. M.A.N. Hendriks	
Instructor	Prof. M. Veljkovic	
Instructor	Dr. M. Pavlovic	
Instructor	Dr. F. Kavoura	
Instructor	O. Karpenko	
Instructor	Dr. T. Tankova	
Responsible for assignments	Dr. T. Tankova	
Contact Hours / Week x/x/x/x	0/0/0/24	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module is composed of 3 parallel units. Unit 1 on Strength and Durability of Steel and Composite Structures and Joints provides in-depth knowledge on how to design, model, analyse and critically assess results of prediction models of structural members and joints. The principles and theory of using experiments and reliability-based assisted design and verification of Steel and Composite Structures (SCS) and joints are explained and practised for monotonic and cyclic loading. Unit 2 on Heavy-duty and Unit 3 on Lightweight steel and composite structures will allow students to apply and extend their knowledge from Unit 1 in real case structures, such as bridges, support structures for renewable energy, lightweight FRP structures, cold-formed steel members, etc. The learning line on Modelling, Data management and sensing, risk analysis is continued from the Faculty Base and Track base module. Students will have the opportunity to participate in monitoring and sensing lab (or virtual) experiments during the module and evaluate data. Ethics related to the design and reporting, construction, ownership and decommission of steel and composite structures are discussed throughout the module. Incidences and accidents which lead to a collapse of structures will be discussed from (un)ethical behaviour of partners in a project. The balance between economy-safety-sustainability of construction work will be addressed, showing future engineers' responsibility. B4-1: The content of the units is defined within 3 themes: Theme 1 (30%): Modelling of Steel and Composite structures and design by experiments Elastic, plastic and damage material modelling of steel and FRP; Modelling of local and global elastic buckling; Anisotropic and layered modelling; Evaluation of design values for resistance of various failure modes of steel members. Theme 2 (30%): Design verification of steel connections and joints- Connection component resistance: resistance of bolts, welds, net cross-section in tension and shear; Shear and axially loaded joints Simple connections; Beam to column moment resistant joints; Steel column base joints. Theme 3 (40%): Fatigue and fracture Design of steel structures against fatigue failure: Welded details; Bolted details; Detail classes; SN curves; HSS curves; Post-welded treatment for welded joints; Local prediction models (Hot spot stress approach, SCF, effective notch stress) by FE analysis on examples of hollow section joints, OSD. Fracture mechanics - theoretical background; Material properties for fracture mechanics; Crack initiation and propagation by analytical models and FEA B4-2: The content of the units is defined within 2 themes: Theme 1: Steel and hybrid bridges - Structural design of steel bridges - Bridge loads, traffic accounting and analysis - Design verifications of steel bridge elements - Advanced bridge analysis - Execution and details of steel bridges - Renovation and maintenance of steel bridges Theme 2: Supporting structures for renewable energy - Hollow section steel joints in jacket structures- Monopile supporting structures for wind turbines and connections. B4-3: The content of the units is defined within 2 themes: Theme 1 (40%): Lightweight and slender steel structures, Stability and resistance of complex columns and frames. Stability of welded and rolled steel members. Cold-formed thin-walled steel members. Theme 2 (60%): FRP Composite Structures Design of FRP structures FRP material behaviour. Analysis and verification of FRP structural members Analysis and verification of FRP joints and details.	
Study Goals	B4-1: Students will be able to: 1. Create, carry out, interpret and document linear and nonlinear finite element analysis of steel members and connections/joints. (B4-LO1) 2. Improve design models for resistance of structural members and joints based on quantification and understanding of scattering of material and structural properties (e.g. partial factors). (B4-LO3, LO4, LO5) 3. Explain the concept of the Monte Carlo method and perform Monte Carlo simulations using Excel and Python programming. (B4-LO3, LO5) 4. Explain the fatigue initiation and crack propagation process and mechanism and determine the main factors influencing the fatigue strength of materials, connections, and structures. (B4-LO4, LO5, LO6) 5. Explain the basics of the fatigue behaviour of critical details (welded connections, bolted connections, tubular joints) and the theoretical background of prediction methods. (B4-LO3, LO4, LO5) 6. Design steel structures exposed to cyclic loading (fatigue strength) and explain design concepts in the presence of structural damage based on the lifetime prediction methodologies. (B4-LO3, LO4, LO5, LO6) 7. Design ductile, damage tolerant and reusable moment resisting and simple joints in steel structures for their beam-to-column and column to base-plate connections. (B4-LO3, LO4, LO5, LO6) 8. Apply the theoretical background to create engineering models for the design verifications of the structural connections/joints	

	exposed to monotonic, cyclic and impact loading. (B4-LO3) B4-2: Students will be able to: 1. Understand the design requirements and principles of designing diverse fixed or movable highway, railway and pedestrian bridges. (B4- LO2,LO3,LO4) 2. Design durable, sustainable and cost-efficient bridges by integrating global structural, cross-section and deck systems. (B4-LO3,LO6) 3. Analyse structural behaviour of steel and steel-concrete bridges in response to construction phases, traffic and environmental loads. (B4- LO4,LO6) 4. Analyse ultimate, serviceability and fatigue limit states of new and existing bridges. (B4-LO3,LO5,LO6) 5. Optimise construction time and costs by combining diverse production and execution methods. (B4-LO3,LO4,LO6) 6. Create strengthening solutions for existing bridges. (B4-LO2,LO3,LO4) 7. Calculate the loads applied in the supporting structure of wind turbine towers and evaluate their structural performance. (B4-LO3,LO6) 8. Validate the design of the tubular and multi-membered towers/jackets and their connections by using calculations according to structural design codes, guidelines and linear and non-linear FE models, including imperfections. (B4-LO1,LO5,LO6) B4-3: Students will be able to: 1. Design lightweight rolled, welded, cold-formed, stiffened and unstiffened cross-sectional steel members of plated and cylindrical steel structures. (B4-LO2,LO3,LO6) 2. analyse the effects of local buckling, imperfections and post-buckling behaviour in thin-walled plated and cylindrical structural members and apply engineering methods to perform the design verification. (B4- LO2,LO3,LO6) 3. Design and analyse steel frames by considering members imperfections and limit state provisions of design codes. (B4-LO5,LO6) 4. Choose optimal combinations of FRP materials and production processes for a specific application in infrastructure and set of loading and environment conditions. (B4-LO4,LO5,LO6) 5. Design FRP structures by considering the structure's specific material behaviour and mechanical and functional requirements. (B4-LO4,LO6) 6. Analyse the mechanical performance and failure modes of FRP structures and joints on the level of plies, laminates and structural components using Classical Laminate Theory, progressive failure analysis and Finite Element models. (B4-LO1,LO4,LO6) 7. Verify designs of FRP structural members and joints by using calculations according to structural design codes. (B4-LO3,LO4,LO6)
Education Method	B4-1: Pre-recorded short video lectures will be embedded into weekly instructions for self-studying aligned with blended learning, live/online lectures during contact hours, problem-based learning through exercises, theoretical problems and home assignments, numerical examples, discussion in teams during contact hours, tutor sessions (FEM software), guest lectures from industry. B4-2: Pre-recorded short video lectures will be embedded into weekly instructions for self-studying aligned with blended learning, live/online lectures during contact hours, problem-based learning through exercises, theoretical problems and home assignments, numerical examples, discussion in teams during contact hours, tutor sessions (individual or with a complete group), role-plays, guest lectures from industry, site visits. B4-3: Pre-recorded short video lectures will be embedded into weekly instructions for self-studying aligned with blended learning, live/online lectures during contact hours, problem-based learning through exercises, theoretical problems and home assignments, numerical examples, discussion in teams during contact hours, tutor sessions (individual or with a complete group), role-plays, guest lectures from industry, site visits.
Assessment	B4-1: LO2,3,4,5,6,7: Application of theoretical background, design and problem-solving skills are tested in open-book written exam with open-ended questions related to steel joints, fatigue resistance prediction by local approach (FEA), composite structures and connection details. LO1,2,3,8: The assessment of design and analysis skills, and collaborative work are tested through an integral report on analysis and calculations connecting content of all 3 modules. Part of report related to unit 1 will focus on analysis and verification specific fatigue detail, local buckling or joint resistance of e.g. a bridge, structure, wind tower or FRP structure from unit 2 and 3. Students will apply engineering tools and verification standards (Eurocodes), create and analyse own finite element models for specific structures and details. B4-2: LO1,2,5,6,7: Application of theoretical background, design and problem-solving skills are tested in open-book written exam with open-ended questions related to steel bridges and wind turbine supporting structures. LO3,4,7,8: The assessment of design and analysis skills and collaborative work are tested through an integral report on analysis and calculations that connect content of all 3 modules. Part of the report related to unit 2 will focus on analysis and verification of limit states (ultimate, serviceability, fatigue) of critical cross section or component of a steel bridge and wind turbine tower. The FE analysis from unit 1 will be implement in the report and compare the engineering tools and design verification standard. B4-3: LO1,3,4,5,6: Application of theoretical background, design and problem-solving skills are tested in open-book written exam with open-ended questions related to FRP composite material, structures and details, and buckling prone, thin-walled steel structure components. LO2,3,5,6,7: The assessment of design and analysis skills and collaborative work are tested through an integral report on analysis and calculations that connect content of all 3 modules. Part of report related to unit 3 will focus on analysis and verification of limit states (ultimate, serviceability, fatigue) of critical cross-section or component of FRP composite structures and slender steel structures. The FE analysis from unit 3 will be implemented in the report, and compared to engineering tools and design verification standards.
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. To be able to follow and efficiently follow this module, pre-knowledge from BSc studies, and preceding modules of MSc study (faculty base module, track base module, A module) are necessary: Basics of structural analysis (incl. principles of second order theory). Basics of continuum mechanics and strength of materials. Design and verification skills for steel cross sections and members class 1, 2 and 3. Basics of quantification and analysis of loads in civil engineering structures. Concepts of limit state analysis and design. Principles of design of structural joints and connections in structures. Basics of the design of steel-concrete composite beams. Basics of stability of 1D and 2D structures. general understanding of numerical methods) Reliability and statistics Basic and advanced structural mechanics Basic knowledge of circularity and sustainability Basic skills in CAD, teamwork, report writing, framework software, hand sketching, information literacy.
Academic Skills	Team-based work, planning a design task, setting up a clear calculation, convincing visual presentation, oral presentation, report writing, sketching, drafting, self-study.
Literature & Study Materials	B4-1: Text books; Reading materials; Readers; Journal/Conference Papers; Pre-recorded Video Lectures; R&D Reports

Judgement	B4-2: Text books; Reading materials; Readers; Journal/Conference Papers; Pre-recorded Video Lectures; R&D Reports
	B4-3: Text books; Reading materials; Readers; Journal/Conference Papers; Pre-recorded Video Lectures; R&D Reports
	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations
Collegerama	Open-book written exam means that all theoretical teaching material is allowed. Numerical examples are not allowed. No

CIEM5250	Building Engineering	15
Responsible Instructor	Dr.ir. H.R. Schipper	
Instructor	Dr. I. Nevzgodin	
Instructor	A.C.B. Schuurman	
Instructor	Ir. E.R. van den Ham	
Instructor	Ir. S. Pasterkamp	
Instructor	Dr.ir. H.R. Schipper	
Instructor	H. Alkisaai	
Instructor	R. Crielaard	
Instructor	Ir. A. Borgart	
Instructor	Prof.dr.ir. P.C. Louter	
Instructor	P. Eigenraam	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	Ir. C.J. Janssen	
Instructor	Dr. E. Brembilla	
Instructor	Dr. F. Kavoura	
Instructor	M.P. Felicita	
Instructor	Ir. J.P.G. Ramler	
Contact Hours / Week x/x/x/x	0/0/0/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module will focus on the engineering of buildings as complex systems in which many facets need to be aligned in an integrated interdisciplinary design process. From an economy point of view, a share of about 70% of construction industry is involved with buildings, which makes this module very relevant for future civil engineers. The module content is grouped in three learning units: LEARNING UNIT 1 (LU1, Architecture, Building Physics and Façades, 4 EC) will provide you with the specific building-related background knowledge that is not directly structural engineering: contemporary and 19th century architectural styles façade design/detailing building physics (heat transport, acoustics, wind flow) building services (heating, ventilation, air-conditioning, vertical transport, water). LEARNING UNIT 2 (LU2, Building Structures, Construction Methods and Foundations, 6 EC) is about the flow of forces in regular building structures and stability in various typologies of buildings (office, commercial, industrial, residential, etc.). Unit 2 is grouped around the following three themes: Theme 1 (Building Structures): advanced knowledge on the flow of forces in building structures stability and lateral resistance and known solutions for various typologies of buildings according to their function (office, commercial, industrial, residential) use and reuse of various building materials, to give you a clear understanding on which options are available and which solutions are appropriate for every part of the new or existing building structure complexity of the design process and responsibilities of different partners in the design and construction chain Theme 2 (Foundations): interaction with the subsoil via the foundation is an important part of the building design, and you will learn how to develop a good foundation in cooperation with the geotechnical advisor. attention will also be paid to deep excavations, that are often required for foundation and parking garages or other underground spaces under a building. Theme 3 (Construction methods): The design of the building is depending on available construction methods, and by studying these, you will get a clear understanding of the limitations and options each of these methods offer. Impact on surroundings and urban context will be addressed, as well as safety of construction, economy and material efficiency (zero-waste) on the construction site. LEARNING UNIT 3 (LU3, Design of Spatial Structures, 5 EC) is an advanced unit in the field of special and spatial building structures, their design process, structural mechanics, structural behaviour and performance, computational modelling, combined with practical use of design- and analysis tools for structural engineering. Topics include structures in nature biomimetics plates, shells and vaults membrane and pneumatic structures cable-nets tensegrity structures (space) frames grid(shell) structures and domes adaptive and deployable structures. Whereas Learning Unit 2 will focus on more regular building types, Learning Unit 3 broadens the scope with an extensive introduction into complex geometries and typologies. Apart from design skills, also academic skills will be trained during this unit since the unit includes a mini conference, including the writing and presentation of scientific papers. Working on an integrating design exercise of a realistic building that connects all three learning units, you will experience the consequences (limitations, options and uncertainties) of available design and construction methods. During the exercise, you will receive feedback from professional building engineers.	
Study Goals	On completion of this module, you will be able to: 1. recognise and discuss architectural styles and construction methods of the 19th, 20th and 21st century;	

	2. distinguish façade typologies and use those in design, while correctly applying the laws of building physics in the design evaluation to meet relevant criteria for climate, energy-efficiency and durability, taking into account stochastic material properties, climate data and user behaviour; 3. discuss the relationship and integration between design of load bearing structure, façade structure and other relevant parts/aspects of the building, such as services, vertical transport, plumbing, climate, ducts; 4. analyse whether reuse of existing structures or structural components in a project is economically and technically feasible and, where it is, reuse such existing building structures in the design of the newly developed building, correctly assessing the reliability of the existing structure or components based on measurements or probabilistic reasoning; 5. correctly identify functional, structural and construction requirements and loads for low-rise and multi-storey buildings with different targeted functions; 6. explain and apply, in the context of a design, the most important typologies and stability systems of low-rise and multi-storey buildings, using different construction materials, and being able to model and simulate the behaviour of such systems both with self-written code and existing FEM software; 7. develop, analyse and evaluate various reliable solutions for building foundations and deep basements in close cooperation with a geoengineer, considering the close interaction between structure and foundation, recognising and limiting risks of uncertain underground soil conditions, surveying and foundation methods; 8. explain how a building design can be constructed in an economic, safe and practical manner, acknowledging what the consequences and risks of construction methods are for the design and vice versa, and acting as a structural engineer with a professional, responsible and ethical attitude; 9. Design spatial structures (such as domes, (grid)shells, inflatables and membrane structures, space frames, cable systems and tensegrity structures), analyse and explain their 3D structural action; 10. Use advanced design tools, numerical analysis tools, and physical models or measurements to analyse and visualise 3D structural behaviour, and develop such models in self-written code.
Education Method	LU1: Lectures, weekly short exercises, discussion in class, design studio LU2: Lectures, individual feedback sessions centred around design exercise LU3: Lectures, hands-on physical modelling workshops, discussion in teams, presentations, tutor sessions (individual or with full group)
Assessment	Each of the three learning units will have a separate written exam, focused on the theory (lecture notes, textbooks) and skills that can best be assessed in a written exam. Since building engineering is very much design-oriented and aiming for a cognitive level that is hard to assess in the limited time of a written exam, many formative assessment moments will be included, for example by expert feedback, discussion sessions, presentations, and peer review of designs. On overall-module level, for each academic year, a different building for a specific clients brief will have to be designed by the students. This design assignment will function as the integrator of the module. Students that only take a single unit from module SE-B5 will design a subset of the total exercise, in combination with the written exam. The final grade will be determined in a single rubric based on the level of understanding displayed at the presentation (oral exam) and the quality of the final assignment reports LU1: individual written exam (2 EC, 13%), design assignment on module level, see below LU2: individual written exam (3.5 EC, 23%), design assignment on module level, see below LU3: individual written exam (2.5 EC, 17%), design assignment on module level, see below The design assignment on module level will comprise 7.0 EC (47%), which ECs are spread over the above 3 learning units (approx. in a proportion 13%, 17%, 17% for LU1/2/3). The assignment will include presentations, poster and paper. The design assignment is likely to be a team exercise, depending on number of participants. Formative assessment will be used throughout the module, in the form of (two)weekly short exercises, quizzes or short tests.
Tags	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Building & Spatial Development Building Engineering Design Ethics Group work Integrated Practicals Soil Mechanics Structural Mechanics Sustainability
Expected prior Knowledge	The module can only successfully be completed with sufficient prior knowledge in the following subjects: - general understanding of optimisation methods and numerical methods (from CFM/MUDE - necessary for LU3) - reliability and statistics (from CFM/MUDE and BSc - necessary for understanding safety of building structures/materials, stochastic loads and soil/foundations) - fluid flows (from Programme Base and BSc - recommended for wind-flow related topics) - energy flow through solids (from Programme Base, necessary for building physics) - basic and advanced structural mechanics (from BSc and SE Track Base) - advanced knowledge on loads on structures (from BSc and SE Track Base) - advanced knowledge of reinforced concrete members and basic knowledge of prestressed concrete and joints/connections (from BSc and SE Track Base) - advanced knowledge of steel members and basic knowledge of joints/connections (from BSc and SE Track Base) - Basic knowledge of timber members and basic knowledge of joints/connections (from BSc and SE Track Base) - basic knowledge of circularity and sustainability (from BSc and SE Track Base) - basic skills in CAD, teamwork, report writing, framework software, hand sketching, information literacy (from BSc) - theory of ethics (from BSc and Programme CE Base)
Academic Skills	Team-based work and collaboration, planning a design task, setting up a clear calculation, convincing visual presentation, oral presentation, academic literature review, academic writing and presenting, sketching, drafting, physical modelling, self-study, effective communication
Literature & Study Materials	LU1: Textbooks, lecture notes, will be announced at start module LU2: Reader Building Structures 1, 2018 ed. K.C. Terwel Reader Building Structures 2 - R. Abspoel 2013 Concrete Building Structures 2016 Structural calculations of High Rise Structures, 2017 Reader Renovation and Reuse, 2022, S. Pasterkamp LU3: To be decided, will be announced at start module
Judgement	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Will be announced on Brightspace at the start of the module.

CIEM5260	Transportation Infrastructures	15
Responsible Instructor	Dr. V.L. Markine	
Responsible Instructor	Dr. X. Liu	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Prof.dr. Z. Li	
Instructor	Dr. K. Anupam	
Instructor	Prof.dr.ir. R.P.B.J. Dollevoet	
Instructor	Dr. Z. Yang	
Instructor	Dr. H. Wang	
Contact Hours / Week x/x/x/x	0/0/0/18	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module provides knowledge on modern tools for design, assessment, performance prediction and optimisation of engineering structures in application to transportation infrastructures, such as roads, railways, bridges, and tunnels. In this process, the entire design-construction-service cycle will be considered. The analysis of structural behaviour commences with defining the loading conditions. Therefore, the dynamic interactions of road/track/bridge/tunnel and moving vehicles will be introduced first. The students will learn how to numerically analyse such interactions and their consequences like rutting, vibrations and noise. The methods for analysis of the long-term behaviour of engineering structures, including wear, (rolling contact) fatigue damage, etc., will be considered as well. Additionally, the students will learn how to apply the structural reliability methods to account for the uncertainties of real life, even including the effects of the weather. The module also focuses on critical issues related to the aging of transportation infrastructures. Topics will cover aspects of material and structural degradation due to increased dynamic loading, environmental factors, and climate change effects, and familiarize with tools to assess and predict structural durability. The use of innovative materials to enhance the performance and durability of transportation infrastructures will be also discussed. The students will get acquainted with modern general-purpose finite element software: ANSYS, ABAQUS, PLAXIS, Discrete element methods, etc., as well as with some dedicated design software, and will use these tools to assess structural behaviour at multi-scale level. To assure the long-term safe performance, the methods of condition monitoring of engineering structures in application to transportation infrastructures will be studied. Several non-destructive testing (NDT) technologies like Digital Image Correlation (DIC), laser scanning, and satellite data used to assess the performance and condition of infrastructure will be covered. Examples of monitoring solutions using static, moving, and crowd sensing technologies will be analysed. Application of the NDT methods for parameter identification of the numerical models will be considered as well. Combined with state-of-the-art data-analysis methods and machine learning, the monitoring information will turn into maintenance strategies. In this module we organise excursions to companies to discuss their engineering approaches for integral design for transportation infrastructures. All knowledge will be applied to real-life case studies to be performed in close collaboration with manufacturers and engineering companies. All these will serve as preparation for MSc theses.	
Study Goals	On successful completion of this module, the student will be able to B6-LO1: Analyse the design-construction-service cycle of transportation infrastructures (SE2, SE3, SE5). B6-LO2: Evaluate state-of-the-art construction processes and techniques for transportation infrastructures (SE2, SE7). B6-LO3: Evaluate transportation infrastructures modelling and design solutions, considering the effects of the inherent dynamics, operational conditions, and relevant sources of uncertainties (SE1, SE4). B6-LO4: Propose physics-based and data-based monitoring and maintenance solutions for transportation infrastructures (SE6, SE8). B6-LO5: Create innovative and ethical solutions and compare results with the state-of-the-art knowledge in real-life challenges from selected case studies from the industry (SE7, SE8). B6-LO6: Practice and further develop effective communication skills (both written and oral) to ethically share results and opinions with peers, engineers, academics, and stakeholders from the transportation infrastructure industry (SE9). B6-LO7: Provide constructive feedback, resolve possible conflicts in diverse teams, and ethically tackle critical transportation infrastructure challenges (SE9). B6-1: After completing this unit, the students will be able to: B6.1-LO1 Analyse and evaluate methods for the construction process of transportation infrastructures. B6.1-LO2 Evaluate design aspects and procedures for transportation infrastructures. B6.1-LO3 Explain the capabilities of different design methods and solutions under different parameters, variations, and uncertainties in properties and loading patterns. B6.1-LO4 Select and apply design methods, and propose improvements in the construction and design of transportation infrastructures by tackling real-life case studies from the transportation industry. B6.1-LO5 Evaluate current practices of maintenance in the transportation infrastructure industry. B6.1-LO6 Ethically and effectively discuss and analyze results with main stakeholders. B6-2: After completing this unit, the students will be able to: B6.2-LO1 Evaluate the most important dynamics and operational conditions of existing transportation infrastructures. B6.2-LO2 Explain degradation mechanisms that can affect transportation infrastructures during their whole life cycle. B6.2-LO3 Assess the performance over time of transportation infrastructures, considering both dynamics and the influence of degradation mechanism. B6.2-LO4 Select and apply methodologies to estimate the remaining lifetime of existing transportation infrastructures. B6.2-LO5 Discuss and present results to different stakeholders effectively and ethically. B6-3: After completing this unit, the students will be able to: B6.3-LO1 Implement state-of-the-art monitoring solutions for transportation infrastructures. B6.3-LO2 Evaluate current practices of maintenance in the transportation infrastructure industry. B6.3-LO3 Design intelligent condition-based maintenance strategies including both physics and data-based characteristics of transportation infrastructures. B6.3-LO4 Assess the performance of transportation infrastructures over their whole life-cycle,	

Education Method Assessment	including sustainability aspects. B6.3-LO5 Effectively and ethically present results and discuss their advantages and disadvantages to different stakeholders. Lectures / Blended Learning / Laboratory and Simulation Experiments / Excursions The module is assessed through four summative assessments: three on the unit level and one on the module level. 25%: B6-1 written exam. A set of questions from this unit will be included in the Module assessment (practical questions on engineering/analytical methods, software for a real-life design problem). 20%: B6-2 written exam. Knowledge of the theoretical background and problem-solving skills are tested in a written exam with open questions. 15%: B6-3 report. The assessment of this unit consists of a final individual report that covers the following topics: state-of-the art monitoring solutions, practical monitoring tools, hands-on experiments, maintenance strategies, life-cycle management, sustainability aspects, and challenges in AI, data analysis, and digitalization. Students will utilize commonly available pavement and railway monitoring tools for infrastructure condition evaluation. Selected questions of this unit will be evaluated as part of the module assessment. Students will select a real-life case study involving transportation infrastructures, summarize current practices and future challenges, present the case to infrastructure stakeholders, and submit a technical report. A presentation for the selected case study will be presented in a colloquium. 40%: Module assessment which consists of both presentation and final report. Designing, analysing, and communication skills are evaluated through this module assessment. A successful participation in the module, requires a minimum evaluation of 5.8 in all units.
Expected prior Knowledge	Students from all preceding SE-A Modules can follow SE-B6. Students from the different SE-A Modules will have different pre-knowledge: - Knowledge about modelling approaches, such as FEM (SE-A1). - Knowledge about material aspects (SE-A2). - Knowledge about aspects of integral design of civil structures including road and rail (SE-A3). From Track-Base, the following are topics we expect students to know about: 305 - Loads on infrastructure engineering, short-term and long-term loads. - Material properties and the corresponding mechanical behaviour. For example, steel, concrete, timber structures. - Sustainability, circularity and LCA approaches used in the construction of infrastructures. - One-dimensional slender structures and plates formulations of the governing equations and fundamental solutions (in both statics and dynamics). From Program-Base and Faculty-Base, the following are topics we expect students to know about: - Coding (Python, Matlab, etc.) tools from MUDE (Faculty-Base) and Program-Base. - Topics from modules in the Program-Base such as deriving, analysing and solving fundamental problems via simpler engineering forms. - Basics of statics, kinetics, mechanics of materials, dynamics, elasticity, etc. - Fundamentals of the Ethics line. We will use specialized software for different examples, including ANSYS. For a further understanding, knowledge about FE is desirable but not mandatory.
Academic Skills	Working with transportation infrastructures is inherently multidisciplinary, making interdisciplinary thinking very relevant in this module. We address challenges from both theory and practice; thus, problem learning and solving, and applying theory to new situations are academic skills the students will use and develop in this module. The assessment considers reports and presentations to stakeholders, so scientific writing, presentation, and time management are essential personal skills.
Literature & Study Materials	B6-1: AASHTO (1986). AASHTO guide for design of pavement structures. Washington D.C., USA. Ansys, VI-Rail, 3D Move manuals. CROW (2002). Design and overlay design for pavements with thin asphalt surfacings (in Dutch). Publication 157. Ede, The Netherlands. Handbook of Railway Vehicle Dynamics. NCHRP 1-37A (2004). Mechanistic-empirical design of new and rehabilitated pavement structures, Transportation Research Board, National Research Council, Washington, D.C., USA. NEN-EN 1991-2:2003 - Eurocode 1 - Actions on structures - Part 2: Traffic loads on bridges, 2003. NEN-EN 1990:2002/a1:2006 - Eurocode: Basis of structural design - Annex 2: Application for bridges, 2002. Online learning material based on Railway Engineering (TUD) ProfEd online courses. Papagiannakis A.T and Masad E.A. (2008). Pavement Design and Materials, 1st edition, Wiley. Shell International Petroleum Company Ltd., Shell pavement design manual (1978), London. UIC. USP project. Modelling method for static loads. UIC. USP project. Modelling method for dynamic loads. Yang H. Huang (2008) Pavement Analysis and Design, 2nd Edition, Pearson Education B6-2: Dawson, A. (2014). Anticipating and Responding to Pavement Performance as Climate Changes. In Climate Change, Energy, Sustainability and Pavements (Eds.: Gopalakrishnan, Steyn and Harvey). Springer. Esveld, C. (2001). Modern railway track, second edition. In Modern railway track, second edition (pp. 1-653). MRT productions. Huang and Di Benedetto (Eds) (2015). Advances in Asphalt Materials: Road and Pavement Construction. Woodhead Publishing Limited. Iwnicki, S., Spiryagin, M., Cole, C., & McSweeney, T. (Eds.). (2019). Handbook of Railway Vehicle Dynamics second edition. CRC Press. https://doi.org/10.1201/9780429469398 Jing, R. (2019). Ageing of bituminous materials : Experimental and numerical characterization, Dissertation TU Delft, 232 p. Lewis, R. Olofsson, U. (2009). WheelRail Interface Handbook. Woodhead Publishing. https://doi.org/10.1016/B978-1-84569-412-8.50031-8. Online learning material based on Railway Engineering (TUD) ProfEd online courses. 311 Srirangam, S.K. (2015). Numerical simulation of tire-pavement interaction, Dissertation TU Delft, 204 p. Varveri, A. (2017). Moisture damage susceptibility of asphalt mixtures: Experimental characterization and modelling, Dissertation TU Delft, 180 p. B6-3: Barriera, M., et al., In Situ Pavement Monitoring: A Review. Farrar, C.R. and Worden K. (2013), Structural Health Monitoring: A Machine Learning Perspective, Wiley, Chichester. Fraden, J. (2010), Handbook of Modern Sensors. Springer, fourth edition. Fwa, T.F., The Handbook of Highway Engineering. Kobbacy, K.A.H. and Murthy, D.P. eds., 2008. Complex system maintenance handbook. Springer. Lajnef, N., Smart Pavement Monitoring System - Federal Highway. Márquez, A.C., (2007). The Maintenance Management Framework: Models and Methods for Complex Systems Maintenance, Springer, London, pp. 1-341. Miradi, M., Knowledge Discovery and Pavement Performance Intelligent Data Mining. Online learning material based on Railway Engineering (TUD) ProfEd online courses. Parmenter, D., 2015. Key performance indicators: developing, implementing, and using winning KPIs. John Wiley & Sons. Sharma, S., A data-driven approach for pavement surface distress classification.
Judgement	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Calculator
Collegegrama	No

Year	2022/2023
Organization	Civil Engineering and Geosciences
Education	Master Civil Engineering

CIE Track Traffic & Transport Engineering (TTE)
--

CIEM6000	Base: Traffic and Transport Engineering	15
Responsible Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Prof.dr.ir. S.P. Hoogendoorn	
Instructor	Ir. A.J. Pel	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Prof.dr. Z. Li	
Instructor	Dr. V.L. Knoop	
Instructor	E. Quaglietta	
Instructor	Dr. O. Cats	
Instructor	Dr.ir. H. Farah	
Instructor	Dr. S. Sharif Azadeh	
Contact Hours / Week x/x/x/x	x/12/6/x	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	The track base provides students with a solid basis of the fundamentals of traffic and transport engineering including planning and design of traffic and transport networks and infrastructure, transport modelling and analysis, traffic modelling and management, and identifying and assessing the ethical and societal implications of traffic and transport engineering solutions. Ethical aspects are important for engineers, practitioners, and researchers in the sector of traffic and transport engineering. In the track base, relevant ethical aspects and societal impacts of traffic and transport engineering solutions (e.g., safety, environment, health, equity, privacy) and the responsibilities when creating a design or developing models will be addressed across the three different units. In addition, the first unit dedicates two lectures to discuss these aspects in more depth.	
Study Goals	On completion of this module, the student will be able to: TTE-TB-LO1: Design the traffic and transport infrastructure and networks. TTE-TB-LO2: Develop a traffic and transport engineering solution and analyse its performance. TTE-TB-LO3: Discuss, analyse and apply traffic and transport models. TTE-TB-LO4: Set-up, perform and report a transport engineering solution/ modelling study. TTE-TB-LO5: Assess the ethical and societal implications of traffic and transport engineering solutions.	
Education Method	Lectures, in class small groups practice problems, in-class discussions, self-assessment quizzes, Group work on the project, feedback on group work. Collegerama is only planned for some (parts of) units; check Brightspace.	
Assessment	Exam 1: Planning and design of Traffic and Transportation Networks and Infrastructure - Written Exam (13,3%) Exam 2: Planning and design of Traffic and Transportation Networks and Infrastructure - Group Assignment (13,3%) Exam 3: Transport Modelling and analysis - Written Exam (33,3%) Exam 4: Traffic Modelling and Management -Written Exam (40%)	
Tags	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Analysis Modelling Project Transport & Logistics Transport phenomena	
Expected prior Knowledge	BSc courses (or equivalent): CTB1002 (Linear Algebra), CTB2000 (Programming), CTB2200 (Probability and Statistics), CCTB2400 (Numerical Math)	
Academic Skills	Critical thinking, problem-based learning and solving, scientific writing and presentation, time management, communication, application of theory to new situations	
Literature & Study Materials	Lecture notes and scientific articles provided by the teachers and are made available through Brightspace.	
Judgement	See Assessment For more information on grading, see article 14 in the Rules and Guidelines (RGE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Varies per unit -- Check Brightspace	
Collegerama	Yes	

CIEM6110	Methods in Traffic and Transport Engineering	9
Responsible Instructor	Dr.ir. S.C. Calvert	
Instructor	Dr.ir. W. Daamen	
Instructor	Dr. M. Snelder	
Instructor	Dr.ir. S.C. Calvert	
Instructor	Prof.dr. J.H.G. Correia	
Instructor	Dr.ir. W.J. Schakel	
Instructor	Dr.ir. A. Gavrilidou	
Instructor	Dr. S. Sharif Azadeh	
Contact Hours / Week x/x/x/x	x/x/12/x	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This module provides the students with further deepening of the three learning lines (data, modelling, and risk) offered in the faculty base with a focus on applications from the Traffic and Transport Engineering track including empirical analysis for transport and traffic engineering, optimisation in transport engineering, simulation and uncertainty in transport engineering.	
Study Goals	On completion of this module, the student will be able to: TTE-A-LO1. Formulate research questions and hypotheses in relation to transport infrastructure, network, operations and behaviour TTE-A-LO2. Design and conduct experiments in the realm of transport infrastructure, network, operations and behaviour TTE-A-LO3. Analyze and interpret empirical and simulation data about transport infrastructure, network, operations and behaviour TTE-A-LO4. Formulate a mathematical optimisation model addressing problems related to transport infrastructure, network, operations and behaviour TTE-A-LO5. Solve optimisation problems using solvers and tailored designed efficient algorithms also considering stochasticity in the inputs. TTE-A-LO6. Develop and implement simulation model functionalities for representing transport infrastructure, network, operations and behaviour TTE-A-LO7. Develop suitable data-driven modelling approaches for verification, calibration and validation of models, input and parameters in the context of transport infrastructure, network, operations and behaviour TTE-A-LO8. Compare models properties and performance for analyzing transport infrastructure, network, operations and behaviour TTE-A-LO9. Apply techniques for risk and uncertainty analysis for transport data and models	
Education Method	Lectures, lab classes with formative assessments where students formulate problems and solve them with a computer.	
Assessment	Exam 1: Empirical Analysis for Transport and Traffic Engineering - Assignment (33,3%) Exam 2: Optimisation in Transport Engineering - Written Exam (33,3%) Exam 3: Simulation and Uncertainty in Transport Engineering - Assignment (33,3%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. 1. BSc courses (or equivalent): CTB1002 (Linear Algebra), CTB2000 (Programming), CTB2200 (Probability and Statistics), CCTB2400 (Numerical Math) 2. Design, organise and implement data collection (MUDE: data) 3. Analyse the collected data using statistics (MUDE: data and uncertainty) 4. Fundamentals of Discrete Choice Models/ Traffic Flow Theory (TTE Base track) 5. Formulate a mathematical problem (MUDE: modelling) 6. Solve continuous and integer optimisation problems (MUDE: modelling) 7. Monte Carlo simulation, random number generators (MUDE: uncertainty)	
Academic Skills	Completion of the MUD and track core	
Literature & Study Materials	Parts of the current reader for CIE4831 (Empirical Analyses for Transport & Planning) Introduction to Operations Research by Hillier and Lieberman. Discrete optimization: Introduction to logistics systems management by Gilbert Laporte. NLP: Numerical optimization by Wright and Nocedal. Lecture slides and a selection of key and relevant papers	
Judgement	See Assessment	
	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations	
Permitted Materials during Exam	Will be communicated during lectures	
Collegerama	No	

CIEM6210	Transport Networks and Systems	15
Responsible Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	Dr. C. Maat	
Instructor	Dr. M. Snelder	
Instructor	Prof.dr. J.H.G. Correia	
Instructor	Ir. A.J. Pel	
Instructor	Prof.dr.ir. L.A. Tavasszy	
Instructor	Dr.ir. A.J. van Binsbergen	
Instructor	N. van Oort	
Instructor	G.K. de Clercq	
Instructor	Dr. S. Fazi	
Instructor	M. Rinaldi	
Contact Hours / Week x/x/x/x	x/x/x/17	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In-depth analysis of the functioning of transport networks for passenger and freight transport, the relevant performance characteristics for the users (private persons, firms) including the resilience, the interaction with economical and spatial development, the broader sustainability effects (societal, economic, environmental) of networks. In-depth analysis of route choice and assignment in road networks, and mode choice and route assignment for passenger and freight transport in intermodal networks, the impact from disruptions and the impact on performance indicators for users and the society. Optimisation and assessment of the design of logistics services, road networks, and intermodal networks, in response and in interaction with socio-economic and spatial planning developments and by balancing the interests of users and society, so by taking into account network performance, resilience and sustainability. The module deepens understanding of the societal impacts of transport networks, and applying methods and approaches to evaluating and design these networks in an ethical and societal responsible way.	
Study Goals	On completion of this module, the student will be able to: B1-LO1. To create integrated solutions for transport networks for various modes, services and user classes to resolve performance challenges. B1-LO2. To evaluate the performance impact of current and (potential) future networks for users and society in different dimensions. B1-LO3. To assess the suitability of standard methods for real-world transport network design problems. B1-LO4. To analyse the current and the expected future performance of transport systems, services and networks and their tuning with land use. B1-LO5. To apply different standard methods for the design of transport networks. B1-LO6. To operationalise performance indicators for transport networks including links with norms and methods of measurement. B1-LO7. To explain which theoretical approaches, optimisation and calculation methods and models are needed for performance estimation, assessment and design steps. B1-LO8. To explain dynamic complexity between transport networks, and land use and other systems (system of systems). B1-LO9. To define road and freight transport networks in the societal, sustainability, perspective and define the main related design problems. B1-LO10. To define road and freight transport networks in their technical and user characteristics, and define the main related design problems.	
Education Method	Lectures, formative assignment (group project) with instruction, practical lecture/workshop, guest lectures	
Assessment	Exam 1: Transport Modelling: Traffic Assignment - Written Exam (20%) Exam 2: Regions, Transport and Networks - Written Exam (26,6%) Exam 3: Freight transport networks and systems - Written Exam (26,6%) Exam 4: Sustainable and Resilient Transport Networks and Systems - Presentation (26,6%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. 1. Basic insights performance, impacts, ethical and social implications and their indicators - of traffic and transport networks (TTE-TB-1, TTE-TB-1) 2. Basic insights in transport economics (individual choice preferences, business perspectives (TTE-TB-1) 3. Basic insights in Transport Modelling and Analysis, including aggregated, disaggregated models, gravity models and discrete choice models for both mobility and freight networks. (TTE-TB-2) 4. Basic knowledge on distribution (OD-estimation), assignment and equilibrium models (TTE-TB-2) Basic knowledge on mixed-integer linear optimisation (TTE-TB-2) 5. Insight in characteristics of unimodal and intermodal networks, physical and service networks for individual passenger transport, collective public transport and for freight transport (TTE-TB-3) 6. Insight in data-driven modelling approaches for verification, calibration and validation of models, input and parameters in the context of transport infrastructure, network, operations and behaviour (TTE-A1)	
Academic Skills	Critical, interdisciplinary and ethical thinking, social-impact aware and solution-oriented learning, information collection and synthesis, communication and responsiveness, scientific writing and presentation, application of theory to new situations.	
Literature & Study Materials	Transport Modelling: Traffic Assignment: - Lecture slides; - Chapter 2 of book: M.S. Fiorenzo-Catalano (2007) 'Choice set generation in multi-modal transportation networks', TRAIL; - Chapters 1 & 2 of paper: Luuk Brederode, Adam Pel, Luc Wismans, Erik de Romph, Serge Hoogendoorn (2019) 'Static Traffic Assignment with Queuing: model properties and applications', Transportmetrica A: Transport Science, Volume 15, Issue 2; - Chapters 1.5 & 3.1 of report: Joe Castiglione, Mark Bradley, John Gliebe (2015) 'Activity-Based Travel Demand Models: A Primer', TRB; - Paper: Menno Yap, Oded Cats, Bart van Arem (2020) 'Crowding valuation in urban tram and bus transportation based on smart card data', Transportmetrica A: Transport Science, Volume 16, 2020 Issue 1; - Paper: Carlo Giacomo Prato (2009) 'Route choice modeling: past, present and future research directions', Journal of Choice Modelling, Volume 2, Issue 1, 2009; pp. 65-100; - Paper: Chris M.J. Tampère, Ruben Corthout, Dirk Cattrysse, Lambertus H. Immers (2011) 'A generic class of first order node models for dynamic macroscopic simulation of traffic flows', Transportation Research Part B: Methodological, Volume 45, Issue 1; - Paper: Bliemer, Michiel C.J., Raadsen, Mark P.H., Brederode, Luuk, Bell, Michael G.H., Wismans, Luc J.J. (2014) 'A unified framework for traffic assignment: deriving static and quasi-dynamic models consistent with general first order dynamic traffic assignment models', working paper University of Sydney; http://hdl.handle.net/2123/19191;	

- Paper: Niels van Oort, Ties Brands, Erik de Romph, Jessica Aceves Flores (2014) 'Incorporating unreliability of transit in transport demand models: Theoretical and practical approach', Prepared for the 93rd Annual Meeting of the Transportation Research Board 2014

Regions, Transport and Networks:

- Lecture slides;
- Book: Jean-Paul Rodrigue (2020) 'The geography of Transport Systems'. Fifth Edition; New York: Routledge, 456 pages. ISBN 978-0-367-36463-2

Freight transport networks and systems:

- Lecture slides;
- Book: Tavasszy, L., G. de Jong (eds) (2014), Modelling Freight Transport, Elsevier (digital edition)
- Report: CE Delft (2020) 'STREAM Freight Transport' (digital edition; annual update)
- Paper: Jin Y. Yen (2015) Finding the K shortest Loopless Paths in a Network, Management Science 17(11)

Sustainable and Resilient Transport Networks and Systems:

- Lecture slides;
- Assignment book and tutorial (dedicated for the course): Sustainable and Resilient Transport Systems: Analysis and Interventions;
- Report: US Department of Transportation/FHWA (2017) 'Travel time reliability: making it there on time, all the time' (digital edition)
- Report: United Nations (2017) 'Transforming our World: the 2030 Agenda for Sustainable Development' (digital edition)
- Report: Department for Communities and Local Government (2009) 'Multi-criteria analysis: a manual'
- Book Chapter: Pascual, U., R. Muradian (eds) (2010) 'The Economics of Valuing Ecosystem Services and Biodiversity'
- Paper: Henry J. Liu, Peter E.D. Love, Michael C.P. Sing, Baozhuang Niu and Jianfeng Zhao (2019) 'Conceptual Framework of Life-Cycle Performance Measurement: Ensuring the Resilience of Transport Infrastructure Assets', Transportation Research D;
- Paper: Karst T. Geurs, W. Boon and B. van Wee (2009) 'Social Impacts of Transport: Literature Review and the State of Practice of Transport Appraisal in The Netherlands and the United Kingdom';
- Handbook of Emission Factors for Road Transport (HBEFA 4.1, 2019 - digital edition);

Judgement

See Assessment

For more information on grading, see article 14 in the Rules and Guidelines (RGBE):

<https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations>

CIEM6220	Road Traffic Systems	15
Responsible Instructor	Dr.ir. H. Farah	
Instructor	Dr.ir. W. Daamen	
Instructor	Prof.dr.ir. S.P. Hoogendoorn	
Instructor	Dr.ir. A. Hegyi	
Instructor	Dr.ir. A.M. Salomons	
Instructor	Dr.ir. W.J. Schakel	
Instructor	Y. Yuan	
Instructor	Dr. M.P. Hagenzieker	
Instructor	Prof.dr.ir. B. van Arem	
Instructor	Dr. V.L. Knoop	
Instructor	Dr.ir. D.C. Duives	
Instructor	Dr.ir. H. Farah	
Instructor	Dr.ir. A. Gavriilidou	
Instructor	M. Rinaldi	
Instructor	I. Martínez Josemaría	
Contact Hours / Week x/x/x/x	x/x/x/22	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module teaches about (road) traffic systems. We consider car traffic, but also active mode traffic (cycling, walking). Note that the module considers the efficiency and safety of the traffic systems, and covers a qualitative and quantitative parts. The module also includes the ethical parts of transport research; these are explicitly included across all four units of the module. The module consists of the following units: - Active modes: Traffic and transport - Traffic safety - Intelligent vehicles for safe and efficient traffic: Design and assessment - Urban and motorway traffic flow modelling and control Active Modes: Traffic and Transport: Contact hours/week: 6h/week - Classroom lectures (4h) + Practicals (2h) List of instructors: Dorine Duives (responsible), Serge Hoogendoorn, Winnie Daamen, Alexandra Gavriilidou Traffic safety: Contact hours/week: 4h/week - Classroom lectures (4h) List of instructors: Haneen Farah (responsible), Marjan Hagenzieker Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment: Contact hours/week: 4h/week - Classroom lectures (4h) List of instructors: Marco Rinaldi (responsible), Bart van Arem, Marjan Hagenzieker, Victor Knoop, Wouter Schakel Urban and Motorway Traffic Flow Modelling and Control: Contact hours/week: 2x2h classroom lectures + 1x4h pc-practical, i.e. 8 h/w List of instructors: Andreas Hegyi (responsible), Victor Knoop, Maria Salomons, Yufei Yuan	
Study Goals	On completion of this module, the student will be able to: TTE-B2-LO1. Analyse, assess, model, predict, and manage road infrastructure systems TTE-B2-LO2. Design and evaluate intelligent vehicle systems to increase traffic safety, efficiency and sustainability TTE-B2-LO3. Participate, contribute and effectively communicate in debates and professional projects involving international, multicultural and interdisciplinary teams. TTE-B2-LO4. Acquire new knowledge and skills to continue operating effectively.	
Education Method	Lectures, formative group assignments + feedback, in-class small groups practice problems, in-class discussions, self-assessment quizzes, guest speakers	
Assessment	Exam 1: Active Modes: Traffic and Transport - Written Exam (20%) Exam 2: Traffic Safety - Assignment (20%) Exam 3: Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment - Assignment (27%) Exam 4: Urban and Motorway Traffic Flow Modelling and Control - Written Exam (33%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. 1. Basic calculus (e.g. vectoren, numeriek, statistiek) 2. Basic modelling techniques (Python/ Biogeme script kunnen schrijven) 3. Basic statistics and regression analysis, basic probability theory, basic notion of optimization 4. Data collection methods 5. Basic knowledge on PDEs (both differential/difference eqs) 6. Traffic variables & fundamental diagrams 7. Shock wave theory 8. Basic Traffic control: static Traffic lights, ramp metering, basics of systems and control 9. SPECIALIST theory (An algorithm to resolve traffic jams) Note: prerequisites 6-9 are covered in the course Traffic modelling and management	
Academic Skills	critical and interdisciplinary thinking, problem-based learning and solving, continuous investigation, scientific writing and presentation, time management and anticipation, application of theory to new situations using the relevant equations.	
Literature & Study Materials	Presentation slides and reading material will be provided after each lecture	
Judgement	See Assessment	

For more information on grading, see article 14 in the Rules and Guidelines (RGBE):
<https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations>

Permitted Materials during Exam

Exam 1 (Active Modes: Traffic and Transport): Simple calculator.
Exam 2 (Traffic safety): assignment.
Exam 3 (Intelligent Vehicles for Safe and Efficient Traffic: Design and Assessment): assignment.
Exam 4 (Urban and Motorway Traffic Flow Modelling and Control): simple calculator; one two-sided A4 paper with own notes, handwritten in own handwriting, with normal font size.

Collegerama

No

CIEM6230	Public Transport and Railway Systems	15
Responsible Instructor	Dr. O. Cats	
Instructor	Prof.dr. R.M.P. Goverde	
Instructor	Dr. D.M. van de Velde	
Instructor	N. van Oort	
Instructor	Dr. O. Cats	
Contact Hours / Week x/x/x/x	x/x/x/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The module Public Transport (PT) and railway systems deals with modelling, design and analysis of public transport and railway operations including the interactions with the customer demand and system governance. Three main units compose this module, namely: i) Public transport planning and operations, which focuses on PT service design, access and egress, PT control and emerging modes. ii) Public transport networks and modelling, which deals with behavioural and network modelling of PT as well as design and analysis of PT systems. iii) Railway Operations and Control, which tackles railway safety and signalling, railway automation and coordination in the railway value chain. The module will add to the learning line on Ethics related to planning, management and control of road- and rail-bound public transport operations. Each unit tackles possible impacts that design/operational decisions of public transport stakeholders and engineers can have on sustainability, public health, user privacy, social inclusion and fairness, presenting methods and solutions to meet international requirements, standards and policies. In addition, lecture material together with the group projects and formative assessment provided in the module, will support shaping the personal and professional responsibility of the students in line with the ethical code of conduct required as a future public transport engineer.	
Study Goals	On completion of this module, the student will be able to: TTE-B3-LO1. Analyse key dependencies among design variables, parameters, and criteria concerning organisation, service, infrastructure, operations and demand of road- and rail-bound public transport. TTE-B3-LO2. Apply and evaluate strategic, tactical, and operational design methods to align public transport/railway services with emerging societal and environmental needs. TTE-B3-LO3. Analyse processes and develop models for capturing interactions among system components including functional relations with other transport systems, society, and environment. TTE-B3-LO4. Design and evaluate intervention plans to build/upgrade, manage and operate public transport and railways in line with customer and stakeholders requirements. TTE-B3-LO5. Define and apply methods to predict and assess service performances of public transport/railway systems and the impacts on other systems, society, and the environment. TTE-B3-LO6. Demonstrate team cooperation, communication, and work management capabilities in three group projects (in unit B3-2) and formative assignments (in B3-1 and B3-3), to support the development of lifelong skills required in the professional world.	
Education Method	On-campus or on-line lectures from instructors, Guest lectures, Active learning activities such as in-class quizzes, games and group exercises, in-class discussions, feedback sessions, pre- and post-lecture material such as reading scientific articles and watching videos, formative assignments.	
Assessment	Exam 1: Public Transport System and Supply Planning and Operations - Written Exam (33,3%) Exam 2: Public Transport Demand and Network Planning and Operations - Group Assignment (33,3%) Exam 3: Railway Operations and Control - Written Exam (33,3%)	
Tags	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. Modelling Policy Analysis Rail & Road Engineering / Planning Sustainability Transport & Logistics Transport phenomena	
Expected prior Knowledge	As a pre-requisite to taking this module the student needs to have previous knowledge about: 1) Transport fundamentals including characteristics of the transport supply and the transport demand, traffic flow properties and attributes, network design variables and key performance indicators for the assessment of transport service quality and reliability. 2) Modelling, simulation and optimisation methods and possible applications to transport systems. 3) Data collection, management and analysis techniques and applications to the transport sector. 4) Statistical and uncertainty analysis with a general understanding of relevant risk mitigation related concepts including robustness, resilience and reliability.	
Academic Skills	The main academic transferable skills that will be developed in this module will be critical thinking, analytical thinking and problem solving, specifically applied to the planning and operations of public transport and railways. The main personal skills that the student will build are organizational, team-building, and interpersonal and presentation abilities by means of concrete cooperation with colleagues in assignments having the scope of emulating projects and interactions in a professional environment.	
Literature & Study Materials	Public Transport System and Supply Planning and Operations: Lecture slides and two papers per week provided during the course Public Transport Demand and Network Planning and Operations: Lecture slides and pre- and post-lecture material including 3-5 reference papers are made available through Brightspace. Railway Operations and Control: A set of papers or book chapters made available on BrightSpace. Each lecture will require reading one paper/book chapter.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Calculator	
Colleggerama	No	

CIEM6240	Road and Railway Engineering	15
Responsible Instructor	Dr. A.A. Nunez Vicencio	
Responsible Instructor	Dr. K. Anupam	
Instructor	Dr.ir. S.M.J.G. Erkens	
Instructor	Ir. C. Kasbergen	
Instructor	Dr. X. Liu	
Instructor	Dr. V.L. Markine	
Instructor	Dr. A.A. Nunez Vicencio	
Instructor	A. Varveri	
Instructor	Prof.dr. Z. Li	
Instructor	Prof.dr.ir. R.P.B.J. Dollevoet	
Instructor	Dr.ir. H. Farah	
Instructor	Dr. H. Wang	
Contact Hours / Week x/x/x/x	x/x/x/16	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module addresses the design of road and railway infrastructures and their management, where the interface between traffic and transport as a service and infrastructure requirements for road and rail are fundamental. In the module, students will learn about classical design criteria for road and railway infrastructures. Moreover, the module covers current developments on topics such as a) robustness and resilience under climate change conditions, b) the use of new and more sustainable materials and design techniques, c) advanced IoT, d) AI-based monitoring and maintenance methods for road and railway infrastructure, and e) the effect of new technologies on design and maintenance of road/railway infrastructure, i.e. automated/smart vehicles and truck platooning. The module also covers aspects of management of road/railway infrastructure networks, focusing on the inter-relation between road/railway traffic management, network performance, and network quality. Particularly, methods to evaluate geometry, surface characteristics, reliability and user safety will be analysed by defining performance criteria, continuous performance monitoring, and different maintenance techniques. The module will enable students to ethically assess the road/railway infrastructure condition and its effect on traffic, capacity, and performance over time.	
Study Goals	On completion of this module, the student will be able to TTE-B4-ILO1: Identify key components for road and railway infrastructure design. TTE-B4-ILO2: Explain fundamental -mechanics and -dynamics of road and railway infrastructures tailored for traffic operation. TTE-B4-ILO3: Assess the road and railway infrastructure condition and predict their performances in terms of reliability, availability, maintainability, safety, sustainability, resilience, serviceability, and robustness for smooth traffic management. TTE-B4-ILO4: Design road and railway infrastructure based on traffic operation demands. TTE-B4-ILO5: Improve the sustainability and reliability of the road and railway infrastructure during the design-construction-operation cycle. TTE-B4-ILO6: Assess the effectiveness of state-of-the-art monitoring solutions and the relation between usage, condition and maintenance strategies for road and railway infrastructure. TTE-B4-ILO7: Evaluate innovative and ethical solutions to current and future challenges of road and railway infrastructure asset management. TTE-B4-ILO8: Propose innovative and ethical solutions to actual challenges of the road and railway infrastructure industries, based on operational traffic demands. TTE-B4-ILO9: Effectively and ethically report and communicate results with stakeholders from the industry.	
Education Method	Lectures / Blended Learning / Laboratory and Simulation Experiments / Excursion	
Assessment	Exam 1: Road Engineering - Written Exam (40%) Exam 2: Railway Engineering - Written Exam (40%) Exam 3: Integrated Project - Report - Group and Individual (20%)	
Expected prior Knowledge	A successful participation in the module, requires a minimum evaluation of 5.8 in all units. The following are pre-knowledge assumed that most students will have: - Geometric design principles of roads, railways and their implications for safety, efficiency, equity and sustainability. - Road and railway components and understanding of its design elements, including key components and infrastructure from subsurface to surface. - Innovations in the field of road and railway infrastructure design, including societal challenges, current trends and future needs. - Transport network design and service planning. - KPIs design for evaluation of impacts of the transportation networks. From Program-Base and Faculty-Base, the following are topics we expect students to know about: - Coding (Python, Matlab, etc.) tools from MUDE (Faculty-Base) and Program-Base. Particularly, probabilistic models, sensing data measurement, collection and processing. - Topics from modules in the Program-Base such as deriving, analysing and solving fundamental problems via simpler engineering forms. - Basics of statics, kinetics, mechanics of materials, dynamics, elasticity, etc., covered in the program-base. - Fundamentals of the Ethics line. We will use specialized software for different examples.	
Academic Skills	Developments in the field of road and railway infrastructures are inherently multi-disciplinary. Students will learn topics including the reliable design of road/railway infrastructures, operations, sensing technologies, IoT, management, etc. As such, interdisciplinary and ethical thinking is a skill required to be further developed in the students. This module considers real case studies coming from challenges currently being faced in the industry. Thus, students will have to apply theory to practice and to develop problem learning and solving skills, while working efficiently and ethically correct. Finally, scientific writing, presentation, and time management skills are critical in this module. Particularly, together with addressing challenges from the industry, the assessment in this module considers reports and presentations to stakeholders.	
Literature & Study Materials	Road Engineering: Barriera, M., et al., In Situ Pavement Monitoring: A Review. Farrar, C.R. and Worden K. (2013), Structural Health Monitoring: A Machine Learning Perspective, Wiley, Chichester. Fraden, J. (2010), Handbook of Modern Sensors. Springer, fourth edition. Fwa, T.F., The Handbook of Highway Engineering. Kobbacy, K.A.H. and Murthy, D.P. eds., 2008. Complex system	

	maintenance handbook. Springer. Lajnef, N., Smart Pavement Monitoring System - Federal Highway. Márquez, A.C., (2007). The Maintenance Management Framework: Models and Methods for Complex Systems Maintenance, Springer, London, pp. 1-341. Miradi, M., Knowledge Discovery and Pavement Performance Intelligent Data Mining. Parmenter, D., 2015. Key performance indicators: developing, implementing, and using winning KPIs. John Wiley & Sons. Sharma, S., A data-driven approach for pavement surface distress classification. Railway Engineering: Esveld, C. (2001). Modern railway track, second edition. In Modern railway track, second edition (pp. 1-653). MRT productions. Farrar, C.R. and Worden K. (2013), Structural Health Monitoring: A Machine Learning Perspective, Wiley, Chichester. Fraden, J. (2010), Handbook of Modern Sensors. Springer, fourth edition. Márquez, A.C., (2007). The Maintenance Management Framework: Models and Methods for Complex Systems Maintenance, Springer, London, pp. 1-341. Online learning material based on Railway Engineering (TUD) ProfEd online courses. Integrated Project: A selection of papers and reports available in Brightspace. Lecture slides and pre- and post-lecture materials.
Judgement	See Assessment
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Calculator
Collegerama	No

H. Alkisaai

Department	Applied Mechanics
Room	23.HG 6.47

Dr. A. Amiri Simkooei

Department	Optical and Laser Remote Sensing
Telephone	+31 15 27 88436

Dr. A. Antonini

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 85263
Room	23.HG 3.88

Dr. K. Anupam

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 82394
Room	23.HG 6.60

Prof.dr.ir. B. van Arem

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 86342

J.A. Arriaga Garcia

Department	Environmental Fluid Mechanics
-------------------	-------------------------------

F. Baart

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering
Room	23.HG 3.77.1

S.C. Barbosa Nunes

Department	Concrete Structures
Telephone	+31 15 27 88058

Dr.ir. I. Barcelos Carneiro M Da R

Department	Applied Mechanics
Telephone	+31 15 27 81340
Room	23.HG 6.54

S. Basu

Unit	Civiele Techniek & Geowetensch
Department	Atmospheric Remote Sensing
Telephone	+31 15 27 89358
Room	23.HG 2.20

Ir. R. Binnekamp

Department	Real Estate Management
Telephone	+31 15 27 89535
Room	23.HG 6.64

Dr.ir. A.J. van Binsbergen

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 81067
Room	23.HG 4.07

Dr.ir. A. Blom

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering
Telephone	+31 15 27 85064
Room	23.S3.00.100

T.M. de Boer

Department	Rivers, Ports, Waterways and Dredging Engineering
-------------------	---

Dr. T.A. Bogaard

Unit	Civiele Techniek & Geowetensch
Department	Water Resources
Telephone	+31 15 27 84286
Room	23.HG 4.92

Ir. L. de Boom

Department	Rivers, Ports, Waterways and Dredging Engineering
-------------------	---

Ir. A. Borgart

Unit	Bouwkunde
Department	Structural Design & Mechanics
Telephone	+31 15 27 84157
Room	08.01WEST130

Dr.ir. J. Bosboom

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 84606
Room	23.HG 3.66

Dr. E. Brembilla

Department	Building Physics
Room	08.01+.West.040

Dr.ir. R.B.J. Brinkgreve

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Telephone	+31 15 27 83327
Room	23.KG 00.490

Dr.ir. W. Broere

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Telephone	+31 15 27 81545
Room	23.KG 00.510

Dr. A. Cabboi

Department	Mechanics and Physics of Structures
Telephone	+31 15 27 83419
Room	23.S2 1.54

M.A. Cabrera

Department	Geo-engineering
Telephone	+31 15 27 83663
Room	23.KG 00.500

Dr.ir. S.C. Calvert

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Room	23.HG 4.02.2

Dr. O. Cats

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 83621
Room	23.HG 4.22

Dr. A. Cicirello

Department	Mechanics and Physics of Structures
Telephone	+31 15 27 83139
Room	23.S2 1.55

G.K. de Clercq

Department	Transport and Planning
Room	23.HG 4.14.6

J.O. Colomes Gene

Department	Offshore Engineering
Telephone	+31 15 27 83461
Room	23.HG 3.65

O. Copuroglu

Unit	Civiele Techniek & Geowetensch
Department	Materials and Environment
Telephone	+31 15 27 81827
Room	23.HG 6.03

Prof.dr. J.H.G. Correia

R. Crielaard

Department	Applied Mechanics
Room	23.HG 6.48

Department	Applied Mechanics
Room	23.HG 6.48

Dr.ir. W. Daamen

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 85927
Room	23.HG 4.37

Dr.ir. K.N. van Dalen

Unit	Civiele Techniek & Geowetensch
Department	Dynamics of Structures
Telephone	+31 15 27 88388
Room	23.HG 3.61

Dr. F. Di Maio

Unit	Civiele Techniek & Geowetensch
Department	Resources & Recycling
Telephone	+31 15 27 88148
Room	23.S2 1.07

Dr.ir. A.A.M. Dieudonné

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Telephone	+31 15 27 86134
Room	23.KG 00.490

Prof.dr.ir. R.P.B.J. Dollevoet

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering

Telephone Room	+31 15 27 82365 23.S2 2.28
---------------------------	-------------------------------

Dr.ir. D.C. Duives

Unit Department	Civiele Techniek & Geowetensch Transport and Planning
Telephone Room	+31 15 27 86304 23.HG 4.24

P. Eigenraam

Unit Department	Bouwkunde Structural Design & Mechanics
Telephone	+31 15 27 87463

Dr.ir. A.P. van den Eijnden

Unit Department	Civiele Techniek & Geowetensch Geo-engineering
Telephone Room	+31 15 27 87443 23.KG 00.470

Dr.ir. S.M.J.G. Erkens

Unit Department	Civiele Techniek & Geowetensch Pavement Engineering
Telephone Room	+31 15 27 85949 23.S2 2.25

R. Esposito

Unit Department	Civiele Techniek & Geowetensch Applied Mechanics
Telephone Room	+31 15 27 82537 23.HG 6.50.1

Dr.ir. H. Farah

Unit Department	Civiele Techniek & Geowetensch Transport and Planning
Telephone Room	+31 15 27 83401 23.HG 4.20

Dr. S. Fazi

Department	Transport en Logistiek
-------------------	------------------------

M.P. Felicita

Department	Bio-based Structures & Materials
-------------------	----------------------------------

Dr. M. Fotouhi

Department	Materials and Environment
-------------------	---------------------------

Dr. K.G. Gavin

Unit Department	Civiele Techniek & Geowetensch Geo-engineering
Telephone Room	+31 15 27 87128 23.KG 00.540

Dr.ir. A. Gavrilidou

Department	Transport and Planning
Room	23.HG 4.40

A.T. Gebremariam

Department	Resources & Recycling
Telephone Room	+31 15 27 81662 23.S2 2.04

Department	Resources & Recycling
Telephone	+31 15 27 81662
Room	23.S2 2.04

Dr. G. Giardina

Department	Geo-engineering
Telephone	+31 15 27 86301
Room	23.KG 00.470

Prof.dr. R.M.P. Goverde

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 83178
Room	23.HG 4.03

Dr. M.P. Hagenzieker

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 86909
Room	23.HG 4.29

Ir. E.R. van den Ham

Unit	Bouwkunde
Department	Building Physics
Telephone	+31 15 27 87267
Room	08.01+.West.040

Dr.ir. H.W.M. van der Ham

Department	Concrete Structures
Telephone	+31 15 27 85452
Room	23.S2 2.04

Dr.ir. A. Hegyi

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 89644
Room	23.HG 4.45

Prof.dr.ing. C.M. Hein

Unit	Bouwkunde
Department	History & Complexity
Telephone	+31 15 27 87196
Room	08.01+.Oost.760

Prof.dr.ir. M.A.N. Hendriks

Unit	Civiele Techniek & Geowetensch
Department	Concrete Structures
Telephone	+31 15 27 86988
Room	23.S2 2.03

Dr.ir. H. Hendrikse

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 88223
Room	23.HG 3.71

Prof.dr. M.A. Hicks

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Telephone	+31 15 27 87433
Room	23.KG 00.060

S. Hiller Bernal

Department	Resources & Recycling
-------------------	-----------------------

Dr.ir. B. Hofland

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Telephone	+31 15 27 85067
Room	23.HG 3.88

Drs. R. ter Hofstede

Department	Rivers, Ports, Waterways and Dredging Engineering
Room	23.HG 3.93

Dr.ir. G. Homem de Almeida Correia

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 81384

Dr.ir. P.C.J. Hoogenboom

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 88081
Room	23.HG 6.45

Prof.dr.ir. S.P. Hoogendoorn

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 85475
Room	23.HG 4.09

A.J. van der Hout

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering
Room	23.HG 3.84

Ir. J.S. Hoving

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 85723
Room	23.HG 3.61.1

Dr. M. Hrachowitz

Unit	Civiele Techniek & Geowetensch
Department	Water Resources
Room	23.HG 4.83

Dr.ir. R.W. Hut

Unit	Civiele Techniek & Geowetensch
Department	Water Resources
Telephone	+31 15 27 84909
Room	23.HG 4.76

A. Jagadeesh

Department	Pavement Engineering
-------------------	----------------------

Ir. C.J. Janssen

Unit	Bouwkunde
Department	Building Physics
Telephone	+31 15 27 81589
Room	08.01+.West.040

Prof.dr.ir. C. Jommi

Unit	Civiele Techniek & Geowetensch
-------------	--------------------------------

Department	Geo-engineering
Telephone	+31 15 27 84173
Room	23.KG 00.140

Dr. H.M. Jonkers

Unit	Civiele Techniek & Geowetensch
Department	Materials and Environment
Telephone	+31 15 27 82313
Room	23.HG 6.07

Prof.dr.ir. S.N. Jonkman

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Telephone	+31 15 27 85278
Room	23.HG 3.73.1

Ir. R.E. Jorissen

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Room	23.HG 3.60

O. Karpenko

Department	Steel & Composite Structures
Telephone	+31 15 27 89626

Ir. C. Kasbergen

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 82729
Room	23.S2 2.36

Dr. F. Kavoura

Department	Steel & Composite Structures
Room	23.S2 2.54

Ir. E. Kementzetzidis

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Room	23.HG 3.45

O. Kirichek

Department	Rivers, Ports, Waterways and Dredging Engineering
Room	23.KG 03.120

Dr. V.L. Knoop

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 88413
Room	23.HG 4.37

Drs. O.C. Koedijk

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering
Telephone	+31 15 27 87323
Room	23.HG 3.84

Dr.ir. D.A. Koleva

Unit	Civiele Techniek & Geowetensch
Department	Materials and Environment
Telephone	+31 15 27 87451
Room	23.HG 6.06.3

Dr.ir. M. van Koningsveld

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering
Room	23.HG 3.77.1

M. Korf

Dr.ir. W.M. Kranenburg

Department	Environmental Fluid Mechanics
Department	Environmental Fluid Mechanics

P.K. Krishnakumari

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Room	23.HG 4.02

Prof.dr.ir. J.W.G. van de Kuilen

Unit	Civiele Techniek & Geowetensch
Department	Bio-based Structures & Materials
Telephone	+31 15 27 82322
Room	23.S2 2.56

Dr.ir. R.J. Labeur

Unit	Civiele Techniek & Geowetensch
Department	Environmental Fluid Mechanics
Telephone	+31 15 27 85069
Room	23.HG 2.94

Ir. E.O.L. Lantsoght

Unit	Civiele Techniek & Geowetensch
Department	Concrete Structures
Telephone	+31 15 27 87449
Room	23.S2 1.04

Dr.ir. R.C. Lanzafame

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Telephone	+31 15 27 85520
Room	23.HG 3.76

Dr. G. Lavidas

Department	Offshore Engineering
Telephone	+31 15 27 83864

L.M. Leclercq

Department	Transport and Planning
-------------------	------------------------

Prof.dr. Z. Li

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 82325
Room	23.S2 2.29

Dr.ir. R.E.F. Lindeboom

Unit	Civiele Techniek & Geowetensch
Department	Sanitary Engineering
Telephone	+31 15 27 82069
Room	23.HG 4.61

Dr. X. Liu

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 87918
Room	23.S2 2.26

Dr. E. Lourens

Department	Dynamics of Structures
Telephone	+31 15 27 87568
Room	23.HG 3.65

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 87568
Room	23.HG 3.65

Prof.dr.ir. P.C. Louter

Department	Applied Mechanics
Telephone	+31 15 27 84450
Room	23.HG 6.50

Department	Applied Mechanics
Telephone	+31 15 27 84450
Room	23.HG 6.50

A.P. Luijendijk

Department	Coastal Engineering
Room	23.HG 3.68

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Room	23.HG 3.68

M. Lukovi

Unit	Civiele Techniek & Geowetensch
Department	Concrete Structures
Telephone	+31 15 27 82320
Room	23.S2 1.04

Dr. C. Maat

Unit	Bouwkunde
Department	Transport and Planning
Telephone	+31 15 27 87640
Room	23.HG 4.25

C. Mai Van

Department	Hydraulic Structures and Flood Risk
-------------------	-------------------------------------

Dr.ir. P. van der Male

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 83473
Room	23.HG 3.71

Dr. P.M.N. Mares Nasarre

Department	Hydraulic Structures and Flood Risk
-------------------	-------------------------------------

Dr. V.L. Markine

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 83206
Room	23.S2 2.33

I. Martínez Josemaría

Department	Transport and Planning
Room	23.HG 4.35

F.P. van der Meer

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 85918
Room	23.HG 6.39

Dr. A.A. Mehrotra

Department	Applied Mechanics
-------------------	-------------------

F. Messali

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 82529
Room	23.HG 6.50.1

Prof.dr. A. Metrikine

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 84749
Room	23.HG 3.59.1

Unit	Civiele Techniek & Geowetensch
Department	Engineering Structures
Telephone	+31 15 27 84749
Room	23.HG 3.59.1

Dr.ir. O. Morales Napoles

Dr. Y. Mosleh

Department	Bio-based Structures & Materials
Telephone	+31 15 27 86776

Dr.ir. E. Mosselman

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering

S. Muraro

Department	Geo-engineering
Room	23.KG 00.510

Dr.ir. G.A. van Nederveen

Unit	Civiele Techniek & Geowetensch
Department	Integral Design & Management
Telephone	+31 15 27 81854
Room	23.HG 6.64

Dr. I. Nevzgodin

Unit	Bouwkunde
Department	Heritage & Values
Telephone	+31 15 27 87742
Room	08.01WEST130

Dr.ir. D.J.M. Ngan-Tillard

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Telephone	+31 15 27 83325
Room	23.KG 00.140

Drs.ir. R.E.P. de Nijs

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Room	23.KG 00.510

Dr.ir. M. Nogal Macho

Department	Integral Design & Management
Telephone	+31 15 27 81829
Room	23.HG 6.62

C. Noteboom

Department	Applied Mechanics
Room	23.HG 6.48

Dr. A.A. Nunez Vicencio

Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 89355
Room	23.S2 2.35

M.C. Onderwater

Department	Coastal Engineering
-------------------	---------------------

N. van Oort

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Room	23.HG 4.10.2

Dr.ir. M. Ottele

Unit	Civiele Techniek & Geowetensch
Department	Materials and Environment
Telephone	+31 15 27 87439
Room	23.HG 6.04

N.G. Ozerkan

Department	Materials and Environment
-------------------	---------------------------

Dr.ir. J.A. Pascoe

Department	Structural Integrity & Composites
Telephone	+31 15 27 86604
Room	62.NB 0.49

Ir. S. Pasterkamp

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 84982
Room	23.HG 6.46

Dr. M. Pavlovic

Unit	Civiele Techniek & Geowetensch
Department	Steel & Composite Structures
Telephone	+31 15 27 83382
Room	23.S2 2.58

S.G. Pearson

Department	Environmental Fluid Mechanics
Room	23.S3.00.120

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Room	23.S3.00.120

Ir. A.J. Pel

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 84874
Room	23.HG 4.35

D.J. Peters

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Telephone	+31 15 27 89030
Room	23.HG 3.87

C. Piccoli

Department	Rivers, Ports, Waterways and Dredging Engineering
-------------------	---

F. Pisanò

Department	Geo-engineering
Telephone	+31 15 27 82844
Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Telephone	+31 15 27 82844

M. Poliotti

Department	Concrete Structures
-------------------	---------------------

Dr. M.A. Popescu

Department	Applied Mechanics
-------------------	-------------------

Dr.ir. B.C. van Prooijen

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 83365
Room	23.00.080

E. Quaglietta

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 82761
Room	23.HG 4.18

E. Ragno

Unit	Civiele Techniek & Geowetensch
Department	Hydraulic Structures and Flood Risk
Room	23.HG 3.75

Ir. J.P.G. Ramler

Department	Integral Design & Management
Department	Integral Design & Management

Ir. G.J.P. Ravenshorst

Unit	Civiele Techniek & Geowetensch
Department	Bio-based Structures & Materials
Telephone	+31 15 27 85721
Room	23.S2 2.57

Dr.ir. A.J.H.M. Reniers

Unit	Civiele Techniek & Geowetensch
Department	Environmental Fluid Mechanics
Telephone	+31 15 27 85426

Room	23.HG 2.93
-------------	------------

M. Rinaldi

Department	Transport and Planning
-------------------	------------------------

Dr.ir. A.M. Salomons

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 88556
Room	23.HG 4.45

Dr.ir. W.J. Schakel

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 84427
Room	23.HG 4.02.2

Dr. R.M.J. Schielen

Department	Rivers, Ports, Waterways and Dredging Engineering
Room	23.S3.00.100

Dr.ir. H.R. Schipper

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 89933

M.A. de Schipper

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 87692
Room	23.HG 3.68

Dr.ir. H.E.J.G. Schlangen

Unit	Civiele Techniek & Geowetensch
Department	Materials and Environment
Telephone	+31 15 27 86535
Room	23.HG 6.17

A.C.B. Schuurman

Department	Applied Mechanics
Room	23.HG 6.48

Dr. S. Sharif Azadeh

Department	Transport and Planning
Telephone	+31 15 27 82307

C. Shen

Department	Railway Engineering
Telephone	+31 15 27 81373
Department	Railway Engineering
Telephone	+31 15 27 81373
Unit	Civiele Techniek & Geowetensch
Department	Railway Engineering
Telephone	+31 15 27 81373

Dr.ir. E.C. Slob

Unit	Civiele Techniek & Geowetensch
Department	Applied Geophysics and Petrophysics
Telephone	+31 15 27 88732

Room	23.HG 3.14
-------------	------------

Dr.ir. C.J. Sloff

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering

Prof.dr.ir. L.J. Sluijs

Unit	Civiele Techniek & Geowetensch
Department	Materials- Mechanics- Management & Design
Telephone	+31 15 27 82728
Room	23.HG 6.42

Dr. M. Snelder

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 84981
Room	23.HG 4.18

Dr. L.M. Stancanelli

Department	Rivers, Ports, Waterways and Dredging Engineering
-------------------	---

Dr.ir. M.J.M.M. Steenbergen

Unit	Civiele Techniek & Geowetensch
Department	Mechanics and Physics of Structures
Telephone	+31 15 27 83385
Room	23.S2 2.32

P. Taneja

Unit	Civiele Techniek & Geowetensch
Department	Rivers, Ports, Waterways and Dredging Engineering
Telephone	+31 15 27 85438
Room	23.HG 3.86

Dr. T. Tankova

Department	Steel & Composite Structures
-------------------	------------------------------

Dr. R. Taormina

Department	Sanitary Engineering
Telephone	+31 15 27 88368
Room	23.HG 4.51

Prof.dr.ir. L.A. Tavasszy

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 83346
Room	23.HG 4.10.1
Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 83346
Room	23.HG 4.10.1

Dr.ir. C.C.J.M. Tiberius

Unit	Civiele Techniek & Geowetensch
Department	Mathematical Geodesy and Positioning
Telephone	+31 15 27 82573
Room	23.HG 2.23

Dr. M.F.S. Tissier

Unit	Civiele Techniek & Geowetensch
Department	Environmental Fluid Mechanics
Telephone	+31 15 27 85122

Room	23.HG 2.92
-------------	------------

A. Tsouvalas

Department	Dynamics of Structures
Telephone Room	+31 15 27 89225 23.HG 3.61
Unit Department	Civiele Techniek & Geowetensch Offshore Engineering
Telephone Room	+31 15 27 89225 23.HG 3.61

Prof.dr.ir. W.S.J. Uijtewaal

Unit Department	Civiele Techniek & Geowetensch Environmental Fluid Mechanics
Telephone Room	+31 15 27 81371 23.S3.00.060

Dr. P.J. Vardon

Unit Department	Civiele Techniek & Geowetensch Geo-engineering
Telephone Room	+31 15 27 81456 23.KG 00.490

A. Varveri

Unit Department	Civiele Techniek & Geowetensch Pavement Engineering
Telephone Room	+31 15 27 84665 23.S2 1.26

Dr.ir. F.A. Veer

Unit Department	Bouwkunde Structural Design & Mechanics
Telephone Room	+31 15 27 81358 26.B2.070

Dr. D.M. van de Velde

Unit Department	Techniek, Bestuur & Management Organisatie en Governance
Room	31.B2.020

Prof. M. Veljkovic

Unit Department	Civiele Techniek & Geowetensch Steel & Composite Structures
Telephone Room	+31 15 27 85816 23.S2 2.52

Dr.ir. A.A. Verhagen

Unit Department	Civiele Techniek & Geowetensch Mathematical Geodesy and Positioning
Room	23.HG 2.16

Ir. F.R.S. Vinke

Unit Department	Civiele Techniek & Geowetensch Rivers, Ports, Waterways and Dredging Engineering
Room	23.HG 3.93

Dr. D.V. Voskov

Unit Department	Civiele Techniek & Geowetensch Reservoir Engineering
Telephone Room	+31 15 27 87829 23.HG 3.28

Ir. P.A. de Vries

Unit	Civiele Techniek & Geowetensch
Department	Steel & Composite Structures
Telephone	+31 15 27 84034
Room	23.S2 2.56

Dr.ir. S. de Vries

Unit	Civiele Techniek & Geowetensch
Department	Coastal Engineering
Telephone	+31 15 27 89220
Room	23.HG 3.70

Dr. H. Wang

Department	Railway Engineering
Telephone	+31 15 27 84019
Room	23.S2 2.31

Ir. J.W. Welleman

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 84856
Room	23.HG 6.55

T.H.G. Wils

Department	Geo-engineering
Room	23.KG 00.060

Prof.dr.ir. A.R.M. Wolfert

Unit	Civiele Techniek & Geowetensch
Department	Engineering Structures
Telephone	+31 15 27 86636
Room	23.HG 6.66

Dr. D. Wüthrich

Department	Hydraulic Structures and Flood Risk
-------------------	-------------------------------------

Dr. Y. Yang

Unit	Civiele Techniek & Geowetensch
Department	Concrete Structures
Telephone	+31 15 27 82277
Room	23.S2 2.05

Dr. Z. Yang

Department	Railway Engineering
Telephone	+31 15 27 81525
Room	23.S2 2.35

G. Ye

Unit	Civiele Techniek & Geowetensch
Department	Materials and Environment
Telephone	+31 15 27 84001
Room	23.HG 6.11

Y. Yuan

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 83632
Room	23.HG 4.24

Dr.ir. M. Zijlema

Unit	Civiele Techniek & Geowetensch
Department	Environmental Fluid Mechanics
Telephone	+31 15 27 83255

Room	23.HG 2.94
-------------	------------

Dr.ir. C. Zwanenburg

Unit	Civiele Techniek & Geowetensch
Department	Geo-engineering
Room	23.KG 00.500

J.A. Álvarez Antolínez

Department	Coastal Engineering
Room	23.HG 3.68

Dr.ir. B. Šavija

Unit	Civiele Techniek & Geowetensch
Department	Materials and Environment
Telephone	+31 15 27 88741
Room	23.HG 6.02
